

When Two Threads Are Better Than One: Modelling HFT Pipeline Latency Under Hawkes Arrivals

Vincent Maciejewski*

September 2026

Abstract

The prevailing view in high-frequency-trading (HFT) system design is that the entire hot path — market-data decode, order-book application, strategy evaluation, order construction — should run on a single-threaded event loop, because each additional thread hop adds wall-clock latency. This paper puts two questions to that view on CME MDP3 market data (NQ front-month, channel 318, regular trading hours, 2025–2026: 276 sessions, 3512 thirty-minute windows, 3.3×10^9 packets). First, what produces the queueing tail behind a single-threaded receiver: the packet rate, the clustering of arrivals, or the number of messages per packet? Second, is a servicing chain cut into two or more stages on separate threads better than one thread, and when?

On the first question: the tail is produced by the self-excitation of the matching engine's transaction stream, and by nothing else that was tested. Not the rate: utilisation is below 0.02 at the service times an HFT hot path occupies, and a window with several times the packet rate has the same tail relative to the service time. Not the message count, which is a separate and smaller mechanism confined to the shortest service times. Not the exchange's packetisation: grouping messages by matching-engine transaction shows that 98.3% of transactions occupy one packet, that the packets of a split transaction arrive $15 \mu\text{s}$ apart, and that 99.9% of the gaps shorter than the feed's $7.5 \mu\text{s}$ tight-gap floor separate different transactions; the packet stream is the transaction stream relayed one packet per event. Transaction starts are self-exciting with a branching ratio of 0.795, equal to the packets' 0.798, and their count variance exceeds Poisson by a factor of 105 at one second. Redrawing transaction times uniformly removes 75–98% of the tail; decoupling transaction sizes from their timing, merging each transaction into one packet, or spreading its packets leaves the tail within 15%. The self-excitation reaches the receiver through two channels whose weight depends on the service time: at $8\text{--}16 \mu\text{s}$ through the surplus of near-simultaneous transactions it produces, four to seventeen times the Poisson share, which a random permutation of the gap sequence preserves (79–88% of the tail survives it); at $64 \mu\text{s}$ and above through runs of transactions, which the permutation destroys (37% and 23% survive). A null that keeps every one-second packet count and randomises within the second has no tail where the real stream has one.

On the second question: yes above a threshold and no below it, and the threshold is a property of the feed. Below the tight-gap floor there is no queue to shorten and a cut adds one $1.7 \mu\text{s}$ hop to every message; above it, one balanced cut removes most of the p_{99} excess for one hop on the median, and pays on every one of the 3512 windows from about $10 \mu\text{s}$. Only the slowest stage matters: a cut that does not shorten the largest stage buys nothing. With the same N cores, a dispatch arrangement that ignores ordering matches or beats the tandem on latency, so the tandem's case is that it keeps the stream in order. At the production operating point of the reference receiver, about $7 \mu\text{s}$, the chain should not be split. We give the analytic framework —

*M2 Tech (16425640 Canada Inc.), Montreal. majeski@gmail.com. KASPAR-HFT is open source at <https://github.com/vincent212/kaspar-hft>.

a burst-limit throughput identity and an exact pathwise reduction of the deterministic-service tandem to a single bottleneck server — test it on the corpus, and compare the arrival statistics against an independent measurement on a live production receiver seven months later, on three instruments; the clustering statistics agree and the production decoder’s service floor coincides with the service time at which the simulated tail appears. Code is in the KASPAR-HFT repository; the packet captures are not redistributable.

Contents

1	Introduction	5
1.1	The design choice, and the safe default	5
1.2	Prior recommendations for single-thread hot paths	5
1.3	Plan of the paper	7
1.4	Contribution and prior-art positioning	8
1.5	Note on the use of AI in this paper	9
1.6	Notation	9
I	Theory	11
2	Analytical framework	11
2.1	Burst-limit throughput theorem	11
2.2	Exact reduction of the tandem to one server, and the scaling bound	13
2.3	The theory in plain terms	16
II	Empirical experiment	19
3	Experiment setup	19
3.1	Data corpus	20
3.2	Packet arrivals are Hawkes-clustered (empirical prerequisite)	21
3.3	Simulator	22
3.4	Stages run concurrently	22
3.5	Sweep design	23
3.6	Null streams and what each removes	23
4	Empirical results	24
4.1	Tail collapse under Hawkes, and its near-absence under Poisson at HFT service times	24
4.2	The tail against three nulls	28
4.3	Transactions, not packets, are the clustered process	28
4.4	The same N cores spent differently	33
4.5	Median cost is linear in N , exactly as $(N - 1)h$	35
4.6	Design equation	35
4.7	Across windows	38
4.8	The tail onset	38
4.9	Conditioning on intensity and criticality	39
4.10	Sensitivity to h	42
5	Evaluation: do empirics match the analytical predictions?	43
6	Cross-validation: the same arrival process, measured live	44
6.1	Two measurements	44
6.2	Statistics that agree	44
6.3	The service floor coincides with the tail threshold	45
6.4	Statistics that disagree, and the reason	46
6.5	Per-packet service is not constant	47

6.6	Span-dependent service	48
III	Best practices: implications for HFT systems design	51
7	The reference instance: KASPAR-HFT and its fast_send pipeline	52
8	Implementation on KASPAR-HFT: two dispatch paths, one handler	54
8.1	Receiver transparency	54
8.2	Measurement, not local reasoning, decides	55
8.3	The production tail: a few large packets	55
8.4	Operational discipline	59
8.5	Separation of code construction from latency optimisation	59
8.6	Contrast with non-fungible frameworks	59
9	Other architectural considerations to re-evaluate under Hawkes	60
10	Recommendations for HFT system designers	61
10.1	Recommendations left unchanged	61
10.2	Recommendations the paper argues against	62
10.3	Recommended design flow	62
IV	Conclusions	64
11	Conclusion	64
12	Future work	66
13	Reproducibility	67
	Appendices	72
A	The Borel batch-size tail at fixed branching ratio	72
B	Numerical check of Theorem 1	72
C	Numerical check of the unequal-stage claims	73
D	A feed on which the tail depends on intensity and criticality	74

1 Introduction

1.1 The design choice, and the safe default

The design of the ingress-to-order pipeline in a high-frequency trading system is a choice between two archetypes: the single-threaded event loop that processes every step of every message on one core (SBE decode, book application, strategy evaluation, order construction, all under one call chain, taking a total service time we denote T), and the multi-actor pipeline — a *tandem* of N servers in series — in which those steps are split across N separate actors, each on its own thread with its own mailbox, performing its part of the work and handing the message to the next across one of $N - 1$ explicit hops whose per-hop wall-clock cost we denote h . The low-latency HFT community is predominantly in the single-thread camp, and single-threaded is generally regarded as the safe choice. The argument cites the cost of a mailbox hop (a mailbox enqueue, a cache-line transfer between the sender’s core and the receiver’s core, and the cost of scheduling or spinning the receiver) and holds that $(N - 1)h$ adds to every message’s end-to-end latency, so that each additional stage is a fixed cost that no throughput or tail benefit recovers.

That argument holds when the single-server throughput $1/T$ exceeds the peak arrival rate. On CME MDP3 the arrival process is strongly self-exciting: packets arrive in Hawkes-clustered bursts [24, 27, 28], and the clustering is the market’s, not the feed’s. The exchange publishes each matching-engine event — a transaction — as, almost always, one packet, and the transaction stream is itself self-exciting with the same branching ratio as the packet stream (Section 4.3); the feed relays the clustering of trading activity without adding to it. During such bursts the instantaneous arrival rate exceeds the single-thread event loop’s service rate $1/T$, and cluster-induced queueing produces a tail one to two orders of magnitude larger than h . The paper’s central empirical observation, on 276 sessions (3512 thirty-minute windows) of NQ front-month packet arrivals, is that this cluster tail dominates end-to-end latency: the 99th-percentile latency (p_{99}) of a constant-service queue driven by real packet arrivals is 2.2 times the service floor at the corpus-median window for a $16\mu\text{s}$ task and 4.6 times for a $32\mu\text{s}$ task, and 2.5 and 6.1 times at the 95th-percentile window. Under a rate-matched Poisson null of the same arrival count the p_{99} still equals the median at $16\mu\text{s}$ and is only 1.3 times it at 32, against the Hawkes arm’s 4.6; over this band the tail is almost entirely a clustering effect.

The question is then whether splitting the pipeline into $N > 1$ stages, each an independent actor on its own core with per-stage service T/N , reduces the tail, and whether the additional median cost $(N - 1)h$ is small relative to the tail reduction. Neither architecture is better unconditionally. Both conditions hold above a service time that depends on the feed’s arrival statistics and fail below it; the paper measures that threshold on one feed, and its recommendation is that designers measure it on theirs.

1.2 Prior recommendations for single-thread hot paths

The single-thread rule has a lineage in the low-latency-systems community, summarised here in reverse chronological order. It has three layers: the HFT-specific codification of the last decade, the general low-latency-systems and event-driven-server culture from which HFT inherited it, and one prominent published counter-argument that the field discarded.

Modern HFT-community codification (2010–present). The founding statement of the current HFT-community formulation is Thompson’s Single Writer Principle [37] — “a single-writer design consistently outperforms a ‘properly’ lock-free or wait-free multi-writer design on most realistic

workloads” — motivated by and applied in the LMAX Exchange architecture [38, 39]. Rigtorp’s widely cited low-latency tuning guide [40] formalises the operational rule (“run a single thread in `SCHED_OTHER` per core and use busy waiting/polling”), and Chronicle Software’s public engineering materials [41] adopt the same architecture for market-data connectors (“within a single event loop . . . busy-spinning and affinity locks”), pinning Chronicle Queue’s design to a single writer per queue. The Aeron messaging library [42] inherits the same lineage. The academic side is thinner, but Kwan et al. [43] codify the pattern as the standard low-latency-HFT design pattern, and recent practitioner posts [44] restate it explicitly (“all critical trading logic runs in a single-threaded event loop, eliminating context switches and lock contention entirely”). Inside HFT specifically, the same recommendation has been reinforced by two decades of private prop-trading arms-race practice among the Chicago desks (Getco, Jump, Optiver, DRW; late 2000s), reported second-hand in industry retrospectives but without an open-literature reference.

Older event-driven-server lineage (1995–2010). HFT did not invent this rule; it inherited it from the event-driven-server culture. The single-threaded event loop as an architectural pattern dates at least to Schmidt’s Reactor pattern [45] in the mid-1990s. It was sharpened by Kegel’s C10K essay [46], which named the OS thread-per-connection model as the bottleneck of scalable network servers and argued for event-driven designs. The pattern went mainstream in production with Sysoev’s nginx web server [47], whose single-threaded worker process is explicitly justified as context-switch avoidance, and later with Dahl’s Node.js runtime [48], which pushed the single-threaded event loop into ordinary application code.

Pro-pipelining minority — outside HFT. A legitimate multi-decade pro-pipelining tradition exists in the datacenter and networking-systems literature; it has been the field’s non-single-thread minority throughout the period covered by the previous two paragraphs. The earliest prominent published argument is the Staged Event-Driven Architecture (SEDA) of Welsh et al. [49], which advocated exactly the multi-stage pipeline of independently scheduled event handlers that the present paper’s tandem design revives. Welsh’s 2010 retrospective [50] is often cited as a walk-back to single-thread, but it is actually a hedged pro-pipelining recommendation: it advocates decoupling stages from thread pools, grouping multiple stages within a single thread-pool domain where latency is critical, and putting a separate thread pool and queue in front of any stage with long or nondeterministic runtime — effectively the present paper’s design rule in different vocabulary. Beyond SEDA the pro-pipelining tradition includes the Click Modular Router [51] for staged packet processing; Naiad [52] for low-latency dataflow; Snap [53] for Google’s multi-engine network stack; and the datacenter microsecond-tail-latency scheduler line — Shinjuku [54], Shenango [55], Caladan [56], Perséphone [57] — which argues for centralised multi-core dispatch specifically to control the microsecond-scale tail. Perséphone in particular is targeted at “wide service-time distributions,” the closest existing framing to our arrival-law claim; but the datacenter line frames the problem as heavy-tailed *service*, not clustered *arrivals*. DPDK explicitly supports both run-to-completion and pipeline modes and its documentation recommends pipeline mode “if some set of packets require longer to process” [58]. Inside HFT specifically the pro-pipelining position is rare in the open literature; a Chinese patent [59] describing a pipelined FAST market-data decoder is one commercial instance.

None of the pro-pipelining sources above ties the design recommendation to Hawkes / self-exciting / near-critical arrival dynamics. The closest theoretical hooks are the MMPP-input tandem work of Kim and Chydzinski [16], discussed further in Section 1.4, and the batch-arrival tandem phase-transition analysis of Vinnikov [60]. The present paper’s mechanism, that the recommendation

reverses when arrivals are near-critically self-exciting, is one we have not found in prior work.

The present author’s earlier work belongs on this list with a qualification.¹ An earlier paper by the present author [35] did not argue for the single-thread principle in its own right. That paper addresses a different objection: the actor model had been considered unsuitable for HFT because of asynchronous-dispatch overhead, and `fast_send` is presented there as the mechanism that removes that overhead for co-located actors, making the actor model usable inside a servicing chain that the prevailing design placed on one thread. Its reference architecture (reader to buffer by asynchronous `send`; decode, book and signal chained by `fast_send` on one thread; signal to order manager by asynchronous `send`) takes the single-thread servicing chain as given and does not compare it against more asynchronous topologies. The present paper revises that design context, not the `fast_send` mechanism: on Hawkes-clustered feeds the servicing chain should be split into asynchronous stages once its service time exceeds the threshold established in Part II. `fast_send` remains the mechanism for chaining actors placed on one thread; what changes is where the thread boundaries inside the chain should be. The design rule is stated in Section 11.

1.3 Plan of the paper

The paper asks two questions and answers each in a named place.

1. *What produces the queueing tail behind a single-threaded receiver?* Three candidates are tested. The packet rate: no, at HFT service times (Section 4.9). The clustering of arrivals: yes (Section 4.2), and Section 4.3 identifies the clustered process as the self-exciting stream of matching-engine transactions, acting through the surplus of near-simultaneous transactions at short service times and through runs of transactions at long ones. The number of messages per packet: a second, separate mechanism, which dominates at or below the tight-gap floor and adds a few percent above it (Sections 6.6 and 8.3).
2. *Are two queues better than one?* Yes above a service-time threshold set by the feed and no below it (Sections 4.1 and 4.6); only the slowest stage matters (Theorem 1); with equal cores a dispatch pool that reorders packets matches the tandem on latency, so the tandem’s advantage is that it preserves order (Section 4.4); at the production operating point the answer is no (Section 7).

The paper proceeds in four parts.

- **Theory.** Two closed-form results (Section 2) predict the pipeline behaviour. Proposition 1 shows that under any arrival process an N -stage tandem has an N -fold throughput speedup on isolated bursts. Theorem 1 shows that a deterministic-service tandem reduces exactly to a single server with the largest stage’s service, and that the end-to-end wait is bounded pathwise by $1/N$ of the single-stage wait plus $(N - 1)h$, at every quantile and under every arrival sequence. Corollary 2 shows that under a rate-matched Poisson null the single-stage wait at the q -quantile is already zero whenever the utilisation satisfies $\rho < 1 - q$ — which at this feed’s rate covers p_{99} for every task up to $16\mu\text{s}$ — so over that band there is nothing to compress and splitting costs exactly $(N - 1)h$. The Hawkes–Oakes cluster representation explains why the single-stage wait on a real feed is one to two orders of magnitude above h . A self-exciting

¹The tail compression reported here had been observed in production before its mechanism was understood: splitting a long task into shorter asynchronously connected stages was known to reduce the tail on a per-strategy basis, and the working assumption at the time was that single-thread implementations were simply faster in the inner loop, not that the recommendation depended on the arrival law. The present paper is the explanation of that observation.

process is equivalent to a Poisson stream of immigrant arrivals, each of which generates a cluster of offspring. Cluster size is governed by the branching ratio n , the mean number of direct offspring per arrival: the mean cluster holds $1/(1 - n)$ packets and the largest clusters reach a scale of $(1 - n)^{-2}$ packets (the cluster-size law is the Borel distribution; Appendix A). At the fitted $n \approx 0.8$ the mean is about five and the scale about twenty-five packets, arriving nearly back-to-back. The last member of a burst of K packets waits behind $K - 1$ predecessors on a single server (Proposition 1), so the single-stage wait grows with cluster size; this is the wait the tandem reduces.

- **Simulation.** We build an N -stage tandem Lindley recursion on real CME MDP3 packet-arrival streams at a hop cost $h = 1.7 \mu s$ (the one-way async mailbox cost derived by halving the round-trip figure in Table 4 of Maciejewski [35]) and a service floor swept across $T \in \{2, 4, 8, 16, 32, 64, 128\} \mu s$, from tasks below the hop cost to tasks far above it. We sweep $N \in \{1, 2, 4, 8\}$ per scenario on each of the 3512 thirty-minute RTH windows of the corpus. Each window is also simulated under a uniform-shuffle Poisson null with the same arrival count and support, differing only in the arrival ordering (Section 3.5); the null isolates the contribution of clustering, and is not a model of what any designer assumes. Section 3.6 adds seven further streams that remove the ingredients of the clustering one at a time, at the packet level and at the level of matching-engine transactions.
- **Design equation for practitioners.** From the empirical result we extract design rules for a latency-critical servicing chain (Sections 7 and 8): split a stage only when its service time is above the feed’s tight-gap floor, about $7.5 \mu s$ on this corpus, where a queueing tail first appears, and in practice from about $10 \mu s$, where the tail excess first exceeds the hop cost on nearly every window; choose the partition with the smallest achievable largest stage $s_{\max}$, since the tail depends only on $s_{\max}$ and a cut that does not lower it adds a hop and nothing else; place the largest stage first, at ingress; keep sub-tasks below the bottleneck together on one thread via `fast_send`, using an asynchronous `send` only at boundaries that lower $s_{\max}$; and settle each boundary by end-to-end measurement against a representative feed rather than by local reasoning at a single call site.
- **Live cross-validation.** The corpus statistics are compared with an independent measurement on a production receiver taken seven months after the corpus ends, on three instruments (Section 6); the clustering statistics agree, the measured decoder floor coincides with the service time at which the simulated tail appears, and the sweep is re-run with the measured per-message decode cost (Section 6.6). The live ZN stream’s far tail is traced to the message count of a few large packets (Section 8.3).

1.4 Contribution and prior-art positioning

A literature search covering foundational tandem theory [6–10], tandem theory under bursty arrivals [13–17], Hawkes single-queue and adjacent work [5, 18–22], and empirical and applied HFT and RPC-pipeline work [30–34] together with the datacenter tail-latency literature found no direct prior art on any of the paper’s three claims. The nearest neighbour on the exchange side is Yoon [29], who reports that a matching engine’s tail latency under load is set by queueing at the serialised match loop during bursts; the present paper is the receiver-side counterpart, with the arrival process measured rather than assumed. The nearest on the arrival-process side is the Hawkes literature on order flow [24–28], which models order-book events for microstructure and forecasting and does not connect them to a receiver’s service queue or to the packet process that carries them. The

first claim is theoretical: that the choice between a single-queue (single-thread) architecture and an N -stage tandem (multi-actor pipeline) is not fixed but *reverses with the arrival law*. At the service times an HFT hot path occupies, single-digit to low-tens of microseconds, Poisson arrivals make the single queue the better choice, because at that utilisation there is no wait to compress and each additional stage adds hop latency only; Hawkes-clustered arrivals at the same mean rate make the tandem better on the tail, because the cluster wait it removes exceeds the hop cost it adds. The reversal is between arrival laws at fixed rate and service time. A Poisson queue also develops a tail at sufficiently long service times, but later and more mildly. We have found no prior work stating that the architectural recommendation reverses between these two arrival laws. The second is empirical: a tandem-queue latency sweep over service floors and stage counts, driven by CME MDP3 packet timestamps, controlled against a rate-matched Poisson null, and calibrated with a hop cost calibrated from the low-load round trip measured on a production HFT actor framework. We have found no prior study that combines a real exchange-calibrated tandem sweep with both that Poisson counterfactual and a measured framework hop cost. The third is the identification of the clustered process. Grouping 3.55×10^9 messages by matching-engine transaction shows that the packet stream is the transaction stream relayed one packet per event, that the transaction stream is self-exciting with the same branching ratio as the packets, and — through eight null streams that remove one ingredient of the clustering at a time (Section 3.6) — that the receiver’s tail is produced by the timing of transactions and not by their size, their division into packets, the packet rate, or second-scale load, acting through a surplus of near-simultaneous transactions at short service times and through runs of them at long ones. We have found no published measurement of the packet process of an exchange feed at this level, and no prior test of whether an exchange’s packetisation or the market’s self-excitation is what a receiver queues on. The full reference list appears in Section 13.

1.5 Note on the use of AI in this paper

This paper was produced with Anthropic’s Claude (via Claude Code) under the author’s direction for code generation (the analysis pipeline in the KASPAR-HFT repository), exploratory analysis, mathematical drafting, prior-art search, prose editing, and figure preparation. The analytical results of Part I, including the proof of Theorem 1, were drafted with the model and then independently checked by the author. The author independently reviewed all mathematical arguments, experimental methodology, numerical results, and citations, and takes sole responsibility for the contents of the paper.

1.6 Notation

The following symbols are used throughout.

Symbol	Meaning
T	total service time per message on a single-stage pipeline, in μs
N	number of stages in the tandem pipeline (one actor per stage)
T/N	per-stage deterministic service time in the N -stage tandem
h	per-hop wall-clock cost of the async mailbox handoff between adjacent stages
k	size of an arrival burst (number of packets in one cluster)
$\tau_k^{(N)}$	exit time of the last message of a k -burst under N stages
$s_i, s_{\max}$	per-stage services of an unequal tandem, $\sum_i s_i = T$, and their maximum
$w_j(s)$	Lindley waiting time of message j at a single deterministic server with service s
$W_j^{(N)}, W_q^{(N)}$	end-to-end wait of message j (latency minus T , hops included) under N stages, and its q -quantile
p_q	q -th percentile of the end-to-end latency distribution (p_{50}, p_{99} , etc.)
$\Delta(N)$	clustering tail excess $p_{99}(N) - p_{50}(N)$ (§4.6)
$\Delta_{\text{single}}(s)$	single-server clustering tail excess at deterministic service s ; $\Delta(N) = \Delta_{\text{single}}(T/N)$ (§4.6)
γ	fitted exponent in $\Delta(N) \approx \Delta(1)N^{-\gamma}$
κ	operator's tail-versus-median weight in the design equation
N^*	stage count minimising the design objective $(N - 1)h + \kappa \Delta(N)$ (§4.6)
μ, α, β	Hawkes background rate, jump size, and decay rate (§3.2)
$n = \alpha/\beta$	Hawkes branching ratio
K	Borel-distributed offspring cluster size (§2.2)
$\bar{\lambda}_{\text{pkt}}$	mean packet-arrival intensity per 30-min window, packets per second
n_{pkt}	branching ratio of the packet-arrival Hawkes fit per window
n_{tx}	branching ratio of the transaction-start Hawkes fit per window (§4.3)
$n_{\text{pkt}}^{\#}$	per-window packet count
ρ	utilisation, $\rho = \bar{\lambda}_{\text{pkt}} \cdot T$
τ	bin width of the Fano factor, in seconds (distinct from the service time T)
$F(\tau)$	Fano factor at bin width τ : $\text{Var}[C_\tau]/\text{E}[C_\tau]$
H	Hurst exponent, from log-log slope of $F(\tau)$
$\sigma_i, \bar{\sigma}$	span of packet i (SBE messages per datagram) and its window mean (§6.6)
S_i	span-dependent service time of packet i (§6.6)

The per-window packet count and the packet-Hawkes branching ratio would both naturally be written n_{pkt} ; the paper reserves n_{pkt} for the branching ratio and writes $n_{\text{pkt}}^{\#}$ for the count.

Part I

Theory

2 Analytical framework

Readers not interested in the formal presentation may skip to Section 2.3, which restates the results in practical terms.

Throughout, the *tandem* is the paper’s model of the multi-actor pipeline: the total per-message service time T is split into N stages placed in series, each stage an independent deterministic-service server running on its own thread that performs its share of the work and passes the message to the next stage across a mailbox hop of cost h . The single-thread event loop is the $N = 1$ case, in which one server performs all of T ; adding stages shortens each server’s service but inserts a hop between neighbours.

A proposition, a theorem and a corollary underlie the simulation. Proposition 1 is the isolated-burst identity. A message’s end-to-end *latency* is its *service time* T plus its *queueing wait*, the time it spends behind predecessors before service begins. Running the N stages concurrently raises throughput from $1/T$ to N/T , so an N -stage tandem clears a back-to-back burst N times faster: the service time is unchanged at T , and the last message’s queueing wait falls by a factor of N . This is the classical pipeline speedup, stated without proof.

For example, split a service- T server into two equal $T/2$ stages and feed it a burst of k back-to-back messages. All of the queueing now happens at the first stage: its last message waits behind $k - 1$ predecessors that each occupy it for only $T/2$, a wait of $(k - 1)T/2$ against $(k - 1)T$ on the single server. The second stage never queues: stage-one departures arrive spaced $T/2$ apart, equal to its service time, so each message passes through in $T/2 + h$. The wait is not spread across two queues; it remains in one bottleneck stage that is now twice as fast. The halving requires an equal split: cutting $T = 10$ as $7 + 3$ leaves a bottleneck of 7, so the wait falls only to $7/10$. In general the tail scales with the largest stage $s_{\max}$ rather than with the number of stages; Theorem 1 makes this exact. Theorem 1 is the general statement: the tandem’s end-to-end wait equals the wait at a single server with the largest stage’s service plus $(N - 1)h$, exactly and on every sample path, and that single-server wait is at most $1/N$ of the wait at service T . Both are elementary and independent of the arrival law; Theorem 1 rests on a reduction theorem of Friedman [11] and on the monotonicity of the Lindley recursion. Corollary 2 treats the Poisson null: the Poisson wait at quantile q is zero until $\rho = 1 - q$, which at the utilisation of a CME packet feed covers p_{99} for every task up to $16\mu\text{s}$; over that band splitting costs $(N - 1)h$ and removes nothing. The arrival law enters only through the size of the single-stage wait to which the factor $1/N$ is applied, and the Hawkes–Oakes cluster representation explains why that wait is large on a real feed. Part II measures the slack in the bound on real arrivals.

2.1 Burst-limit throughput theorem

Proposition 1 (Burst-limit N -factor speedup). *Consider an isolated cluster of k arrivals whose j -th member arrives no later than $(j - 1)T/N$ after the first, so that no server idles during the burst (an instantaneous batch is the canonical case). Let $\tau_k^{(1)}$ denote the exit time of the last arrival under a single-stage deterministic-service queue with service T , and let $\tau_k^{(N)}$ denote the exit time under an N -stage tandem with per-stage deterministic service T/N and per-stage hop delay h . Then*

$$\tau_k^{(1)} = kT, \quad \tau_k^{(N)} = T + (k - 1)\frac{T}{N} + (N - 1)h, \quad (1)$$

and in particular

$$\lim_{k \rightarrow \infty} \frac{\tau_k^{(1)}}{\tau_k^{(N)}} = N. \quad (2)$$

Remark 2.1. *Proposition 1 is the classical pipelined-tandem throughput result [8, 9]; we state it without proof and use it only to fix the baseline against which the arrival-dependent tail reduction of Theorem 1 is measured. What is paper-specific is the numerical calibration of T and h on a production HFT actor framework and the observation that under real CME MDP3 arrivals the ratio $\tau_k^{(1)}/\tau_k^{(N)}$ is what governs the observed tail-latency compression.*

Reading Equation (1). In the paper’s HFT setting we take $T = 10\mu\text{s}$ as a typical HFT task on a single-thread event loop, and k is the size of a Hawkes-cluster burst that hits the pipeline back-to-back. The single-stage formula $\tau_k^{(1)} = kT$ says that in an event-loop design the last message of a k -burst waits behind $k - 1$ predecessors, each of which spends the full T on the one server; a 10-packet cluster on a $10\mu\text{s}$ server therefore has a last-message latency of $100\mu\text{s}$. The tandem formula $\tau_k^{(N)} = T + (k - 1)(T/N) + (N - 1)h$ decomposes the last-message latency into three physical pieces: (i) T , the total service work every message must traverse regardless of N ; (ii) $(k - 1)(T/N)$, the pipelined wait behind $k - 1$ predecessors at throughput N/T rather than $1/T$; and (iii) $(N - 1)h$, the one-time hop-cost tax paid to traverse $N - 1$ mailbox boundaries on the way through. The single message ($k = 1$) pays only (i) and (iii). Under Hawkes arrivals term (ii) is large on a single stage and falls by a factor of N under a tandem, while term (iii) is an additive cost of $(N - 1)h$.

Reading Equation (2). The ratio $\tau_k^{(1)}/\tau_k^{(N)}$ tends to N as the cluster size k grows, meaning that in the burst-saturation limit a tandem of N actors clears the burst N times faster than an event loop. This is the classical pipeline throughput speedup restated for HFT: a 4-actor pipeline drains a large burst in one-quarter of the wall-clock time that a single-thread event loop would take, and a 12-actor pipeline in one-twelfth. The speedup is obtained with the same total service work per message: the pipeline does not reduce the work per event, it overlaps the work across cores so that the queue behind a cluster clears in parallel. The arrival law does not enter Equation (2); what the arrival law controls is *how often* the burst regime is entered, and Theorem 1 shows the same factor governs every message on every sample path.

On what T is, and on the natural stage boundaries. The paper adopts $T = 10\mu\text{s}$ as a ballpark typical HFT single-thread event loop. Any real HFT pipeline of that duration is already a sequential chain of internally atomic sub-tasks (SBE decode, order-book application, strategy touch, order construction on CME MDP3, per Section 8), so the N stages of the tandem are placed at the existing boundaries between those sub-tasks; the paper is not subdividing any single sub-task, and in particular is not subdividing the SBE decode, which is a linear pass over the wire bytes of a single message and cannot be split. Ten microseconds is chosen for exposition; none of the paper’s qualitative results depend on it. The relevant knob for the design equation is the ratio T/h : doubling T at fixed h doubles the cluster-wait terms of Equation (1) and leaves the hop tax unchanged. The floor below which splitting cannot pay (Section 4.6) turns out to be set by the feed rather than by h : it is the arrival stream’s tight-gap floor, and on this corpus it binds well above $2h$. The paper writes “per-stage service T/N ” as a simplifying modelling convention that assumes equal-sized stages; in reality the per-stage services s_i sum to T but are unequal, and the achievable pipeline throughput is $1/\max_i s_i$ rather than N/T . Theorem 1 handles the unequal case exactly: the factor becomes

$s_{\max}/T$ in place of $1/N$, the tail still contracts under Hawkes, and the median still grows linearly at slope h per hop.

The mechanism applies to any deterministic-service tandem regardless of the arrival law; the arrival law determines how often large- k clusters occur. The next subsection makes the per-burst factor a pathwise bound.

2.2 Exact reduction of the tandem to one server, and the scaling bound

Proposition 1 treats one isolated burst. The result that carries the paper's design claim is that the same N -factor holds pathwise for every message under every arrival sequence, at the level of the end-to-end wait. Fix an arrival sequence $a_1 \leq a_2 \leq \dots$ (any point process, clustered or otherwise, with batches allowed as coincident times) and write $\delta_j = a_{j+1} - a_j \geq 0$. For a single FIFO server with deterministic service s fed by this sequence, the waiting time of message j obeys the Lindley recursion [12]

$$w_1(s) = 0, \quad w_{j+1}(s) = (w_j(s) + s - \delta_j)^+. \quad (3)$$

Define the end-to-end wait of message j in the N -stage tandem as $W_j^{(N)} = (\text{exit time}) - a_j - T$, i.e. latency in excess of the total service work; under $N = 1$ this is $W_j^{(1)} = w_j(T)$.

Theorem 1 (Reduction and scaling bound). *Consider an N -stage tandem of FIFO servers with deterministic per-stage services $s_1, \dots, s_N$ summing to T , deterministic hop delay h between consecutive stages, unbounded buffers, and an arbitrary arrival sequence. Let $s_{\max} = \max_i s_i$. Then for every message j and every sample path,*

$$W_j^{(N)} = w_j(s_{\max}) + (N - 1)h, \quad (4)$$

$$w_j(s') \leq \frac{s'}{s} w_j(s) \quad \text{for all } 0 < s' \leq s, \quad (5)$$

and consequently

$$W_j^{(N)} \leq \frac{s_{\max}}{T} W_j^{(1)} + (N - 1)h = \frac{W_j^{(1)}}{N} + (N - 1)h \quad \text{under equal stages } s_i = T/N. \quad (6)$$

Since Equation (6) holds pathwise, it holds for every quantile of the stationary distribution whenever one exists: $W_q^{(N)} \leq W_q^{(1)}/N + (N - 1)h$ for all q . The same inequality holds for the empirical (sample) quantiles computed over any finite trace, directly from the pathwise bound and without invoking stationarity; the sample-quantile form is what Part II measures, and it should be read as a statement about the observed windows rather than about a stationary-distribution quantile or a finite-sample maximum.

Proof. Equation (4), equal stages. Departures from a FIFO server with deterministic service s are spaced at least s apart, because each departure requires s of uninterrupted service after the previous one. A FIFO server with deterministic service s whose arrivals are spaced at least s apart never queues: by induction, message j departs at its arrival time plus s , which is at or before the arrival of message $j + 1$. Stage 1 is a single server with service T/N fed by the original sequence, so message j leaves it at $a_j + w_j(T/N) + T/N$. The arrival sequence at stage 2 is the stage-1 departure sequence shifted by h , spaced at least T/N apart, so stage 2 adds exactly $h + T/N$ and its departures are again spaced at least T/N apart; the same holds at each subsequent stage. Summing, the exit time is $a_j + w_j(T/N) + N \cdot T/N + (N - 1)h$, and subtracting $a_j + T$ gives Equation (4) with $s_{\max} = T/N$.

Equation (4), unequal stages. By the order-invariance theorem of Friedman [11], for a tandem of FIFO stages with constant service times, unlimited queues, and any arrival sequence, each customer's time in system is independent of the order of the stages. Deterministic hop delays preserve this: a hop is a constant shift of one stage's departure sequence into the next stage's arrival sequence, and constant shifts commute with the Lindley recursion, so reordering stages together with their hops leaves every exit time unchanged. Reorder so that the stage with service $s_{\max}$ comes first. Its departures are spaced at least $s_{\max}$ apart, and every subsequent stage has service $s_i \leq s_{\max}$, so by the no-queueing argument above each of them adds exactly $s_i + h$ to every message. The exit time is $a_j + w_j(s_{\max}) + \sum_i s_i + (N - 1)h$, and subtracting $a_j + T$ gives Equation (4).

Equation (5). By induction on j with $w_1(s') = 0 = (s'/s)w_1(s)$. The map $x \mapsto (x + s' - \delta_j)^+$ is nondecreasing, so

$$w_{j+1}(s') = (w_j(s') + s' - \delta_j)^+ \leq \left(\frac{s'}{s}w_j(s) + s' - \delta_j\right)^+ \leq \left(\frac{s'}{s}w_j(s) + s' - \frac{s'}{s}\delta_j\right)^+ = \frac{s'}{s}w_{j+1}(s),$$

where the second inequality uses $\delta_j \geq (s'/s)\delta_j$, which holds because $\delta_j \geq 0$ and $s' \leq s$. Combining Equations (4) and (5) with $s' = s_{\max}$, $s = T$ gives Equation (6). $\square$

Remark 2.2 (What the theorem says and what it does not). *Equation (4) says all queueing in an equal-stage deterministic tandem happens at stage 1; stages $2, \dots, N$ contribute exactly $T/N + h$ each to every message. Proposition 1 is the isolated-burst special case, where Equation (6) holds with equality. Equation (6) is a bound rather than an identity because the tandem runs its bottleneck stage at utilisation ρ/N while the single stage runs at ρ : the induction step in the proof of Equation (5) gives away $(1 - 1/N)\delta_j$ at every message j that the single server is still serving when message $j + 1$ arrives, so the tandem clears each inter-arrival gap N times more effectively than the scaled bound credits it for. The paper's empirical γ (Section 4.6), defined by $W_q^{(N)} - (N - 1)h \approx N^{-\gamma} W_q^{(1)}$, therefore satisfies $\gamma \geq 1$ in the model, and the excess $\gamma - 1$ measures that slack. Nothing in the theorem depends on the arrival law; what the arrival law determines is the size of $W_q^{(1)}$ that the factor $1/N$ is applied to, which is the subject of the next remark. The hop cost h enters every quantile as exactly $(N - 1)h$, in the median and in the tail alike.*

Remark 2.3 (Unequal stages: only the largest stage matters). *Real cut points do not give equal stages. Equation (4) states the consequence: the end-to-end wait of any deterministic tandem equals the wait at a single server with service $s_{\max}$, plus h per hop. Three consequences follow.*

1. *The tail factor is $s_{\max}/T$, and the stage count enters only through the hop tax. Splitting a $10\mu\text{s}$ task as $7 + 3$ compresses the tail excess to at most 70% of single-stage, against 50% for $5 + 5$, and both pay one h .*
2. *The order of the stages is irrelevant to end-to-end latency: $7 + 3$ and $3 + 7$ give identical latencies on every sample path. Order determines only where the queue physically sits (below).*
3. *A cut that does not lower $s_{\max}$ adds a hop and nothing else. Refining $7 + 3$ into $7 + 2 + 1$ leaves every quantile of the wait unchanged and raises end-to-end latency by h . Conversely, sub-tasks can be merged onto one thread (via **fast_send**) without any tail cost as long as the merged service does not exceed $s_{\max}$. The design problem is therefore to choose the partition with the smallest $s_{\max}$ reachable at the natural cut points, and then the fewest hops that realise it.*

Appendix C checks all three numerically.

Remark 2.4 (Where the burst goes: the bottleneck absorbs the clustering). *The proof of Equation (4) rests on one fact: departures from a deterministic server with service s are spaced at least s apart. Applied to a Hawkes input, that fact says what the bottleneck stage does to the clustering. Upstream of the bottleneck the stream is the raw feed, with intra-cluster inter-arrival times far below $s_{\max}$ and instantaneous intensity far above $1/s_{\max}$. Downstream of the bottleneck, every cluster still leaves as the same group of K messages, so the count-level branching structure is unchanged (the Borel cluster sizes, and the Fano factor at time scales beyond $Ks_{\max}$, are exactly those of the input), but the intra-cluster spacing has been regularised to $s_{\max}$ and the instantaneous intensity is capped at $1/s_{\max}$. The clustering is not removed; it is regularised to a spacing that any consumer at least as fast as the bottleneck can absorb, which is why stages with service $s_i \leq s_{\max}$ never queue. If the bottleneck is stage 1, all of the waiting in the pipeline sits in its mailbox and every downstream mailbox holds at most one message; if the bottleneck is stage $b > 1$, the stages before it queue partially (the raw burst is still bursty relative to their smaller services) and stage b absorbs the remainder, with the same total by Equation (4).*

Four practical consequences follow. First, place the largest stage first, at ingress: end-to-end latency is unchanged by the order, but this confines all queueing to one mailbox, keeps the downstream mailboxes at depth one, and places the burst-absorbing queue where its receiver is continuously busy during a cluster (the regime in which the load-dependent hop cost of Section 4.10 is smallest). Second, any burst-aware logic, such as load shedding, coalescing of superseded book updates, or a cancel-on-burst risk rule, can only observe the raw burst upstream of the bottleneck; downstream it sees a rate- $1/s_{\max}$ stream and cannot tell a cluster from steady flow except by counting. Third, a packet-Hawkes fit run on the bottleneck's departure stream would report a lower intensity peak and a longer effective kernel while recovering the same branching ratio in counts; operators instrumenting a pipeline should fit on the ingress stream. Fourth, deterministic hop delay is what keeps downstream stages queue-free; hop jitter of amplitude ϵ lets a downstream stage queue by at most ϵ per message, so the reduction holds up to that jitter in practice.

Fact (Hawkes–Oakes 1974 cluster representation [2]). *A stationary Hawkes process with background rate $\mu > 0$ and branching ratio $n \in [0, 1)$ is stochastically equivalent to a Poisson-cluster process: a rate- μ Poisson process of immigrant arrivals, each of which independently spawns an offspring cluster whose total size K (including the immigrant) follows the Borel distribution*

$$\Pr(K = k) = \frac{(nk)^{k-1} e^{-nk}}{k!}, \quad k = 1, 2, 3, \dots \quad (7)$$

with mean $E[K] = 1/(1 - n)$ and variance $\text{Var}[K] = n/(1 - n)^3$ [3].

Remark 2.5 (Where Hawkes enters: the size of the single-stage wait). *Under Poisson arrivals at the utilisation of a CME packet feed ($\rho = \bar{\lambda}_{\text{pkt}}T \ll 1$), the single-stage wait is zero for almost every message, so $W_q^{(1)} = 0$ for every quantile q with $\rho < 1 - q$ and Equation (6) has nothing to compress; that is Corollary 2, and at this corpus's rate it holds at p_{99} for all $T \leq 16 \mu\text{s}$. Under Hawkes arrivals the cluster representation puts K near-coincident messages behind one immigrant, and in the instantaneous-batch picture the last of them waits $(K - 1)T$ on a single stage. Stirling's formula applied to Equation (7) gives, for large k ,*

$$\Pr(K = k) \approx \frac{1}{n\sqrt{2\pi}} k^{-3/2} (ne^{1-n})^k, \quad (8)$$

and $ne^{1-n} < 1$ for every $n < 1$, with $-\log(ne^{1-n}) \approx (1 - n)^2/2$ near criticality. The Borel batch size is therefore light-tailed at every fixed $n < 1$: its survival function follows the critical $k^{-1/2}$ power

law only in the window $k \ll k^* = (1 - n)^{-2}$ and decays geometrically beyond it (Appendix A). The consequences for the paper are, first, that the largest clusters, and hence the single-stage cluster wait they produce, have a characteristic scale of order $Tk^* = T(1 - n)^{-2}$, which at the corpus-median $n \approx 0.8$ (Section 3.2) is $\approx 25T$ and at $n = 0.9$ would be $100T$, one to two orders of magnitude above the service floor and the hop cost; second, that heavy-tail results built on subexponential batch sizes, such as the single-big-batch principle for $M^{[K]}/G/1$ [4] and the monotone-separable tail asymptotics of Baccelli et al. [14], do not apply to this batch distribution at any fixed $n < 1$, so no closed-form power-law exponent for the tail probability should be expected; and third, that the design claim does not depend on any of this, because Theorem 1 applies the factor $1/N$ to whatever $W_q^{(1)}$ the feed produces. Part II measures $W_q^{(1)}$ on real arrivals.

Corollary 2 (Poisson null). *Under homogeneous Poisson arrivals at rate $\bar{\lambda}$ with $\rho = \bar{\lambda}T < 1 - q$, the q -quantile of the single-stage wait is $W_q^{(1)} = 0$, because $\Pr(W^{(1)} > 0) = \rho$ for $M/D/1$ by PASTA. By Theorem 1, $W_q^{(N)} \leq (N - 1)h$, and by Equation (4) $W_q^{(N)} \geq (N - 1)h$, so*

$$W_q^{(N)} = (N - 1)h \quad \text{for all } N \text{ and all } q \leq 1 - \rho. \quad (9)$$

At the utilisations of a CME packet feed (corpus-median 452 packets per second, so $\rho = 0.0036$ at $T = 8\mu\text{s}$) this covers p_{99} at every $T \leq 16\mu\text{s}$ in the paper's sweep: end-to-end latency is $T + (N - 1)h$ at p_{50} and at p_{99} alike, and splitting costs $(N - 1)h$ with nothing to compress. Where ρ/N exceeds $1 - q$, Equation (6) still holds and the tandem reduces the Poisson wait as well; splitting then helps under Poisson for the ordinary utilisation reason rather than for clustering. The sweep contains six such p_{99} cells, all at the long-service end: $(T, N) = (32, 1), (64, 1), (64, 2), (128, 1), (128, 2)$ and $(128, 4)$, spanning $\rho/N = 0.0144$ to 0.0578 . It contains one such p_{95} cell ($T = 128, N = 1, \rho = 0.0578 > 0.05$) and many $p_{99.9}$ cells, and Section 4.1 finds the wait in those cells and in no others.

2.3 The theory in plain terms

This subsection restates Proposition 1, Theorem 1 and Corollary 2 informally. The formal statements above are the reference.

Components of latency. A message's end-to-end latency is the sum of three terms: the work done on it (T in total, however many stages that work is spread over), the hop tax (h for every mailbox boundary it crosses, so $(N - 1)h$ in an N -stage pipeline), and the time it spends sitting in queues waiting for a busy actor. The third term is the wait, W . The work is fixed by the code and the hop cost by the framework; the wait is the only term the architecture changes, and on a bursty feed it dominates the tail.

Proposition 1: one burst. Suppose k messages land at once. On a single thread the last of them waits for the $k - 1$ ahead of it, each costing T : it waits $(k - 1)T$. On an N -stage pipeline the first stage takes only T/N per message, so the last message waits $(k - 1)T/N$ at stage 1 and then flows through the remaining stages without waiting again, because each of them is fed at exactly the rate it can serve. The pipeline clears a burst N times faster for the same total work, because N cores process N different messages at the same instant.

Theorem 1: every message, any arrival sequence. Two facts.

1. In a pipeline where each stage does a fixed amount of work per message, only the slowest stage ever builds a queue. Every other stage receives messages no faster than it can process them, because the slowest stage upstream (or, if the slowest stage is downstream, the stages before it) meters them out. The pipeline therefore behaves as one thread whose per-message work is $s_{\max}$, plus the hop cost. With equal stages, $s_{\max} = T/N$.
2. Take any recording of arrival times, real or synthetic, and replay it through two single-threaded servers: one doing T of work per message and one doing T/N . The second server's queue wait for message number j is never more than $1/N$ of the first server's wait for the same message. The reason is that in the faster server every backlog is N times smaller and every idle gap between arrivals clears at least as much of it.

Combined: for every message in every replay, the pipeline's wait is at most the single-thread wait divided by N , plus $(N - 1)h$. Because this holds message by message, it holds for any summary statistic: median, p_{99} , $p_{99.9}$, or maximum. The bound is met with equality on an isolated instantaneous burst and is strict whenever arrivals are spread out, because the pipeline's first stage is idle more often than the single thread and drains more of its backlog in each idle gap. The measured compression is written $N^{-\gamma}$; the theorem guarantees $\gamma \geq 1$, and Section 4 measures $\gamma = 3.3$ and 2.7 at $T = 64$ and $128 \mu s$. At shorter service times the tandem reduces the excess to zero and no exponent is defined.

Role of the arrival law. The arrival law does not enter either theorem. It enters only through the size of the single-thread wait to which the factor $1/N$ is applied. Under Poisson traffic at the load of a CME packet feed, the single thread is idle when almost every message arrives, so for tasks up to $16 \mu s$ its p_{99} wait is zero; there is nothing to divide by N , and the pipeline is simply $(N - 1)h$ slower (Corollary 2). Under the real feed, packets arrive in self-exciting clusters, the single thread's p_{99} is 2.2 to 41.5 times T in the paper's data for tasks of 16 to $128 \mu s$, and the pipeline divides the excess by N or better. The Hawkes cluster representation explains the cluster sizes: they follow the Borel distribution, whose large clusters have of order $(1 - n)^{-2}$ packets, about 25 at the corpus's $n \approx 0.8$.

Design rules.

- The tail improvement is set by the slowest stage, $s_{\max}/T$, and by nothing else about the decomposition. Balanced stages are the best case for a given number of hops.
- A cut that does not reduce the slowest stage costs h at every quantile and improves nothing. Sub-tasks can be merged onto one thread until their combined work reaches $s_{\max}$ at no cost to the tail.
- The order of the stages does not change any message's latency. It changes only where the queue physically sits; putting the slowest stage first keeps all queueing in one mailbox.
- Every hop costs h on every message, at the median and in the tail.
- Whether to split reduces to one comparison: the single-thread wait at the quantile of interest, times $(1 - s_{\max}/T)$, against h per hop added. On a Poisson feed at HFT service times the left side is zero; on a clustered feed at the same rate it is tens to thousands of microseconds.

The guarantees above are exact within a specific model. Its assumptions follow.

Fixed work per message. Each stage is assumed to spend the same amount of time on every message. That is what makes only the slowest stage queue: a stage that sometimes takes longer than its upstream neighbour would occasionally fall behind and build its own queue, and then the pipeline’s wait would exceed the single-server wait at $s_{\max}$. The simulator of Part II uses constant per-packet service, so the theorems describe it exactly; Section 6.6 relaxes this for the message count per packet. Real actors also have work that varies with book state, and for them the theorems are a model to be checked by measurement (Section 8).

No drops, no bounds, in-order. Mailboxes are unbounded, messages are never discarded or coalesced, and each stage processes in arrival order. A framework that drops or merges messages under load changes the arithmetic (in the pipeline’s favour, at the cost of lost messages); a priority queue changes which message pays the wait but not the total.

Fixed hop delay. The hop is modelled as a constant h . A jittery hop lets a downstream stage queue by at most the jitter amplitude per message, so the reduction holds up to that amplitude. The paper’s calibration uses the low-load, wake-from-sleep value of h , which Section 4.10 argues is an upper bound inside a burst.

Equal stages, no cross-core cost beyond h . The N stages are taken to split the work evenly and to exchange nothing but the message: moving book state or cache lines between the cores that run adjacent stages is charged nothing beyond h . Unequal stages are handled by Remark 2.3; a cross-core cost would add to h and is a matter for measurement (Section 4.10).

Proposition 1 needs a dense burst. The one-burst identity assumes the k messages arrive faster than the first stage can drain them, i.e. faster than one per T/N . For a burst that trickles in, the ratio in Equation (2) is $kN/(k + N - 1 + N(N - 1)h/T)$, which climbs toward N as the burst grows. Theorem 1 does not need this assumption; it covers every arrival pattern.

Limits of the theory. It does not say how large the single-thread wait is on a given feed. That number depends on the feed’s clustering and utilisation and, as the remark after the Borel fact explains, has no clean closed form at a fixed branching ratio: the Borel cluster size is light-tailed for every $n < 1$, so the heavy-tail formulas that would give a power-law exponent do not apply. The paper measures the single-thread wait on real CME arrivals instead (Part II). Nor does the theory predict how far below the $1/N$ bound the pipeline lands; that slack is also measured. What the theory guarantees is that the pipeline’s wait is at most $1/N$ of the single-thread wait plus the hop cost, on any arrival sequence.

Part II

Empirical experiment

3 Experiment setup

This section describes the empirical experiment that tests the analytical predictions of Section 2: the data corpus, the confirmation that packet arrivals are Hawkes-clustered (the reason the single-stage wait that Theorem 1 divides by N is large at all), the tandem-Lindley simulator, and the sweep design that produces the numbers reported in Section 4. Figure 1 shows the end-to-end data flow of a single 30-minute window; Algorithm 1 states the full per-window procedure in pseudocode.

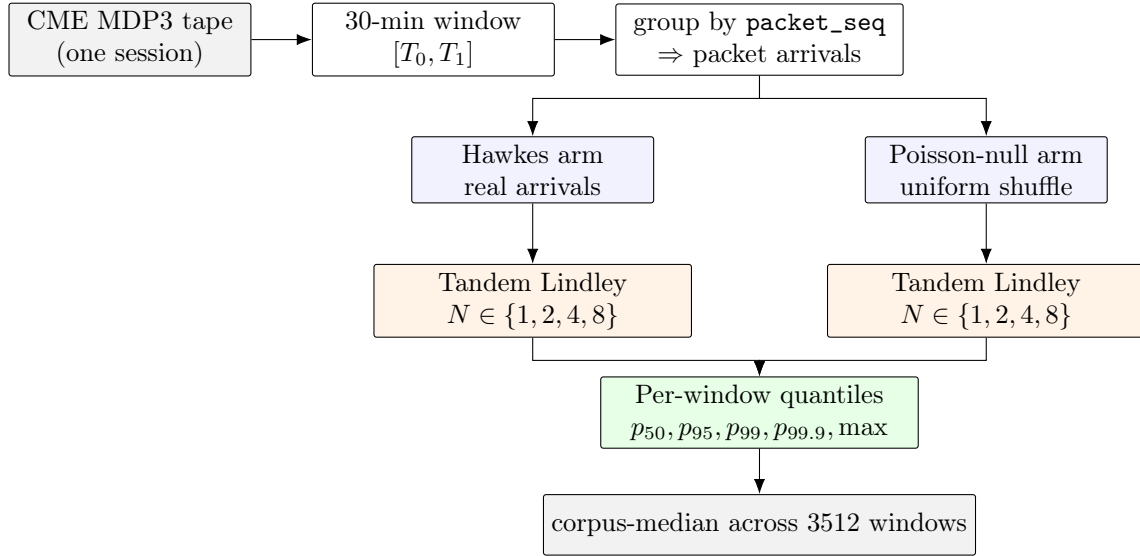


Figure 1. Per-window data flow of the empirical experiment. A single session tape is chopped into 30-minute RTH windows. Each window’s messages are grouped by `packet_seq` into a real packet-arrival stream (Hawkes arm). A rate-matched Poisson-null arm is generated by uniformly shuffling the same arrival count over the same window. Both arms are run through the N -stage tandem Lindley simulator at $N \in \{1, 2, 4, 8\}$. Per-window quantiles are recorded and aggregated by corpus median across all 3512 windows in the 276-session corpus.

Algorithm 1 Per-window procedure of the empirical experiment.

```
1: input: session tape  $S$ , service  $T$ , hop  $h$ , stage counts  $\mathcal{N}$ 
2: for each 30-min RTH window  $W$  in  $S$  with  $[T_0, T_1]$  do
3:    $\{m_i\} \leftarrow$  messages in  $W$ 
4:    $\{(\text{seq}, t)\} \leftarrow \{(\text{packet\_seq}(m_i), \text{sendingTime}(m_i))\}$ 
5:    $\text{arr}_H \leftarrow \text{sort}(\{\min_i t_i : \text{seq}_i = s\}_s)$   $\triangleright$  Hawkes arm (real arrivals)
6:    $n_{\text{pkt}}^\# \leftarrow |\text{arr}_H|$ 
7:    $\text{arr}_P \leftarrow \text{sort}(\text{Uniform}(\min \text{arr}_H, \max \text{arr}_H)^{n_{\text{pkt}}^\#})$   $\triangleright$  Poisson-null arm (seeded per window)
8:   for regime  $\in \{H, P\}$ ,  $N \in \mathcal{N}$  do
9:      $\text{lat} \leftarrow \text{TANDEMLINDLEY}(\text{arr}_{\text{regime}}, N, T, h)$ 
10:    record  $(S, W, \text{regime}, N, \{p_{50}, p_{95}, p_{99}, p_{99.9}, \max\}(\text{lat}))$ 
11:   end for
12: end for
13: return corpus-median of each recorded quantile across all  $(S, W)$  pairs
14: function TANDEMLINDLEY( $\text{arr}, N, T, h$ )
15:    $\text{dep} \leftarrow \text{arr}$ 
16:   for stage  $k = 1, \dots, N$  do
17:     if  $k > 1$  then  $\text{dep} \leftarrow \text{dep} + h$ 
18:   end if
19:    $\text{dep} \leftarrow \text{dep} + \text{LINDLEY}(\text{dep}, T/N)$ 
20: end for
21:   return  $\text{dep} - \text{arr}$ 
22: end function
```

3.1 Data corpus

The corpus is 281 sessions of CME MDP3 channel 318 (NQ) front-month pcap-derived tapes covering 2025-01 through 2026-02, RTH only, holidays and half-days excluded. Each MBO record carries a `packet_seq` (the UDP packet sequence number) and two nanosecond timestamps: `transactTime`, the CME matching-engine event time carried in the MDP3 message body, and `sendingTime`, the gateway’s per-packet UDP header timestamp. Messages are grouped into packets by `packet_seq`, and a packet’s arrival time is its `sendingTime`.

`transactTime` is not a suitable arrival clock. A single matching-engine transaction is routinely split across several UDP packets that leave the gateway microseconds apart, so all of those packets share one `transactTime`; keying arrivals on it collapses distinct packets onto one instant and produces spurious coincident arrivals. Measured across 178.7 million packets, 3.18% carry a `transactTime` shared with at least one other packet, in groups of up to 66; the true emission spread within such a group has a median of $15.4 \mu\text{s}$ and reaches 51 ms. Because the quantity under study is the spacing of closely packed arrivals, this artefact inflates every clustering and tail statistic. `sendingTime` is emitted once per packet at nanosecond resolution and is the receiving pipeline’s input clock; over the same 178.7 million packets, ordered by `packet_seq`, it produces no ties and no sequence inversions. All results below are keyed on `sendingTime`.

Each session tape is chopped into non-overlapping 30-minute RTH windows; for each window we compute the packet-arrival stream and the per-window quantities listed in Section 1.6. The corpus yields 3512 per-window records from the 276 sessions that contain at least one complete RTH window, roughly 13 windows per session; the remaining five of the 281 tapes are holiday or half-day sessions with none.

3.2 Packet arrivals are Hawkes-clustered (empirical prerequisite)

The paper’s simulator input is the packet-arrival stream, so the first empirical question is whether that stream is clustered or Poisson. Three model-free and one model-based diagnostic on the 276-session, 3512-window NQ front-month corpus reject Poisson. The fraction of interarrival gaps shorter than one-tenth of the mean gap has a corpus median of 73% per 30-minute window, a 5th–95th percentile range of 54–78% and a full range of 38–85%, versus the Poisson benchmark $1 - e^{-0.1} \approx 9.52\%$ that would obtain for any exponential-gap process regardless of rate; not one of the 3512 windows comes within a factor of three of the Poisson value. The Fano factor $F(\tau) = \text{Var}[C_\tau]/\text{E}[C_\tau]$ of packet counts in bins of width τ rises from $F(0.1\text{ s}) = 51.7$ to $F(60\text{ s}) = 1027$ across the corpus median, against $F(\tau) \equiv 1$ for Poisson at every τ ; the log-log fit gives a Hurst exponent $H = 0.74$, against $H = 0.5$ for Poisson. These are within-window figures, corpus-median over the 3512 windows; pooling a whole session instead raises the Fano level by roughly a factor of four without changing the shuffle ratio, for reasons taken up in Section 6.4. Running the identical estimator over the paper’s own Poisson-null arm returns $F(\tau) = 1.00$ at $\tau = 0.1$ through 5 s, 0.94 at 60 s, and $H = 0.50$, which confirms that the estimator is unbiased at every bin width used here and that the Hawkes figures above are not an artefact of binning or of finite window length. Fitting the standard exponential-kernel Hawkes process [1] $\lambda(t) = \mu + \sum_{t_i < t} \alpha e^{-\beta(t-t_i)}$ to each window by MLE [23] (Ogata log-likelihood with the standard recursive intensity, closed-form compensator, Nelder–Mead over (μ, α, β)) gives a branching ratio $n = \alpha/\beta$ with a corpus median of 0.798 and a 5th–95th percentile range of 0.761–0.832 across the 3512 windows (extremes 0.566 and 1.000, the latter a single window pinned at the fit boundary), with every fit converged; the same optimum is recovered by a reparameterised L-BFGS-B fit to four decimals on spot-checked windows. This is the same near-critical regime that Filimonov and Sornette [24] and Hardiman et al. [27] document on ES, at a somewhat lower level than their $n \approx 0.9$; they fit trade arrivals rather than packet arrivals, and a packet aggregates several book events, which is expected to lower the fitted self-excitation. Filimonov and Sornette [25] show that an exponential-kernel fit with a constant background over a fixed window overstates n when the background intensity varies within the window, and that ES estimates fall well below criticality once that variation is modelled; the n reported here is descriptive and subject to the same caveat, and none of the paper’s conclusions rests on its value. The model-free diagnostics above, the Fano-against-shuffle comparison of Section 4.3 and the gap histograms there hold whatever the fitted n ; the nonparametric estimates of Achab et al. [26] on EUREX order books give the same near-critical picture without a kernel assumption. A Fisher–Yates shuffle of the interarrival gap sequence within each window, which preserves the marginal gap distribution exactly and destroys only the ordering, removes most but not all of the excess: the corpus-median Fano at $\tau = 5\text{ s}$ falls from 259.7 to 17.6, and at $\tau = 1\text{ s}$ from 108.4 to 17.5, i.e. 94% and 85% of the excess over the Poisson value of 1. What the shuffle removes is specifically the *scaling*: the shuffled Fano is flat at 16.1–17.6 across every bin width from 0.1 to 60 s, giving $H = 0.50$ against the unshuffled $H = 0.74$. All of the long-range dependence therefore lives in the ordering, which is what distinguishes a self-exciting process from a renewal process. The flat residual of ≈ 17 is a scale-free over-dispersion contributed by the gap marginal itself, which is heavy-tailed; so the corpus is neither a renewal process nor a pure Hawkes process with light-tailed gaps, but a self-exciting process with heavy-tailed gaps. This split bounds what the Hawkes fit describes: the fitted n characterises the ordering component, and the simulator is driven by the recorded arrivals, not by the fitted process. The simulator input is therefore an arrival process far from Poisson, and the matched-rate Poisson null is a strict counterfactual. The Hawkes fit is used only as a description of the observed clustering; the tandem reduction of Section 2 does not depend on the Hawkes assumption, since any arrival process with the same clustering drives the same single-server queue. The model-free diagnostics above, not the fit, establish the empirical

premise.

3.3 Simulator

Packet arrivals $t_1 < t_2 < \dots$ enter stage 1. Each stage k is a single-server FIFO queue with deterministic service T/N (own actor, own thread), and departures from stage k are delayed by h before arriving at stage $k + 1$. Standard Lindley recursion per stage:

$$W_{i+1}^{(k)} = \max\left(0, W_i^{(k)} + \frac{T}{N} - (a_{i+1}^{(k)} - a_i^{(k)})\right), \quad (10)$$

with $a_i^{(k+1)} = a_i^{(k)} + W_i^{(k)} + T/N + h$.

3.4 Stages run concurrently

Each stage’s actor runs on its own thread, so stage 2 can be processing message $k - 1$ while stage 1 is processing message k . The simulator captures this because Lindley is applied separately per stage, each with its own busy-idle timeline. What the split buys is not the overlap as such but the shorter queue in front of each stage: a burst that one server of service T holds in a queue of length k is held by the first stage at T/N and drained N times faster.

Worked example. Three packets arriving at $t = 0, 1, 2 \mu\text{s}$; single-stage total service $T = 8 \mu\text{s}$ (a service floor in the sweep of Table 2); two-stage tandem with per-stage service $T/2 = 4 \mu\text{s}$ and hop $h = 1.7 \mu\text{s}$:

	msg 1	msg 2	msg 3
single-stage arrival	0	1	2
single-stage latency (μs)	8.0	15.0	22.0
tandem arrive stage 1	0.0	1.0	2.0
tandem wait stage 1	0.0	3.0	6.0
tandem leave stage 1	4.0	8.0	12.0
tandem arrive stage 2 (+ h)	5.7	9.7	13.7
tandem wait stage 2	0.0	0.0	0.0
tandem leave stage 2	9.7	13.7	17.7
tandem latency (μs)	9.7	12.7	15.7

At wall-clock $t \approx 7$, stage 1 is processing message 2 while stage 2 is processing message 1. Message 1 pays the $1.7 \mu\text{s}$ hop cost (latency $8.0 \rightarrow 9.7$); message 3, at the end of the burst, falls from 22.0 to 15.7 μs . The median rises and the tail falls, at the same message count, with throughput $2/T$ rather than $1/T$. This is the mechanism of Proposition 1.

A chain of synchronous `fast_sends` on one thread shortens no queue and pays $(N - 1)h$; only stages on their own threads do (Section 8).

Calibration and cost. The paper reports numerical results at $h = 1.7 \mu\text{s}$ — the one-way async mailbox cost obtained by halving the round-trip figure in Table 4 of Maciejewski [35] — swept across $T \in \{2, 4, 8, 16, 32, 64, 128\} \mu\text{s}$ to cover the sub-hop regime through the slow-servicing regime. The simulator is deterministic given the tape and (T, h, N) ; one $O(n)$ pass per stage per window makes its wall-clock cost negligible next to the tape read.

3.5 Sweep design

The design asks whether an N -stage tandem reduces p_{99} latency under real Hawkes packet arrivals and whether the same architecture achieves the same compression under a rate-matched Poisson null. For each 30-minute window we run the tandem Lindley of Section 3.3 at $N \in \{1, 2, 4, 8\}$ under two arrival regimes: the real packet-arrival timestamps from the tape (Hawkes arm) and a uniform-shuffle of the same $n_{\text{pkt}}^{\#}$ arrivals over the same support (Poisson-null arm); the further null streams that separate the ingredients of the clustering are defined in Section 3.6. The support is the observed packet span $[\min \text{arr}_H, \max \text{arr}_H]$ rather than the nominal window $[T_0, T_1]$, so the null is rate-matched to the arrivals it replaces rather than to wall-clock; the two coincide on busy windows and the former is the stricter counterfactual on sparse ones. The null is therefore a *conditional homogeneous-Poisson* process: a homogeneous Poisson process at rate $n_{\text{pkt}}^{\#}/(\max \text{arr}_H - \min \text{arr}_H)$ conditioned on the observed count over the observed support. Per (window, N , regime) we record $p_{50}, p_{95}, p_{99}, p_{99.9}$, max of end-to-end latency. Aggregation is corpus-median across all windows; the two headline curves per scenario are $p_{99}(N)$ and $p_{50}(N)$ under Hawkes and Poisson overlaid.

The paper’s empirical calibration is $h = 1.7 \mu\text{s}$ (one-way async cross-thread mailbox cost, per Table 4 of Maciejewski [35], halved for one-way) swept over $T \in \{2, 4, 8, 16, 32, 64, 128\} \mu\text{s}$ so the scenarios span from the sub-hop regime through the slow-servicing regime. Theorem 1 predicts $p_{99}^H(N) - (N - 1)h - T \leq (p_{99}^H(1) - T)/N$ on every window, i.e. a fitted exponent $\gamma \geq 1$ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$; Corollary 2 predicts $p_{99}^P(N) = p_{50}^P(N) = T + (N - 1)h$ exactly in every cell whose per-stage utilisation satisfies $\rho/N < 0.01$, and a strictly positive Poisson wait in every cell where it does not; the prediction is two-sided, so a Poisson p_{99} sitting on the floor at $\rho/N > 0.01$ would falsify it just as a Poisson p_{99} above the floor at $\rho/N < 0.01$ would. Any of (i) a window on which the Hawkes bound is violated, (ii) a Poisson $p_{99}(N)$ on the wrong side of the $\rho/N = 0.01$ boundary, or (iii) Hawkes $p_{50}(N)$ growing at slope $\gg h$ per stage falsifies the paper’s model of the simulator.

3.6 Null streams and what each removes

The claim under test is causal: the clustering of arrivals produces the queueing tail. A tail in the Lindley recursion can be built by four distinct properties of an arrival stream, and the real stream has all four at once:

1. the *rate*, the number of packets per second;
2. the *gap distribution*: how often two consecutive packets land closer together than the service time T , regardless of the order in which those tight gaps occur;
3. the *ordering*: whether the tight gaps come in runs (a burst of ten packets $5 \mu\text{s}$ apart) or lie singly among long gaps;
4. *slow rate variation*: busy and quiet stretches at the scale of seconds within a window.

A null stream is a synthetic stream that keeps some of these and destroys the rest. Running the identical simulator on the real stream and on the null, and comparing the tails, measures the share of the tail that the destroyed property was carrying. Each null is a controlled subtraction of one ingredient, and the design is factorial in the sense that the four properties are removed in different combinations by different nulls. The reference quantity throughout is the single-stage p_{99} excess, $p_{99} - T$, and the statistic reported is the fraction of the real stream’s excess that survives on the null, corpus-median over the 3512 windows. All nulls are built per window from that window’s own real stream, with seeds derived from the session and window identifiers so that every run is reproducible.

Packet-level nulls. The uniform Poisson null P of Section 3.5 redraws every packet time uniformly over the window: it keeps (1) and destroys (2), (3) and (4) together, so a tail that vanishes under it is attributable to clustering in some form but not to any particular form. The *gap shuffle* G takes the window’s sequence of interarrival gaps, permutes it at random (Fisher–Yates, the renewal null already used for the Fano diagnostic of Section 3.2), and re-accumulates arrival times: the same packet count, the same multiset of gaps and therefore the same tight-gap fraction and the same $7.5\ \mu\text{s}$ floor, with the ordering destroyed. G keeps (1) and (2) and removes (3) and (4). A tail that survives G is set by how many tight gaps the stream contains; a tail that G removes needs the runs. The *1 s-binned Poisson* B cuts the window into one-second bins, keeps each bin’s real packet count, and redraws the arrivals uniformly inside the bin: it keeps (1) and (4) and removes (2) and (3). A tail that survives B is a busy-second load effect, not a microsecond one. Section 4.2 reports all three.

Transaction-level nulls. The packet nulls cannot distinguish two accounts of the tight-gap surplus: that the exchange splits one matching-engine event across back-to-back packets, or that the events themselves arrive clustered and the packets relay them. Section 4.3 therefore groups every message by its `transactTime`, the identifier of one matching-engine event (a *transaction*), and builds five further streams in which each transaction is a block — its packets, their message counts and their internal spacing — and exactly one ingredient is removed. Service in these runs is charged per packet at its measured cost, $S_i = T(1 + r(\sigma_i - 1))$ with σ_i the message count and r the live decoder’s slope-to-floor ratio, so that a large packet adds work rather than redistributing it, and the size of a transaction can matter if it does. Table 1 lists all eight streams.

The logic of the two families is the same and they are read together. If P and TP remove the tail, the tail is the timing of the transaction process. If TS , TM and TW leave it, the size of transactions, their division into packets, and the spacing of those packets are not the mechanism, and the exchange is relaying events rather than creating bursts. G and TG remove the same ingredient — the arrangement of events into runs — at the packet and at the transaction level respectively; agreement between them identifies the packet-level ordering channel with the clustering of transactions. What survives G is the surplus of near-simultaneous events over a rate-matched Poisson stream, which Table 5 measures directly and which is itself a product of the self-excitation, since a renewal stream at the same rate carries a fraction of it.

4 Empirical results

4.1 Tail collapse under Hawkes, and its near-absence under Poisson at HFT service times

This section presents the main result. It runs the simulated pipeline over the whole corpus and reports, for each task length and each stage count, how long a typical message takes and how long the slowest one in a hundred takes. Each case is run twice: once on the real packet arrivals, which come in bursts, and once on a control in which the same number of packets is scattered uniformly at random over the window. The comparison isolates the effect of bursts. The findings are these. On randomly scattered arrivals there is almost no slow tail at the task lengths a trading hot path uses, so splitting a task across threads buys nothing and costs a hop per stage. On the real, bursty arrivals a slow tail appears once a task is longer than about seven microseconds, and it grows quickly with task length, until the slowest messages take tens of times longer than the task itself. There, splitting the task into two or more stages cuts the tail sharply for a small, fixed cost on the typical message, and the longer the task, the more stages pay off. In short, it is the bursts, not the average load, that create the tail, and that is why splitting helps on real data and not on the random control.

stream	kept	destroyed	question answered
H	the real stream	—	reference
P	packet count, support	gap distribution, ordering, slow rate variation	is the tail due to clustering in any form?
G	count; the multiset of gaps, hence the tight-gap fraction and the floor	ordering of gaps into runs; slow rate variation	is the tail set by how many tight gaps there are, or by their arrangement?
B	count per one-second bin	microsecond gap structure and ordering	is the tail busy-second load?
TG	every transaction intact, in its original order	the idle gaps between transactions, permuted	does the clustering of transactions into runs build the tail?
TP	every transaction intact	transaction start times, redrawn uniformly	does transaction timing matter at all?
TS	real start times	which transaction shape sits at which start (shapes permuted across starts)	do large transactions arriving at particular moments matter?
TM	real start times	the split of a transaction into packets (each transaction becomes one packet)	does the exchange's packetisation matter?
TW	real start times, the split into packets	the tight spacing inside a split transaction (packets spread by the median idle gap)	does back-to-back spacing inside a transaction matter?

Table 1. The eight arrival streams. P , G and B act on packet times with constant service and are reported in Table 3; TG – TW act on transaction blocks with span-dependent service and are reported in Table 7. Reading the results together: $P \approx 0$ and $TP \approx 0$ say the tail is the timing of the transaction process; TS , TM , $TW \approx 1$ say sizes, packetisation and intra-transaction spacing are not the mechanism; G and TG , which remove the same thing at two levels of description, give the same profile across T and split the clustering into the two channels named in Section 4.3.

Table 2 reports corpus-median end-to-end latency quantiles for every (T, N) cell of the sweep at $h = 1.7 \mu\text{s}$, on 276 sessions and 3512 thirty-minute windows (five of the 281 session tapes contain no complete RTH window and are dropped). Figure 2 plots the same numbers. The Poisson-null columns are the uniform-shuffle counterfactual of Section 3.5 with the same packet count per window.

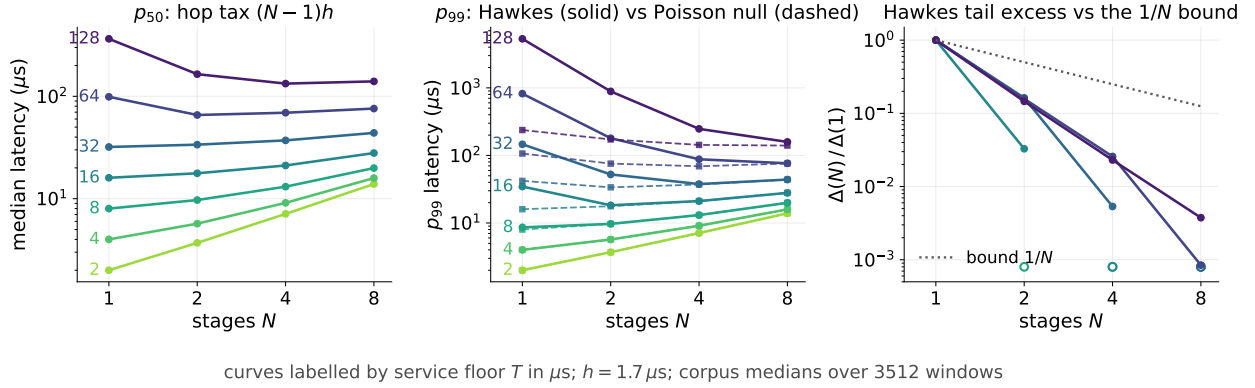


Figure 2. Corpus-median latency versus stage count N for each service floor T (one shade per T , each curve labelled at its $N = 1$ end). Left: median. Centre: p_{99} , Hawkes solid and Poisson null dashed. Right: the Hawkes tail excess ratio $\Delta(N)/\Delta(1)$ against the Theorem 1 bound $1/N$ (dotted); open markers are cells where the tandem drives $\Delta(N)$ to exactly zero, drawn at the panel floor because zero cannot be placed on a log axis. The median rises by exactly h per hop wherever the queue is empty at the median, which is every cell except the three longest-service Hawkes cells. The Hawkes p_{99} falls with N only at $T \geq 16 \mu\text{s}$; at $T \leq 8 \mu\text{s}$ there is no tail to compress and it rises with the hop count, tracking the median. The Poisson p_{99} likewise tracks the median up to $T = 16 \mu\text{s}$ and separates from it at $T \geq 32 \mu\text{s}$, where utilisation queueing begins.

Under the Poisson null the queue is empty when almost every message arrives, so every reported quantile equals the service floor, $p_{50} = p_{99} = T + (N - 1)h$, in every cell with $T \leq 16 \mu\text{s}$. The first exception is $T = 32 \mu\text{s}$, $N = 1$, where Poisson p_{99} is $42.0 \mu\text{s}$ against a floor of 32 : at the corpus-median packet rate of 452 packets per second the utilisation there is $\rho = 452 \times 32 \mu\text{s} = 0.0145$, above the 0.01 at which Corollary 2 says the Poisson wait must first appear at p_{99} , and it does; at $N = 2$ the per-stage utilisation halves to 0.007 and the wait vanishes again. Beyond that the null arm is a perfectly ordinary queue: at $T = 64$ and $128 \mu\text{s}$ the single-stage Poisson p_{99} reaches 106.7 and $237.7 \mu\text{s}$, and splitting reduces it, to 75.7 and $172.4 \mu\text{s}$ at $N = 2$. Under Poisson, splitting is therefore not useless in general; it is useless in the regime an HFT hot path occupies. The $p_{99.9}$ column shows the same pattern one decade lower: the Poisson wait is visible at $N = 1$ for every $T \geq 4 \mu\text{s}$ (single-stage $\rho > 10^{-3}$) and at $N = 2$ for $T \geq 8 \mu\text{s}$, and absent wherever the per-stage ρ/N falls below 10^{-3} . Every one of these switch-on points is the $\rho = 1 - q$ boundary of Corollary 2, which the corpus reproduces at $q = 0.99$ and 0.999 (Section 5). The null arm is a queue with a small, predictable wait, and over $T \leq 16 \mu\text{s}$ splitting under Poisson costs $(N - 1)h$ at every reported quantile.

Under Hawkes a tail appears only above a threshold. At $T = 2$ and $4 \mu\text{s}$ there is no tail: single-stage p_{99} and $p_{99.9}$ both equal the service floor, the clustered feed is indistinguishable from the Poisson null, and splitting costs $(N - 1)h$ and removes nothing. The tail appears between $T = 4$ and $8 \mu\text{s}$ and then grows rapidly: single-stage p_{99} is 1.00 , 1.00 , 1.08 , 2.16 , 4.57 , 12.8 and 41.5 times the service floor at $T = 2, 4, 8, 16, 32, 64, 128 \mu\text{s}$, and single-stage $p_{99.9}$ reaches $97.6T$ at $T = 128$. Splitting into two stages cuts the p_{99} from 34.59 to $18.31 \mu\text{s}$ at $T = 16$, from 146.1 to 52.3 at $T = 32$, from 820.7 to 179.8 at $T = 64$, and from 5316.9 to 886.4 at $T = 128$, in each case for a median cost of $1.7 \mu\text{s}$ per hop; the two-stage p_{99} is below the single-stage p_{99} on 100% of the 3512 windows

T	N	Hawkes (real arrivals)			Poisson null		
		p_{50}	p_{99}	$p_{99.9}$	p_{50}	p_{99}	$p_{99.9}$
2	1	2.00	2.00	2.00	2.00	2.00	2.00
2	2	3.70	3.70	3.70	3.70	3.70	3.70
2	4	7.10	7.10	7.10	7.10	7.10	7.10
2	8	13.90	13.90	13.90	13.90	13.90	13.90
4	1	4.00	4.00	4.00	4.00	4.00	5.78
4	2	5.70	5.70	5.70	5.70	5.70	5.70
4	4	9.10	9.10	9.10	9.10	9.10	9.10
4	8	15.90	15.90	15.90	15.90	15.90	15.90
8	1	8.00	8.61	10.14	8.00	8.00	13.79
8	2	9.70	9.70	9.70	9.70	9.70	11.48
8	4	13.10	13.10	13.10	13.10	13.10	13.10
8	8	19.90	19.90	19.90	19.90	19.90	19.90
16	1	16.00	34.59	53.97	16.00	16.00	29.84
16	2	17.70	18.31	19.84	17.70	17.70	23.49
16	4	21.10	21.10	21.10	21.10	21.10	22.88
16	8	27.90	27.90	27.90	27.90	27.90	27.90
32	1	32.00	146.14	259.11	32.00	42.02	62.02
32	2	33.70	52.29	71.67	33.70	33.70	47.54
32	4	37.10	37.71	39.24	37.10	37.10	42.89
32	8	43.90	43.90	43.90	43.90	43.90	45.68
64	1	98.88	820.70	1548.68	64.00	106.73	126.72
64	2	65.70	179.84	292.81	65.70	75.72	95.72
64	4	69.10	87.69	107.07	69.10	69.10	82.94
64	8	75.90	76.51	78.04	75.90	75.90	81.69
128	1	364.45	5316.85	12487.72	128.00	237.65	289.07
128	2	164.58	886.40	1614.38	129.70	172.43	192.42
128	4	133.10	247.24	360.21	133.10	143.12	163.12
128	8	139.90	158.49	177.87	139.90	139.90	153.74

Table 2. Corpus-median end-to-end latency quantiles in μs at $h = 1.7 \mu\text{s}$, for each service floor T (μs) and stage count N ; 276 sessions, 3512 thirty-minute RTH windows of CME MDP3 NQ front-month packet arrivals. The Poisson null is the uniform shuffle of the same arrival count per window. What the table shows: the median column equals $T + (N - 1)h$ in every cell but three (Hawkes $N = 1$ at 64 and 128 μs and $N = 2$ at 128), so each stage costs exactly one hop at the median. Under real arrivals the single-stage p_{99} sits on the floor at $T \leq 4 \mu\text{s}$, is $1.08T$ at $8 \mu\text{s}$, and then grows faster than the service time: $2.2T$, $4.6T$, $12.8T$ and $41.5T$ at $T = 16, 32, 64$ and $128 \mu\text{s}$, with $p_{99.9}$ roughly twice the p_{99} excess again. Under the Poisson null the single-stage p_{99} equals the median up to $16 \mu\text{s}$ and is $1.3T$, $1.7T$ and $1.9T$ at 32, 64 and 128. One cut ($N = 2$) takes the Hawkes p_{99} from 34.6 to 18.3 at $16 \mu\text{s}$, from 146 to 52 at 32, from 821 to 180 at 64 and from 5317 to 886 at 128, for a median cost of $1.7 \mu\text{s}$; further cuts continue to lower the tail until the per-stage service falls below the floor, after which every quantile rises by h per stage. What follows: below about $8 \mu\text{s}$ there is no queue to shorten and splitting is cost only, under either arrival law; from $16 \mu\text{s}$ the real stream carries a tail that the Poisson stream at the same rate does not, and splitting removes most of it for one hop; the Poisson tail that appears from $32 \mu\text{s}$ is ordinary utilisation queueing ($\rho = \bar{\lambda}T$ reaching 0.01–0.05) and is 11 to 47 times smaller than the real one. The reason is the arrival process: a deterministic server queues only when consecutive gaps are shorter than its service time, the real stream has four to seventeen times the Poisson share of such gaps (Table 5) and arranges them in runs, and a cut halves the service each stage must fit into a gap, so the number of gaps that queue, and the length of the runs that compound, both fall with N (Theorem 1).

at every $T \geq 16 \mu\text{s}$. The stage count that minimises p_{99} rises with T : $N = 1$ at $T \leq 8$, $N = 2$ at $T = 16$, $N = 4$ at $T = 32$, $N = 8$ at $T = 64$ and 128 . The $T = 2$ and $4 \mu\text{s}$ rows, both below the floor, are included to demonstrate that boundary: splitting there is dominated under both arrival laws. Section 4.8 resolves the $4\text{--}8 \mu\text{s}$ onset to $1 \mu\text{s}$ and Section 4.9 explains what sets its location.

4.2 The tail against three nulls

The Poisson null of Table 2 redraws every arrival uniformly over the whole window. That removes three things at once: the microsecond-scale self-excitation, the slower rate variation across the window, and the shape of the gap distribution, including the tight-gap floor of Section 4.9. The tail difference in Table 2 is credited to the first of these, but the mechanism of Section 4.9 runs on the third: a queue forms when consecutive gaps are shorter than the service time, and that is a property of the gap distribution, which a re-ordering of the same gaps leaves untouched. Two further nulls separate the three. The *gap shuffle* permutes each window’s interarrival gaps at random (the renewal null of Section 3.2): same count, same gap distribution, same tight-gap floor, no ordering. The *1 s-binned Poisson* keeps each one-second bin’s own packet count and redraws the arrivals uniformly inside the bin: same second-scale rate variation, no microsecond clustering.

Table 3 gives the corpus-median p_{99} under all four arrival arms. The share of the single-stage p_{99} excess over T that survives the gap shuffle is 88% at $T = 8 \mu\text{s}$, 79% at 16 and 59% at $32 \mu\text{s}$, falling to 37% at 64 and 23% at $128 \mu\text{s}$. The 1 s-binned null keeps none of it at 8 and $16 \mu\text{s}$ and 13%, 6% and 2% at 32, 64 and $128 \mu\text{s}$, within a few points of the uniform null. Two readings follow. The tail is not second-scale load: a null that keeps every one-second packet count has no tail where the real stream has one. And at the service times of an HFT hot path the tail is set mainly by the gap *distribution* — the surplus of gaps shorter than the service time over what a Poisson stream at the same rate carries — with the *ordering* of those gaps into runs supplying the remaining fifth at $16 \mu\text{s}$ and the majority only from $64 \mu\text{s}$ upward. Section 4.3 traces both the surplus and the ordering to one source, the self-excitation of matching-engine transactions, so the attribution to clustering stands, with the mechanism made precise: clustering acts on a short service time through the excess of near-simultaneous arrivals it produces and on a long one through the runs.

4.3 Transactions, not packets, are the clustered process

The nulls of Section 4.2 act on packet times. A packet is the exchange’s unit, not the market’s. CME MDP3 publishes each matching-engine event — a *transaction*, identified by the common `transactTime` carried by every message it generates — as one or more UDP datagrams, and a datagram may carry the end of one transaction and the start of the next. Two accounts of the tight-gap surplus are therefore open. Either the exchange splits transactions across back-to-back packets, in which case the clustering is an artefact of packetisation and a receiver that reassembled transactions would see none of it; or the transactions themselves arrive clustered and the packets relay them. The corpus decides between the two. Every message in the 3512 windows was grouped by `transactTime` and every packet labelled with the transactions it carries: 3.30×10^9 packets, 3.55×10^9 messages, 3.24×10^9 transactions; 3.0% of packets (median window) carry parts of two transactions.

T	N	real	uniform	gap shuffle	1 s-binned
2	1	2.00	2.00	2.00	2.00
2	2	3.70	3.70	3.70	3.70
2	4	7.10	7.10	7.10	7.10
2	8	13.90	13.90	13.90	13.90
4	1	4.00	4.00	4.00	4.00
4	2	5.70	5.70	5.70	5.70
4	4	9.10	9.10	9.10	9.10
4	8	15.90	15.90	15.90	15.90
8	1	8.61	8.00	8.54	8.00
8	2	9.70	9.70	9.70	9.70
8	4	13.10	13.10	13.10	13.10
8	8	19.90	19.90	19.90	19.90
16	1	34.59	16.00	30.75	16.00
16	2	18.31	17.70	18.24	17.70
16	4	21.10	21.10	21.10	21.10
16	8	27.90	27.90	27.90	27.90
32	1	146.14	42.02	99.61	47.36
32	2	52.29	33.70	48.45	33.70
32	4	37.71	37.10	37.64	37.10
32	8	43.90	43.90	43.90	43.90
64	1	820.70	106.73	340.62	112.62
64	2	179.84	75.72	133.31	81.06
64	4	87.69	69.10	83.85	69.10
64	8	76.51	75.90	76.44	75.90
128	1	5316.85	237.65	1306.63	246.03
128	2	886.40	172.43	406.32	178.32
128	4	247.24	143.12	200.71	148.46
128	8	158.49	139.90	154.65	139.90

Table 3. Corpus-median p_{99} in μs under the real arrivals and the three nulls of Section 3.6, for every service floor T and stage count N of the sweep. Uniform is the Poisson null of Table 2; the gap shuffle keeps the multiset of gaps and destroys their ordering; the binned null keeps each one-second packet count and destroys the microsecond structure. What the table shows: at $T \leq 4 \mu\text{s}$ every arm sits on the floor $T + (N - 1)h$, because no gap in the stream is shorter than $4 \mu\text{s}$ and no queue can form. At $N = 1$ and $T \geq 8 \mu\text{s}$ the real stream carries an excess over T (0.61, 18.6, 114, 757 and $5189 \mu\text{s}$ at $T = 8, 16, 32, 64, 128$) that the binned null and the uniform null remove almost entirely (both at or within 2–15% of the excess in every cell, and identical to each other at 8 and $16 \mu\text{s}$), while the gap shuffle retains 88%, 79%, 59%, 37% and 23% of it. At $N \geq 2$ the four arms converge on $T + (N - 1)h$ as soon as the per-stage service T/N falls below the floor (at $T = 16, N = 2$: 18.3, 17.7, 18.2, 17.7). What follows: the tail is not second-scale load, since a stream with the real busy seconds and random microsecond placement has none of it; at $8\text{--}32 \mu\text{s}$ the tail is set mainly by how many gaps shorter than T the stream contains, since a stream with the same gaps in random order keeps most of it; and from $64 \mu\text{s}$ the ordering of those gaps into runs carries the majority. The reason is the Lindley recursion itself: a single gap shorter than T costs the next packet one wait, which the following long gap drains, so a short service time is hit by every tight gap whether or not its neighbours are tight, whereas a long service time needs a sustained run of tight gaps before the backlog exceeds one service, and runs are what the shuffle removes. The tandem's benefit does not depend on which channel carries the tail: a cut that takes T/N below the floor removes the excess under every arm.

	1	2	3	4	5	6–10	11–20	21+
messages per packet	95.897	2.439	0.869	0.401	0.188	0.171	0.027	0.009
packets per transaction	98.295	1.689	0.011	0.002	0.001	0.001	0.000	0.000
messages per transaction	95.362	2.729	0.822	0.558	0.195	0.280	0.038	0.016

Table 4. Size distributions, percent of the pooled corpus (3.30×10^9 packets, 3.24×10^9 transactions). A transaction is one packet 98.3% of the time and two 1.7% of the time; three or more packets is one transaction in ten thousand. The message count per transaction is the message count per packet with the split transactions folded back in.

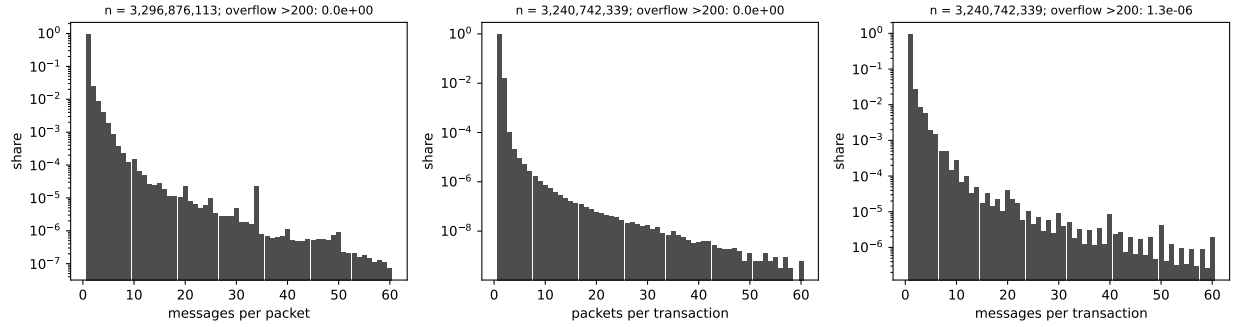


Figure 3. The three size distributions of Table 4 on a logarithmic share axis, counts 1–60. Messages per packet and per transaction have the same geometric decay; packets per transaction has none beyond two.

Packetisation does not produce the tight gaps. Table 4 gives the three size distributions. A split transaction is rare, and when it occurs its packets are not back-to-back: the gap between two packets of the same transaction has a corpus-median p_{50} of $15 \mu s$, 1.6% of such gaps are shorter than $10 \mu s$ and 0.1% shorter than the $7.5 \mu s$ floor; a multi-packet transaction spans $15 \mu s$ at the median and $110 \mu s$ at p_{99} from first to last packet (Figure 4, left, and Figure 5, right). The tight gaps sit between transactions (Table 5).

gap shorter than	share of all packet gaps	of which inside one transaction	Poisson at the same rate
$7.5 \mu s$	2.15%	0.1%	0.56%
$16 \mu s$	20.6%	5.0%	1.24%
$32 \mu s$	34.1%	4.6%	2.46%

Table 5. Tight packet gaps, pooled over the corpus (3.30×10^9 gaps; bin edges at 7.9 , 17.8 and $35.5 \mu s$ on the logarithmic grid of Figure 4). What the table shows: 2.15% of all packet gaps are shorter than the $7.5 \mu s$ floor, 20.6% shorter than $16 \mu s$ and 34.1% shorter than $32 \mu s$; a Poisson stream at the same rate would carry 0.56%, 1.24% and 2.46%, so the real stream has four, seventeen and fourteen times the Poisson share of gaps below the three service times of interest. Of the gaps below the floor, 0.1% lie between two packets of the same transaction; below 16 and $32 \mu s$ the share is 5%. What follows: the tight gaps that make a deterministic server queue are overwhelmingly gaps between different matching-engine events, not the exchange’s division of one event into several datagrams, so a receiver that reassembled transactions before decoding would remove at most 5% of the gaps that queue and none below the floor; and the surplus of tight gaps over Poisson, which Table 3 shows carries most of the tail at 8 – $16 \mu s$, is a property of when transactions occur relative to each other. The reason the surplus exists is the self-excitation established in Table 6: a transaction raises the probability of another within tens of microseconds, which puts far more mass at short gaps than any renewal stream at the same mean rate can produce.

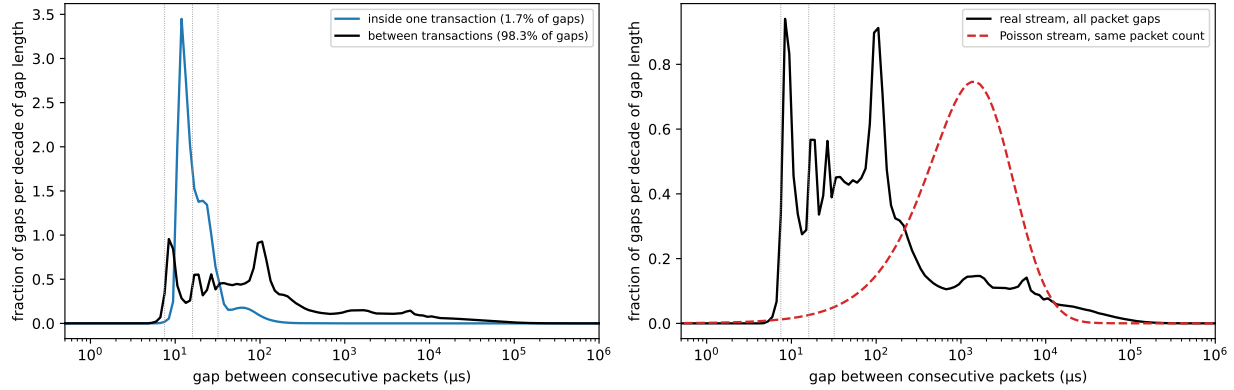


Figure 4. Packet interarrival gaps, pooled over the corpus. The gap axis is logarithmic and the vertical axis is the fraction of gaps per decade of gap length (twenty bins per factor of ten; the area under a curve between, say, 10 and 100 μs is the fraction of that curve's gaps in that range, and the total area is one). Dotted lines at 7.5, 16 and 32 μs . Left: gaps between two packets of the same transaction ($n = 5.6 \times 10^7$ gaps) against gaps between packets of different transactions ($n = 3.24 \times 10^9$ gaps). Each curve is normalised to its own total, so heights compare shapes, not counts (the blue gaps are 1.7% of all gaps, the black 98.3%). What it shows: the intra-transaction distribution is a single narrow peak at 12–16 μs with almost no mass below 10 μs and none below the floor — the pieces of one event are paced by the exchange at about 15 μs , slower than the closest that two packets of different events arrive, so the tightest gaps on the feed are never between pieces of the same event; the inter-transaction distribution has a sharp edge at the 7.5 μs floor, a broad plateau from the floor to about 30 μs , a second mode near 100 μs , and a long tail to seconds. Right: all packet gaps of the real stream (black; since 98.3% of gaps are between transactions this curve is the left panel's black curve with the blue one folded in) against the gaps of a Poisson stream with the same packet count over the same window (red, dashed), the distribution a receiver would face if the market's events were independent of one another at the observed rate. What it shows: the Poisson density is a single hump centred near 1 ms with negligible mass below 100 μs ; the real density puts a third of its mass below 32 μs , with the floor edge and the 100 μs mode absent from the null. The two curves are not two versions of the same shape: the real feed is not a Poisson stream, and the region in which it differs is exactly the region of gaps shorter than an HFT service time. What follows: the edge at 7.5 μs is a property of the gateway — no two packets of any origin are published closer than that — and it sets where a queue first forms; everything between the floor and 32 μs is gaps between different transactions, so the queueing regime of an HFT receiver is populated by distinct matching events arriving within one service time of each other, not by fragments of one event; and the whole of that region is absent from a Poisson stream at the same rate, which is why the uniform null in Table 2 has no tail below 32 μs . The 100 μs mode is the characteristic spacing of successive transactions within a burst, the time scale on which the self-excitation of Table 6 acts.

Transactions are self-exciting. Treating each transaction as one event at the time of its first packet, the transaction-start process carries the same clustering as the packet process (Table 6): the Fano factor of transaction counts rises from 7.6 at 1 ms to 105 at 1 s, against 5.5 to 17 once the idle gaps between transactions are shuffled and 1 for Poisson, and the exponential-kernel MLE gives a branching ratio of 0.795 on transaction starts against 0.798 on packets, with every window's fit converged. The two gap densities of Figure 5 (left) are indistinguishable from each other and unlike the Poisson curve at every scale. The self-excitation belongs to the transaction stream, which is to say to the market: the exchange publishes each matching event as it occurs, and the packet process inherits the clustering of the events.

count window τ	transactions	transactions, idle gaps shuffled	packets
1 ms	7.6	5.5	7.8
10 ms	21.6	10.8	22.2
100 ms	50.5	15.8	51.8
1 s	105.4	17.1	108.4
branching ratio n	0.795	—	0.798

Table 6. Fano factor $\text{Var}[C_\tau]/\text{E}[C_\tau]$ of event counts, corpus median over 3512 windows, for transaction starts, for the same starts with the idle gaps between transactions randomly permuted (transactions kept intact), and for packets; Poisson gives 1 at every τ . Last row: exponential-kernel Hawkes branching ratio, corpus median, all fits converged. The within-window Fano values are lower than the whole-session values of Section 3.2 for the reason given there.

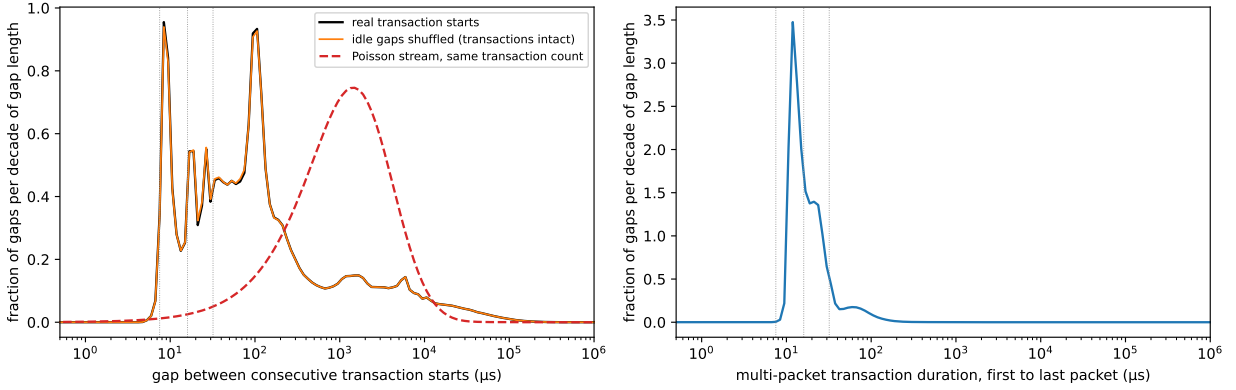


Figure 5. Left: gaps between consecutive transaction starts, pooled over the corpus ($n = 3.24 \times 10^9$), for the real stream, for the stream with the idle gaps between transactions shuffled, and for a Poisson stream with the same transaction count. Right: duration of a multi-packet transaction from its first to its last packet ($n = 5.5 \times 10^7$). Axes as in Figure 4: logarithmic gap axis, fraction of gaps per decade of gap length; dotted lines at 7.5, 16 and 32 μs . The real and shuffled curves coincide, which is the statement that permuting the order of the idle gaps does not change their distribution; the Poisson curve is a different shape, with no mass below 100 μs .

The tail follows the transaction process. To separate the parts of the transaction process that build the queue, each window is re-simulated at $N = 1$ with service charged per packet at its measured absolute cost, $S_i = T(1 + r(\sigma_i - 1))$ with σ_i the packet’s message count and r the slope-to-floor ratio of the live decoder (0.043 on NQ book, 0.142 on ZN book, Section 6.3), so that a large packet adds work rather than redistributing it. Five counterfactual streams are built from the real one (H), each removing one ingredient. TG keeps every transaction intact — its packets, their sizes and their spacing — and permutes the idle gaps between transactions, removing the clustering of transactions and nothing else. TP keeps the transactions intact and redraws their start times uniformly over the window. TS keeps the real start times and permutes the transactions’ shapes among them, decoupling size from timing. TM merges each transaction into a single packet at its start. TW keeps the starts and spreads a split transaction’s packets by the median idle gap instead of their real spacing. Table 7 reports the corpus-median p_{99} of each stream and the share of the real stream’s excess over T that it retains.

T	H	TG	TP	TS	TM	TW
16	34.8	31.0 (80)	20.7 (25)	37.4 (114)	34.9 (101)	35.6 (104)
32	146.8	100.5 (60)	52.9 (18)	155.9 (108)	146.3 (100)	148.1 (101)
64	832	345 (37)	118 (7)	839 (101)	795 (95)	833 (100)
128	5460	1319 (22)	251 (2)	5218 (95)	5035 (92)	5459 (100)

Table 7. Single-stage p_{99} in μs , corpus median over 3512 windows, span-dependent service at the NQ slope ratio, packet-weighted; in parentheses the share in percent of the real stream’s $p_{99} - T$ that the counterfactual retains. TG: clustering of transactions removed; TP: transaction timing random; TS: transaction sizes decoupled from timing; TM: each transaction one packet; TW: intra-transaction spacing removed. At $T \leq 8 \mu\text{s}$ the excess is under $1 \mu\text{s}$ and the shares are not meaningful. The ZN slope ratio and message weighting move every cell by at most a few points.

Three things follow. The tail is the transaction process: randomising transaction times removes 75–98% of it. Sizes and packetisation do not enter: decoupling transaction sizes from timing, merging each transaction into one packet, or removing the spacing inside a split transaction leaves the tail within 15% of its value at every service time, at the ZN slope as at the NQ slope. And the clustering of transactions — the ordering that TG destroys — carries 20% of the tail at $16 \mu\text{s}$, 40% at 32, 63% at 64 and 78% at $128 \mu\text{s}$, the same profile the packet-level gap shuffle gave in Table 3. The remainder, dominant at HFT service times, is the surplus of near-simultaneous transactions over a Poisson stream at the same rate (Table 5); that surplus is itself a product of the self-excitation, since a renewal stream with the same rate has a quarter to a seventeenth as many tight gaps. The attribution in the abstract is therefore the following. The queueing tail behind a single-threaded receiver on this feed is produced by the clustering of matching-engine transactions; it is not produced by packetisation, by the message count per packet, or by the packet rate; and it acts on a short service time through how often transactions land within the service time of each other and on a long one through how long the runs of them last.

4.4 The same N cores spent differently

The tandem of Table 2 runs on N cores and the single thread on one, so part of what it buys is capacity: its throughput is N/T against $1/T$, and more capacity shortens queues under any arrival law. The design question is whether a chain of stages is the right way to spend the same N cores, and the two alternatives are the ones HFT systems actually use. *Sharding* runs N independent single-thread loops, each owning a disjoint slice of the stream, by instrument or by channel: full service T per message and no hops, so it wins the median outright, and its tail depends on whether a burst lands on one shard or spreads over several. *Dispatch*, an M/D/ N queue, puts one queue in front of N identical servers each taking a whole packet at service T : the same capacity N/T as the tandem and no hops, so a burst of k packets drains in about the same time, while each packet’s own latency stays T rather than $T + (N - 1)h$.

Neither is available on this feed, for a reason the chain of stages does not share. A CME burst is one instrument’s order flow, so sharding by instrument sends the whole burst to one core at service T and relieves nothing; the corpus is a single instrument, on which sharding by instrument is degenerate by construction, and sharding packets round-robin breaks the per-instrument ordering the book requires. Dispatch breaks that ordering for the same reason and needs a resequencing stage behind the servers, whose cost is a hop. The chain of stages is the one arrangement that spends N cores on a single ordered stream without reordering it. Table 8 puts the dispatch alternative beside the tandem on the real arrivals with ordering ignored, so it is a lower bound on what dispatch could achieve.

T	N	tandem		M/D/ N dispatch	
		p_{50}	p_{99}	p_{50}	p_{99}
2	1	2.00	2.00	2.00	2.00
2	2	3.70	3.70	2.00	2.00
2	4	7.10	7.10	2.00	2.00
2	8	13.90	13.90	2.00	2.00
4	1	4.00	4.00	4.00	4.00
4	2	5.70	5.70	4.00	4.00
4	4	9.10	9.10	4.00	4.00
4	8	15.90	15.90	4.00	4.00
8	1	8.00	8.61	8.00	8.61
8	2	9.70	9.70	8.00	8.00
8	4	13.10	13.10	8.00	8.00
8	8	19.90	19.90	8.00	8.00
16	1	16.00	34.59	16.00	34.59
16	2	17.70	18.31	16.00	16.00
16	4	21.10	21.10	16.00	16.00
16	8	27.90	27.90	16.00	16.00
32	1	32.00	146.14	32.00	146.14
32	2	33.70	52.29	32.00	48.04
32	4	37.10	37.71	32.00	32.00
32	8	43.90	43.90	32.00	32.00
64	1	98.88	820.70	98.88	820.70
64	2	65.70	179.84	64.00	171.38
64	4	69.10	87.69	64.00	72.50
64	8	75.90	76.51	64.00	64.00
128	1	364.45	5316.85	364.45	5316.85
128	2	164.58	886.40	128.00	868.98
128	4	133.10	247.24	128.00	223.06
128	8	139.90	158.49	128.00	128.00

Table 8. Real (Hawkes) arrivals, corpus medians in μs : N cores as an N -stage tandem against the same N cores as one queue feeding N whole-packet servers, with no resequencing charged to the latter.

At $T = 16\text{--}128\mu\text{s}$ the dispatch arm’s p_{99} is at or below the tandem’s in every cell, by a margin that is small where the tandem already removes most of the excess ($N = 2$: 16.0 against $18.3\mu\text{s}$ at $T = 16$, 48.0 against 52.3 at 32 , 171 against 180 at 64 , 869 against 886 at $128\mu\text{s}$) and closes to zero at $N = 8$, where both arrangements have spare capacity. On the median the difference is exactly the hop cost: dispatch holds $p_{50} = T$ while the tandem pays $T + (N - 1)h$. With the same N cores, then, a design free to reorder packets matches or beats the tandem on the tail and wins the median by $(N - 1)h$ — before its own costs, which the arm does not charge: an ingress stage that assigns packets to servers, a hop to and from the servers, and a resequencer that holds each packet until its predecessors have been released, whose wait behind a slower predecessor is a queue of its own on a span-dependent service. The costed dispatch model is listed in Section 12. What the table settles is that the tandem’s case does not rest on latency: it is preferred because it keeps an ordered stream in order without a resequencing stage.

4.5 Median cost is linear in N , exactly as $(N - 1)h$

This section looks at the typical message rather than the slow one. Its point is that splitting a task into more stages makes the typical latency worse by a fixed, predictable amount, exactly one mailbox hop per added stage, and nothing more. The reason is that the typical message arrives when the queue in front of it is empty, so it pays only the service time plus the hops. The simulation confirms this to the reported precision in almost every cell of the sweep. The few exceptions are at the longest service times, where the queue never empties and even the typical message waits; those sit outside the range the paper recommends, and splitting cures them anyway.

The p_{50} columns of Table 2 match the single-message formula of Proposition 1

$$p_{50}(N) = T + (N - 1)h \quad \text{with } h = 1.7 \mu\text{s},$$

in all 28 cells of the Poisson arm and in 25 of the 28 Hawkes cells — for instance 8.00, 9.70, 13.10, 19.90 at $T = 8$. Under Poisson the identity holds because the queue is empty; under Hawkes it holds because the median message arrives in a quiet inter-cluster stretch where the queue is also empty. The hop tax is therefore exactly h per boundary on the median, with no hidden component, as Theorem 1 says it must be at every quantile.

The three exceptions are at the long-service end of the sweep: $(T, N) = (64, 1)$, $(128, 1)$ and $(128, 2)$, where the Hawkes p_{50} is 98.88, 364.45 and 164.58 μs against floors of 64.00, 128.00 and 129.70. There the service time is long enough that the median message no longer arrives in a quiet stretch: clusters merge into a persistent backlog and the typical message waits. This is the regime boundary appearing in the median rather than in the tail, and it lies outside the range the paper’s recommendations address. Splitting restores the identity once the per-stage service is below the threshold; at $T = 128$ the median is on the floor by $N = 4$.

4.6 Design equation

This section turns the measurements above into a rule for choosing how many stages to split a task into. The idea is simple. Splitting a task across more threads has a fixed price: every added stage costs one mailbox hop, and that hop is paid on every message, fast or slow. It also has a benefit: when packets arrive in bursts, messages queue behind one another, and shorter stages drain those queues faster, so the slow tail shrinks. The question is when the benefit outweighs the price.

The section does four things. It writes the median and the tail latency as two short formulas, a hop-cost term plus a tail term. It tabulates the tail term from the corpus and shows that it depends only on how long the slowest stage is, and that it disappears entirely once every stage is short enough. It combines cost and benefit into one expression the operator can minimise, with a table of where the optimum falls. And it draws the practical conclusion: the goal is not many threads but a short slowest stage, and a task already shorter than the feed’s tight-gap floor, about seven and a half microseconds here, should not be split at all.

From the corpus:

$$p_{99}^{\text{Hawkes}}(N) = T + (N - 1)h + \Delta(N), \tag{11}$$

$$p_{50}^{\text{Hawkes}}(N) = T + (N - 1)h, \tag{12}$$

where $\Delta(N) \equiv p_{99}^{\text{Hawkes}}(N) - p_{50}^{\text{Hawkes}}(N)$ is the “clustering tail excess”. Table 9 lists it, together with the ratio $\Delta(N)/\Delta(1)$ that Theorem 1 bounds by $1/N$.

T	$\Delta(N)$ (μs)				$\Delta(N)/\Delta(1)$			γ
	$N=1$	$N=2$	$N=4$	$N=8$	$N=2$	$N=4$	$N=8$	
2	0.00	0.00	0.00	0.00	—	—	—	—
4	0.00	0.00	0.00	0.00	—	—	—	—
8	0.61	0.00	0.00	0.00	0.000	0.000	0.000	—
16	18.59	0.61	0.00	0.00	0.033	0.000	0.000	—
32	114.14	18.59	0.61	0.00	0.163	0.005	0.000	—
64	721.82	114.14	18.59	0.61	0.158	0.026	0.001	3.32
128	4952.41	721.82	114.14	18.59	0.146	0.023	0.004	2.68
bound					0.500	0.250	0.125	≥ 1

Table 9. Corpus-median clustering tail excess $\Delta(N) = p_{99}(N) - p_{50}(N)$ under Hawkes arrivals at $h = 1.7 \mu\text{s}$, its ratio to the single-stage value, and the least-squares exponent γ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$. The last row is the Theorem 1 bound.

Two features of Table 9 follow from the theorems. First, Δ depends on (T, N) only through the per-stage service T/N : every diagonal of constant T/N carries one number — 0.00 at $T/N \leq 4 \mu\text{s}$, 0.61 at $T/N = 8$, 18.59 at 16, 114.14 at 32, 721.82 at 64 — to the reported precision, across up to four different (T, N) pairs, without fitting. This is the reduction identity Equation (4) on real arrivals: the tandem’s tail is the single-server tail at $s_{\max}$, and the table is a table of $\Delta_{\text{single}}(s)$ at $s \in \{0.25, 0.5, \dots, 64, 128\} \mu\text{s}$, read off the diagonals.

Second, every ratio is at or below the bound $1/N$, and the pathwise check of Section 5 finds no message on any window above it. The margin is large, because of the threshold. Because $\Delta_{\text{single}}(s)$ is identically zero for $s \lesssim 7.5 \mu\text{s}$ (Section 4.8), a split that pushes the per-stage service below that floor does not merely divide the tail excess, it removes it. At $T = 32$, Δ falls $114.14 \rightarrow 18.59 \rightarrow 0.61 \rightarrow 0.00$ as N goes $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$: a ratio of 0.005 at $N = 4$ against a bound of 0.25. The fitted exponent γ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$ is correspondingly steep, 3.32 at $T = 64$ and 2.68 at $T = 128$, well above the $\gamma = 1$ that the bound alone would permit; at $T \leq 32$ the tandem drives $\Delta(8)$ to exactly zero and no power law is fitted, since the least-squares slope would be reading the numerical floor rather than the data. The decay of the excess is therefore not a $1/N$ division on this corpus; it terminates at zero once the per-stage service is below the floor. The design consequence is that the objective is not a large N but an $s_{\max}$ below the feed’s tight-gap floor. The utilisation slack described after Theorem 1 contributes as well, and Appendix B reproduces it on synthetic Borel clusters.

The operator’s stage-count optimum is

$$N^* = \arg \min_N \left[(N-1)h + \kappa \Delta(N) \right]$$

for the operator’s tail weight κ (the number of microseconds of median the operator will spend to remove one microsecond of p_{99} excess). Over the simulated grid $N \in \{1, 2, 4, 8\}$ at $h = 1.7 \mu\text{s}$ the breakpoints are:

T (μs)	$N^* = 1$	$N^* = 2$	$N^* = 4$	$N^* = 8$
2	always	—	—	—
4	always	—	—	—
8	$\kappa < 2.8$	$\kappa \geq 2.8$	—	—
16	$\kappa < 0.095$	$0.095 \leq \kappa < 5.6$	$\kappa \geq 5.6$	—
32	$\kappa < 0.018$	$0.018 \leq \kappa < 0.19$	$0.19 \leq \kappa < 11$	$\kappa \geq 11$
64	$\kappa < 0.0028$	$0.0028 \leq \kappa < 0.036$	$0.036 \leq \kappa < 0.38$	$\kappa \geq 0.38$
128	$\kappa < 0.0004$	$0.0004 \leq \kappa < 0.0056$	$0.0056 \leq \kappa < 0.071$	$\kappa \geq 0.071$

A dash means that stage count is never optimal at any κ : at $T \leq 8 \mu\text{s}$ the tail excess is already zero by $N = 2$, so further splitting can only add hop cost. An operator who values tail and median equally ($\kappa = 1$) lands on $N^* = 1$ at $T = 2, 4$ and 8 , 2 at $T = 16$, 4 at $T = 32$, and 8 at $T = 64$ and $128 \mu\text{s}$, which is also the ordering of the raw p_{99} minima in Table 2. Over the service times an HFT hot path occupies, an operator must weight the tail several times more than the median before splitting pays; above the threshold the required weight is small.

At real cut points the stages are unequal, and by Remark 2.3 the tail excess depends on the partition P of the servicing chain only through its largest stage, $\Delta(P) = \Delta_{\text{single}}(s_{\max}(P))$, i.e. the single-stage tail excess measured at service $s_{\max}$. The optimum over partitions is then

$$P^* = \arg \min_P \left[\text{hops}(P) h + \kappa \Delta_{\text{single}}(s_{\max}(P)) \right],$$

which Table 9 lets the operator evaluate directly, since its diagonals are $\Delta_{\text{single}}(s)$ on a power-of-two grid of s . Two rules follow. A cut is worth its hop only if it lowers $s_{\max}$ by enough that $\kappa [\Delta_{\text{single}}(s_{\max}) - \Delta_{\text{single}}(s'_{\max})] > h$; and among partitions with the same $s_{\max}$ the one with the fewest hops is preferred, so sub-tasks downstream of the bottleneck should be merged onto one thread until their sum reaches $s_{\max}$.

Design principle (only $s_{\max}$ matters). For a servicing chain cut at fixed boundaries, the tail-latency question is not “how many threads?” but “what is the smallest achievable largest stage $s_{\max}$?” By Theorem 1 the clustering tail excess of any partition is $\Delta_{\text{single}}(s_{\max})$, independent of the stage count; a cut that does not lower $s_{\max}$ compresses no tail and still costs one hop h . Choose the partition with the smallest reachable $s_{\max}$, then the fewest hops that realise it.

Floor on T : short tasks cannot be usefully split. Equations (11)–(12) imply a lower bound on the tasks the recommendation applies to. Splitting a task of total service T into N stages spends $(N - 1)h$ at every quantile to compress a tail excess $\Delta(1)$ that is itself a strongly increasing function of T (0 at $T \leq 4 \mu\text{s}$, rising to $39T$ at $T = 128$, Table 9). When T is at or below the feed’s tight-gap floor there is no tail excess to remove at all, and the design equation prescribes $N^* = 1$; a hop there adds cost only. The sweep confirms this: at $T = 2$ and $4 \mu\text{s}$, $\Delta(1) = 0$, so a split removes nothing and p_{99} rises by the full hop, from 2.00 to 3.70 and from 4.00 to 5.70. Above the floor: at $T = 16 \mu\text{s}$, $\Delta(1) = 18.59 \mu\text{s}$, of which the split removes 17.98 for the same 1.7, and p_{99} falls from 34.59 to 18.31 on every window.

Two separate necessary conditions apply. The first is a property of the *framework*: a split cannot pay unless the tail excess it removes exceeds the hop cost it adds, which on this framework’s shipping BQueue path ($h \approx 1.7 \mu\text{s}$) puts a floor at roughly $T \gtrsim 2h \approx 3.4 \mu\text{s}$. The second is a property of the *feed*: there is no tail excess to remove at all until T exceeds the arrival stream’s tight-gap floor, about $7.5 \mu\text{s}$ on this corpus (Section 4.9). The binding constraint is whichever is larger, and on CME

MDP3 channel 318 with a BQueue mailbox it is the feed’s, by roughly a factor of two; this is why $T = 4\mu s$, although above $2h$, has $\Delta(1) = 0$ and should not be split. A framework with a much more expensive mailbox would find its own floor binding instead. Throughout the rest of the paper, “the floor” means the larger of the two. Sub-microsecond tasks, for instance a single SBE integer decode or a small handler that fires from a strategy fast path, belong on `fast_send` rather than on a cross-thread mailbox, regardless of feed statistics.

4.7 Across windows

The results so far are for a typical half-hour window. This section asks whether they hold on every window, including the busiest ones, or only on average. The answer matters because a busy window has a far worse single-thread tail than the typical one, so an operator cannot size a pipeline from the average alone. The finding is that the split wins on essentially every window from about ten microseconds up, and on no window at eight microseconds or below, where there is no tail to remove and the split only adds hop cost, with a sharp crossover between. The recommendation is therefore not an artefact of averaging.

The corpus medians understate the tail on the busiest windows. Across the 3512 windows the single-stage p_{99}/T has a median of 1.0, 1.0, 1.1, 2.2, 4.6, 12.8, 41.5 and a 95th percentile of 1.0, 1.0, 1.1, 2.5, 6.1, 19.9, 81.7 at $T = 2, 4, 8, 16, 32, 64, 128\mu s$, so one window in twenty at $T = 32$ has a single-stage p_{99} above $195\mu s$ and at $T = 128$ above 10.5 ms. The tandem’s win is not confined to the median window: $p_{99}(2) < p_{99}(1)$ and $p_{99}(4) < p_{99}(1)$ both hold on 100% of windows at every $T \geq 16$, and on no window at all at $T \leq 8$, where there is no tail to compress and the split pays only hop cost. The crossover is sharp and sits just below $T = 10\mu s$: on the refined sweep of Section 4.8, $p_{99}(2) < p_{99}(1)$ on 0% of windows at $T = 8\mu s$, 67.0% at $T = 9\mu s$ and 99.7% at $T = 10\mu s$. The per-window fitted branching ratio has median 0.798 and a 5th–95th percentile range of 0.761–0.832 (Section 3.2); how the tail excess depends on it, and on the window’s packet rate, is examined in Section 4.9. The intervening p_{95} quantile is recorded per window in the sweep (Section 3.5) and omitted from the tables above only for brevity; it lies between the reported p_{50} and p_{99} in every cell by construction, and under the Poisson null it equals the service floor wherever p_{99} does, since $\rho/N < 1 - 0.99$ implies $\rho/N < 1 - 0.95$. The 3512 windows are treated as evaluation units for the descriptive statistics reported here; they are nested within 276 sessions and are not independent, so any statistical uncertainty on the aggregates would be clustered at the session level rather than treating 3512 as an independent sample size.

4.8 The tail onset

The coarse sweep brackets the onset of the clustering tail between $T = 8$ and $T = 16\mu s$, too wide for a design rule. We reran the single- and two-stage arms on the same 3512 windows at $T \in \{5, 6, 7, 8, 9, 10, 12, 14, 16\}\mu s$:

T (μs)	p_{99} Hawkes	excess over T	windows with a tail	$p_{99}(2) < p_{99}(1)$
5	5.00	0.00	0.0%	0.0%
6	6.00	0.00	0.0%	0.0%
7	7.00	0.00	4.5%	0.0%
8	8.61	0.61	98.6%	0.0%
9	10.85	1.85	100.0%	67.0%
10	13.52	3.52	100.0%	99.7%
12	19.83	7.83	100.0%	100.0%
14	26.68	12.68	100.0%	100.0%
16	34.59	18.59	100.0%	100.0%

Two distinct thresholds are visible. The first is the *tail* threshold: between $T = 7$ and $T = 8 \mu\text{s}$ the fraction of windows with any p_{99} excess jumps from 4.5% to 98.6%. The second is the *splitting* threshold, about $1.5 \mu\text{s}$ later, between $T = 8$ and $T = 10 \mu\text{s}$: a tail must exist *and* exceed the hop cost before a split repays it. At $T = 8 \mu\text{s}$ nearly every window has a tail, but its median excess is $0.61 \mu\text{s}$, well under $h = 1.7 \mu\text{s}$, and splitting loses on every window. By $T = 10 \mu\text{s}$ the excess is $3.52 \mu\text{s}$, above h , and splitting wins on 99.7%.

The first threshold is set by the feed’s tight-gap structure rather than by its mean rate; the next section establishes this.

4.9 Conditioning on intensity and criticality

One would expect the slow tail, and with it the case for splitting, to depend on how busy the market is and on how strongly one packet triggers the next. If it did, the rule for splitting would have to change with market conditions. This section tests that expectation and finds that, over the task lengths a trading hot path uses, it does not hold: a window with four to five times the packet rate has the same tail, relative to the task length, as a quiet one. The reason is that a busy market produces more bursts, not tighter bursts. The spacing between packets inside a burst is set by how the exchange gateway emits them, about seven and a half microseconds on this feed, and that spacing does not change with the market state. Since it is the tightest spacing that starts a queue, the point at which a tail appears is a constant of the feed. Market load only matters for very long tasks, where the ordinary effect of a busy server takes over. The practical consequence is that the rule for splitting needs no live estimate of market conditions; the section closes by confirming the result on five-minute windows, and by noting that a rule based on the average gap between packets, rather than the tightest, would be wrong by two orders of magnitude.

The results above are conditioned on the service time T alone. A Hawkes tail depends on two further quantities that the corpus median averages over: the window’s mean packet rate $\bar{\lambda}$, and its branching ratio $n = \alpha/\beta$.

We sort the 3512 windows into quintiles, first by $\bar{\lambda}$ and then by the fitted n , and report the corpus median of the normalised single-stage tail p_{99}/T within each quintile. Normalising by T removes the service floor, so a value of 1.0 means no tail at all and the quintile-to-quintile ratio isolates the conditioning effect.

quintile	median	p_{99}/T , corpus median				
		$T=8$	$T=16$	$T=32$	$T=64$	$T=128$
<i>by packet rate $\bar{\lambda}$ (pkt/s)</i>						
Q1	202	1.08	2.29	4.74	12.25	31.76
Q3	451	1.07	2.12	4.48	12.68	42.81
Q5	942	1.08	2.11	4.51	13.92	55.67
Q5/Q1		1.00	0.92	0.95	1.14	1.75
<i>by branching ratio n</i>						
Q1	0.769	1.08	2.18	4.51	11.61	30.73
Q3	0.798	1.08	2.16	4.61	13.17	42.83
Q5	0.825	1.07	2.15	4.61	14.65	58.73
Q5/Q1		1.00	0.98	1.02	1.26	1.91

The result divides at $T \approx 32\text{--}64\mu\text{s}$. For $T \leq 32\mu\text{s}$ the normalised tail is flat in both variables: a $4.66\times$ change in packet rate moves it by 0–8%, and in the wrong direction at $T = 16$ and $T = 32$, where the *busiest* quintile has the slightly *smaller* normalised tail. A $0.769 \rightarrow 0.825$ change in branching ratio moves it by at most 2%. For $T \geq 64\mu\text{s}$ both variables have an effect, in the expected direction: by $T = 128\mu\text{s}$ the busiest quintile’s tail is $1.75\times$ the quietest and the most critical quintile’s is $1.91\times$ the least. This is the boundary between the clustering regime and the utilisation regime found elsewhere in the paper: for $T \geq 64\mu\text{s}$ the Poisson null has a tail too (Section 4.1), utilisation $\rho = \bar{\lambda}T$ is the governing quantity, and ρ is proportional to $\bar{\lambda}$. Intensity matters where utilisation queueing applies and not below it.

The invariance is not a simulator artefact. The Poisson arm, run on the same windows through the same simulator, shows the intensity dependence that utilisation queueing requires: at $T = 16$ and $32\mu\text{s}$ its normalised p_{99} is $1.34\times$ and $1.68\times$ higher in the busiest quintile than in the quietest, because $\rho = \bar{\lambda}T$ crosses the 0.01 switch-on of Corollary 2 within the quintile range. The Hawkes arm on the same windows at the same T gives $0.92\times$ and $0.95\times$. The simulator therefore resolves intensity; the clustered tail does not depend on it at these service times. The result is the same unbinned: the per-window Spearman correlation of p_{99}/T with $\bar{\lambda}$ is -0.19 at $T = 16$ and -0.09 at $T = 32\mu\text{s}$, and with n is -0.05 and $+0.02$; at $T = 128\mu\text{s}$ both are $+0.48$ and $+0.54$. Comparing the top and bottom 2% of windows by rate, a $26\times$ range, gives a p_{99}/T ratio of 1.02 at $T = 32$ and 2.57 at $T = 128$.

The mechanism is that at $\rho \ll 1$ clusters do not overlap. Raising $\bar{\lambda}$ raises the number of clusters per window, each with the same internal spacing and, at fixed n , the same size distribution, so the position of the p_{99} packet within its cluster, and hence p_{99}/T , is unchanged. Intensity enters once clusters begin to overlap, and n enters once T is long enough for many members of one cluster to fall within a single service time; on this corpus both occur at $T \geq 64\mu\text{s}$.

The reason intensity has no effect below the crossover. The null result has a mechanical explanation in the arrival stream. Taking the same $\bar{\lambda}$ quintiles and reporting the corpus median of the per-window interarrival-gap quantiles:

quintile	$\bar{\lambda}$ (pkt/s)	p_1 gap	p_5 gap	p_{25} gap	p_{50} gap	mean gap
Q1	202	7.45	8.85	21.30	87.10	4944
Q2	320	7.52	8.75	21.87	84.52	3128
Q3	451	7.53	8.70	22.70	82.16	2215
Q4	620	7.51	8.66	23.36	81.98	1612
Q5	942	7.43	8.58	22.55	76.52	1062
Q5/Q1	4.66×	0.997×	0.969×	1.059×	0.878×	0.215×

All gaps in microseconds. The mean gap falls by exactly the reciprocal of the rate increase, $0.215 \approx 1/4.66$, which is definitional. Every other column is approximately invariant, and the tightest is the most invariant: the p_1 gap moves from 7.45 to 7.43 μs , a ratio of 0.997, while the rate more than quadruples.

Raising the intensity of a self-exciting process raises the rate at which clusters are generated; it does not change the spacing of events inside a cluster. The intra-cluster spacing is set by the exchange’s packet-emission granularity, a property of the CME gateway rather than of the market state, and on this corpus that floor is about 7.5 μs . A busy window contains more bursts, not tighter bursts.

The queueing consequence: a deterministic server of service time T begins to build a queue when it is handed two arrivals separated by less than T , so the service time at which a tail appears is set by the *tight* end of the gap distribution, not by its mean. The tight end is invariant to $\bar{\lambda}$; therefore the threshold is invariant to $\bar{\lambda}$. The two numbers agree: the measured p_1 gap of 7.43–7.53 μs across every intensity quintile brackets the measured tail onset of Section 4.8, between $T = 7$ and $T = 8 \mu\text{s}$, where the fraction of windows carrying any tail jumps from 4.5% to 98.6%.

Robustness to window length. A 30-minute window averages $\bar{\lambda}$ over a long interval, and the quintile split above may be comparing time-averaged rates that have smoothed away the relevant intensity variation. We therefore reran the entire pipeline — packet extraction, per-window Hawkes MLE, Poisson null and the full (T, N) grid — on *five*-minute windows, giving 21,060 windows in place of 3512. Shorter windows do resolve more intensity variation, as expected: the corpus $\bar{\lambda}$ spread widens from 6.8×

	$\bar{\lambda}$ Q5/Q1	$T=8$	$T=16$	$T=32$	$T=64$	$T=128$
30-min windows ($n = 3512$)	4.66×	1.00	0.92	0.95	1.14	1.75
5-min windows ($n = 21,060$)	4.98×	1.00	0.93	0.96	1.15	1.74

Entries are the Q5/Q1 ratio of the normalised single-stage tail p_{99}/T . Resolving intensity six times more finely, over a wider realised range, moves no entry by more than 0.01. The result is a property of the arrival process, not of the window length.

Consequence for the design rule. The rule (split a stage whose service time exceeds the feed’s tight-gap floor; do not split below it) can therefore be stated as a constant of the feed rather than as a function of market state. It requires no runtime intensity estimate and does not change between a quiet and a volatile session. This holds because of the invariance measured above, and should

be rechecked on any feed whose packet-emission behaviour differs from CME MDP3’s, since the emission granularity sets the constant. Appendix D constructs a feed on which both dependences appear at $T = 8\mu s$, and identifies the packed-gap fraction as the quantity that separates it from CME.

The tight-gap floor is not the mean interarrival gap. On this corpus the mean gap is $2215\mu s$ at the median window, about $300\times$ the p_1 gap. A rule stated on the mean gap would recommend never splitting an HFT stage, and would be wrong by two orders of magnitude.

4.10 Sensitivity to h

The rule depends on the cost of a mailbox hop, and this section asks what changes if that cost is different. A cheaper hop makes splitting worthwhile over a wider range of task lengths, but it does not move the point below which there is no tail to remove, since that point is set by the feed and not by the framework. Any other framework can therefore measure its own hop cost, put it into the same equation, and read off its own answer; a framework with a very expensive hop should not split at all. The section also explains why the hop cost used throughout is a cautious overestimate. It was measured with the receiving thread asleep, so it includes the cost of waking it. Inside a burst the receiver never sleeps, because messages arrive faster than it can process them, and the true hop cost there is a few hundred nanoseconds rather than nearly two microseconds. The benefit of splitting during a burst is therefore, if anything, understated.

The design equation is monotone in h : at fixed T and fixed corpus $\Delta(N)$, every κ breakpoint in the table above scales linearly with h , so a smaller h lowers every breakpoint and can only raise N^* at a fixed κ . At $h = 1.7\mu s$ the p_{99} -minimising N is 1, 1, 1, 2, 4, 8, 8 across the seven service floors; at a hypothetical $h = 0.5\mu s$ busy-poll mailbox it would be 1, 1, 2, 2, 4, 8, 8 — the change is a single notch, at $T = 8\mu s$, because below the tight-gap floor no reduction in h can make a split worth taking when there is no tail to remove. Reducing the hop cost widens the band of service times over which splitting pays; it does not move the threshold, which is set by the feed. For any other framework the procedure is the same: measure the one-way asynchronous hop cost h on its own dispatch path, put it into Equation (11) with the corpus Δ of Table 9, and read off N^* ; a framework whose h exceeds the tail excess a split could remove gets $N^* = 1$, and the recommendation reverses.

h is load-dependent; $1.7\mu s$ is a conservative upper bound. The $h = 1.7\mu s$ figure is the one-way cost of the shipping KASPAR-HFT BQueue mailbox measured at low load, where the receiver is sleeping on a condition variable and the send path must pay the kernel wake latency to bring it back onto CPU. That wake latency is the dominant term in the $\sim 3.4\mu s$ round-trip of Table 4 in Maciejewski [35], and the general fact that a spinning receiver delivers a message an order of magnitude faster than a wake-from-condvar receiver is standard in the microsecond-latency systems literature: David et al. [61] measure the primitive costs directly on commodity Linux hardware, Belay et al. [62] exploit this in the IX data plane by pinning busy-polling network threads, and Rigtorp’s operational guide [40] formalises the “run one thread per core and busy-wait” rule that flows from it; the `fast_send` paper’s own grouped-`send` row (Table 4, ~ 90 ns) is the same effect on the same framework — when the receiver never actually sleeps, the send path collapses. Under sustained load, and specifically inside a Hawkes cluster where the receiver’s queue is never empty, the receiver never sleeps: successive messages arrive faster than the receiver’s per-message service time, so it stays awake in a tight run-loop and the enqueue path reduces to a mutex acquire, a queue push, and a condition-variable notify against a receiver that is not waiting (a near-no-op). The effective h in that regime is dominated by the mutex + cache-line-transfer cost, typically a few hundred nanoseconds rather than $1.7\mu s$. This is the regime in which the tail reduction is largest, so

the reported effect underestimates the benefit whenever the pipeline is inside a burst. Characterising the load-conditioned hop cost $h(\rho)$ is left to future work.

5 Evaluation: do empirics match the analytical predictions?

This section compares the empirical numbers of Section 4 against the closed-form predictions of Section 2.

Proposition 1 (burst-limit throughput). The single-message identity $p_{50}(N) = T + (N - 1)h$ is predicted exactly by Proposition 1 for any arrival law when the median message finds the queue empty. Table 2 confirms it in all 28 (T, N) cells of the Poisson arm and in 25 of the 28 Hawkes cells at $h = 1.7 \mu\text{s}$, to the $0.01 \mu\text{s}$ reporting precision; the three Hawkes exceptions are the longest-service cells, where the median message itself waits (Section 4.5).

Theorem 1 (reduction and scaling bound). The prediction is $\Delta(N) \leq \Delta(1)/N$ on every window, and more strongly $W_j^{(N)} \leq W_j^{(1)}/N + (N - 1)h$ for every individual message. The corpus-median ratios in Table 9 sit at or below $1/N$ in every cell, with substantial slack everywhere: the tightest ratio in the sweep is 0.163 at $T = 32$, $N = 2$, against a bound of 0.5. The slack is largest where a split drives the per-stage service under the tight-gap floor and removes the excess outright rather than dividing it, which is why the fitted γ is 3.32 and 2.68 at $T = 64$ and 128 rather than near 1. The pathwise form was checked directly by the simulator: across all 3512 windows, all seven service floors, both arms, and $N \in \{2, 4, 8\}$, no message violates the bound (the check allows N nanoseconds for the rounding of T/N ; the largest excess over that tolerance is 0). The reduction identity Equation (4) is visible in Table 9 without any fitting: Δ is constant along every diagonal of constant T/N , so the tandem’s tail is the single-server tail at $s_{\max}$ on real arrivals exactly as on synthetic ones (Appendices B and C).

Corollary 2 (Poisson null). The prediction is $p_q^P(N) = T + (N - 1)h$ wherever the per-stage utilisation satisfies $\rho/N < 1 - q$, and a visible wait wherever it does not. Table 2 matches both halves: Poisson p_{99} equals the floor in 22 of the 28 cells and exceeds it in exactly the six cells whose per-stage utilisation clears 10^{-2} — $(T, N) = (32, 1), (64, 1), (64, 2), (128, 1), (128, 2)$ and $(128, 4)$, at $\rho/N = 0.0144$ to 0.0578 — and in no cell below it. Poisson $p_{99.9}$ likewise exceeds the floor exactly in the cells with $\rho/N > 10^{-3}$ (single stage at $T \geq 4$, two stages at $T \geq 8$, four stages at $T \geq 16$, eight stages at $T \geq 32$) and nowhere else. The null arm is therefore a working queue whose small wait appears where PASTA says it must, and splitting under it costs $(N - 1)h$ at every reported quantile.

Summary. All three analytical statements hold on the full corpus. The pathwise bound of Theorem 1 holds on every message of every window, and its slack is measured: $\gamma = 3.32$ and 2.68 at $T = 64$ and $128 \mu\text{s}$, and at $T \leq 32 \mu\text{s}$ the tandem removes the excess outright rather than dividing it.

6 Cross-validation: the same arrival process, measured live

The results of Part II so far are measurements on recorded data: pcap-derived message tapes replayed offline. That design gives scale and repeatability at the cost of external validity. A replayed tape cannot show whether the fitted clustering survives the network path to a receiver, whether it is specific to the 2025–2026 sample, or whether the service times the sweep assumes correspond to production decode times.

This section addresses all three with instrumentation of the KASPAR-HFT production system. The two measurements were built for different purposes and share only the exchange they observe.

6.1 Two measurements

The live measurement instruments the KASPAR-HFT market-data path in production: a timestamp t_0 when a UDP datagram is handed up from the socket, a timestamp t_1 when the resulting book update is published, and one 16-byte record per admitted message carrying the pair together with the ring-buffer depth and the message’s position inside its datagram. It covers 8.29 million messages on six streams — ES, NQ and ZN futures, book and trade — over 53 minutes of a single afternoon. Its latency results are reported in Maciejewski [35]; we draw on them here, and on the underlying per-message records, as a control on Part II.

	Part II corpus	production measurement
source	pcap tapes, replayed offline	live multicast, production recorder
period	2025-01 to 2026-02	2026-09-16 (7 months out of sample)
instruments	NQ front month	ES, NQ, ZN $\times$ book, trade
arrival clock	CME gateway <code>sendingTime</code>	receiver t_0 at socket read
estimator	per-window, median of 3512 windows	pooled over one 53-min capture
sample	3512 windows, 276 sessions	8.29 M messages, 6 streams

Table 10. The two measurements compared in this section: different instruments, a period that does not overlap, a different arrival clock, and a different estimator.

The arrival-clock difference is the most significant. Part II keys arrivals on `sendingTime`, the timestamp the CME gateway writes into the UDP header as the packet leaves the exchange. The live measurement keys them on t_0 , taken by our own process after the packet has crossed the network, the NIC and the kernel. If the clustering reported in Section 3.2 were an artefact of exchange-side emission scheduling, the intervening path, with its own queueing, coalescing and interrupt moderation, would be expected to attenuate it. It does not.

6.2 Statistics that agree

This section checks that the bursty arrival pattern the paper is built on is real, and not something produced by replaying recordings. It takes four numbers that each measure how far packet arrivals are from evenly spread. The first is the share of gaps between packets that are tiny, under a tenth of the average gap. The second is how much the variability of packet counts drops when the order of the packets is scrambled, which removes any bunching. The third is a measure of whether busy stretches persist rather than fading at once. The fourth is how strongly one packet tends to trigger the next. Each of the four is computed twice, independently: once on the recorded corpus that the rest of the paper uses, and once on the live production receiver, seven months later, on different instruments, with a different clock. Three of the four match directly, to within the spread seen across

the live streams. The fourth is measured in two different ways, one an estimate and one an upper bound, and the estimate falls inside the bound. The conclusion is that the bursts are a property of the exchange feed itself: they show up at the same strength whether one looks at a recording or at the live wire.

Table 11 puts the four statistics that both measurements compute side by side. The NQ column is the like-for-like comparison, since Part II’s corpus is NQ front month; the third column gives the range across the three live book streams.

	Part II (NQ, 3512 win)	live NQ book	live, three books
$P(\text{gap} < \text{mean}/10)$	73.2%	63.1%	63.1–84.8%
Fano(5s) collapse under shuffle	14.8×	12.2×	12.2–18.5×
Hurst exponent H	0.740	0.702	0.702–0.743
branching ratio n	0.798 (MLE)	$\leq 0.850\text{--}0.972$ (Fano ceiling)	

Table 11. The same four Poisson-rejection statistics, computed independently on recorded and on live data. The Poisson values are 9.52%, 1×, 0.5 and 0 respectively.

Three of the four are direct comparisons and agree to within the spread across live streams. The fourth compares different estimators. Part II fits the branching ratio by maximum likelihood, per window, using the Ogata recursion (Section 3.2); the live measurement does not fit a Hawkes process, and bounds n from above by inverting the Fano ceiling, $n \leq 1 - 1/\sqrt{F}$. The MLE’s 0.798 lies within every bound the live streams imply, which run from 0.850 on NQ trade to 0.972 on NQ book. The bound is loose, and the agreement is weak evidence on its own; its value is that it is independent evidence and is consistent.

The arrival process modelled in Part II is therefore not an artefact of replay, of the sample period, or of the exchange-side clock; it is present at the same strength in a live receiver seven months later on other instruments.

6.3 The service floor coincides with the tail threshold

The live system supplies one number that a recording cannot: how long the production decoder actually takes to handle a message when nothing is queued in front of it. This section measures that time three independent ways and finds they agree, at about seven microseconds. The notable result is where that number falls. Part II found, from the recorded corpus, that a tail appears once a stage takes longer than the tightest gap between packets, about seven and a half microseconds. The production decoder sits just under that, within a few hundred nanoseconds, by two measurements that share nothing. The production system therefore operates right at the edge of the tail. The section then shows what pushes it over: a packet carrying a single message is handled below the threshold, and a packet carrying two or more at or above it, on every stream measured. Whether a packet’s service crosses the threshold is decided by whether it carries one message or more.

Isolating messages that arrive to an empty ring and first in their datagram, so that neither queueing nor in-packet serialisation contributes, gives the hot path directly. Three estimators agree:

method	ES	NQ	ZN
direct median, empty ring and first in packet	7.01	7.24	6.89
fitted intercept of the median position ladder	6.98	7.23	6.83
floor at single-message packets during an FOMC release	7.1	7.5	7.0

Table 12. Three independent estimates of the KASPAR-HFT socket-to-book floor, in μs , on the book streams. The three differ in sample, estimator and load regime. The third was taken while the packet rate rose 4.6 to $9.8\times$ within a second; the floor rose by at most $0.27\mu\text{s}$ against either other estimator.

Across all six streams the fitted floor spans 6.81 to $7.23\mu\text{s}$, a range of $0.42\mu\text{s}$ across six multicast channels on one host and one afternoon.

Compare this with Section 4.9. The service time at which a deterministic stage on this feed begins to accumulate a tail is the feed’s tight-gap floor, its 1st-percentile interarrival gap, measured at 7.43 to $7.53\mu\text{s}$ and invariant across intensity quintiles; the refined sweep of Section 4.8 locates the switch-on between $T = 7$ and $T = 8\mu\text{s}$. The production decoder’s floor is 6.81 – $7.23\mu\text{s}$.

The production system operates at the threshold, to within a few hundred nanoseconds, by two measurements made on different data with different estimators. The agreement is close to definitional for a p_{99} target, since a deterministic server leaves its floor at p_{99} once about one packet in a hundred arrives within T of the previous one; for a $p_{99.9}$ target the relevant floor is the 0.1st-percentile gap, which is lower, and Table 2 already shows a $p_{99.9}$ excess at $T = 8\mu\text{s}$.

The live measurement locates the crossing. Per-packet service is the floor plus one slope for each message beyond the first (Section 6.5), so the service time of a datagram carrying span messages is floor + (span – 1) $\times$ slope. Evaluating that on each stream’s own fitted pair:

	span 1	span 2	span 3
ES book	6.98	7.55	8.11
NQ book	7.23	7.54	7.85
ZN book	6.83	7.80	8.76
ES trade	6.81	7.52	8.24
NQ trade	7.14	7.67	8.19
ZN trade	6.81	7.77	8.74
range	6.81–7.23	7.52–7.80	7.85–8.76

Table 13. Per-packet service time in μs by datagram span, from each stream’s measured floor and slope, against the feed’s tight-gap floor of 7.43 – $7.53\mu\text{s}$ (Section 4.9). Every stream is below the threshold at span 1 and at or above it at span 2; ES trade, at 7.52, sits inside the 7.43 – 7.53 band.

All six streams are below the threshold when a datagram carries one message and at or above it when it carries two. All six crossings fall in the same unit interval of span, from floors and slopes fitted separately on six multicast channels against a threshold measured on a different corpus. For this system, whether a packet’s service is above or below the threshold is determined by whether it carries one message or more.

6.4 Statistics that disagree, and the reason

Two numbers do not match between the recorded and the live measurements, and this section explains why neither is a cause for concern. In both cases the disagreement is about how the number was computed, not about the feed, and in both cases one of the two measurements can diagnose

the other. The first is the raw level of burstiness, which reads about four times higher live. That is because the live figure was taken over one uncut stretch that mixes a quiet period and a busy one, and a rate that drifts across the stretch inflates the measure whether or not the arrivals are genuinely bursty; the recorded corpus computes the same quantity inside each half-hour window, which removes that drift. The change in burstiness when the packets are scrambled, which is the effect the paper actually uses, agrees closely between the two. The second is a control value that should come out at exactly one half after scrambling, and reads a little above it live. That is a known small-sample bias of the estimator on a short capture; on the much larger recorded corpus the same control comes out at one half, as it should. Neither discrepancy touches any quantity the paper’s argument rests on, and for the level of burstiness the recorded corpus gives the more conservative value, which is the one reported.

The absolute Fano factor: pooling inflates it. Part II measures a corpus-median Fano at a 5-second window of 259.7; the live streams give 1024 to 1260. The ratio under the gap shuffle agrees closely (Table 11), so the discrepancy is in the level, not in the effect. The cause is the estimator row of Table 10. The live figure is computed over one uncut 53-minute capture that spans a quiet stretch and a busy one; a rate that drifts between the two inflates the count variance and reads as clustering whether or not the process is self-exciting. Part II computes the Fano factor *within* each 30-minute window and then takes the corpus median, which removes between-window rate variation by construction. Pooling raises the level by roughly a factor of four; the windowed estimate is the conservative one and is the value reported.

The shuffled Hurst null: finite-sample bias. A Fisher–Yates shuffle of a gap sequence destroys the ordering while preserving the marginal exactly, so the shuffled series is a renewal process and its Hurst exponent must be 0.5. The live measurement recovers 0.600 to 0.660 instead, and attributes the excess to finite-sample bias in the Hurst estimator at the number of windows available in a 53-minute capture and states its result as the gap between 0.74 and 0.62 rather than the distance from 0.5. Part II confirms the attribution: over 3512 windows the same shuffle returns $H = 0.501$, and the uniform-shuffle Poisson null returns $H = 0.496$ (Section 3.2). The bias is finite-sample and vanishes with more, shorter series.

6.5 Per-packet service is not constant

The simulator in Part II assumed that every packet takes the same time to process. The live measurement shows that it does not, and this section sets out what that means. A packet on this feed carries a variable number of messages, usually one but sometimes dozens, and they are decoded one after another, so a packet takes longer the more it carries. Large packets are rare but expensive: on the most fragmented stream, ZN trade, one packet in nine carries two thirds of the messages and consumes six sevenths of the decode time. The task length used throughout the paper is therefore the work for a typical packet, not a fixed constant of the system. Two points follow, set out at the end of the section: the results survive, because the cost per message is constant and only the bundling varies; and the number of messages per packet is a second, separate source of slow messages, distinct from the bursts studied so far. It acts precisely where bursts do not, on tasks shorter than the tightest gap between packets, and fades where bursts take over. The next section re-runs the sweep with the message count included to show that the two mechanisms are almost independent.

The sweep of Section 3.5 treats each packet as one arrival with a constant service time T . Per-packet service is not constant. A UDP datagram on this feed carries a variable number of SBE

messages — call it the packet’s *span* — which are decoded in order, so a message at position k waits for k decodes before its own. Median latency follows

$$\text{median} = \text{floor} + \text{slope} \times \text{position},$$

with the floor of Section 6.3 and a slope of 0.31 to $0.97 \mu\text{s}$ per message depending on the stream. Span is not close to constant:

stream	packets	messages	p_{50}	p_{90}	p_{99}	max
ES book	2,123,042	2,415,610	1	1	7	45
NQ book	2,506,205	3,010,036	1	2	11	36
ZN book	824,536	965,334	1	1	6	45
ES trade	83,140	235,601	3	8	17	83
NQ trade	50,279	83,233	2	7	16	85
ZN trade	22,864	100,659	2	11	71	85

Table 14. Messages per UDP datagram (span), packet-weighted quantiles. The median book packet carries one message; the 99th percentile carries 6 to 11 on the book streams and 71 on ZN trade.

Because service is charged per message, large packets account for a disproportionate share of processing time:

span ≥ 10	% of packets	% of messages	% of decode time
ES book	0.52	4.55	5.34
NQ book	1.68	4.87	2.57
ZN book	0.47	7.29	7.29
NQ trade	5.34	19.31	16.16
ES trade	6.69	26.53	40.10
ZN trade	11.31	66.18	85.54

Table 15. Concentration of work in large datagrams. On ZN trade, 11.3% of packets carry 66% of the messages and consume 86% of decode time.

The constant-service assumption is therefore violated at the packet level, and Part II’s T is the per-packet work at a typical span rather than a constant of the system. Two points follow.

First, at the message level the assumption approximately holds: the slope is fixed within a stream to the precision of the fit, and it is the composition of packets, not randomness in per-message work, that makes per-packet service vary. Theorem 1 is pathwise and arrival-law independent, and Remark 2.3 already handles unequal stages through $s_{\max}$; what the live data adds is a measured magnitude for the variability that Section 2.3 lists as an assumption.

Second, span is a separate tail mechanism on the same stream, active where the queueing mechanism is not. Section 6.6 re-runs the sweep with span-dependent service. The serialisation channel supplies the tail below the tight-gap floor, where clustering supplies none, and is negligible above it, where clustering supplies all of it; the two channels are close to independent on this corpus.

6.6 Span-dependent service

The previous section found that a packet takes longer to process the more messages it carries. This section re-runs the whole simulation with that fact built in, to find out what it changes. The processing time is modelled three ways: fixed per packet, as in Part II; growing with message count

at the rate measured on the live system; and, as a deliberately exaggerated upper bound, fully proportional to message count. All three do the same total work on average, so any difference in the slow tail comes from how the work is spread across packets. There are four findings. First, message count and bursts produce slow messages in different places. Message count matters only for short tasks, at or just above the tightest gap between packets, where bursts build little or no queue; for longer tasks the bursts dominate and message count adds a few percent at $16\mu\text{s}$ and one or two beyond. Second, at the simulated task length nearest the production system’s seven microseconds, eight, message count accounts for roughly two thirds of the slow tail and bursts for one third, and the live system reports the same ordering. Third, the recommendation on how many stages to split a task into is unchanged at every task length. The in-packet part of that tail is the plain cost of decoding several messages one after another, which splitting the work into stages does not remove; the queue a large packet opens behind itself is taken up in Section 8.3. Fourth, the two mechanisms do not compound: large packets are no more likely to arrive inside a burst than outside one, checked two ways, so their effects simply add, and the tables of Part II, which measure the burst mechanism alone, stand as they are.

We re-extracted the per-packet span for all 3512 windows — aligned element-wise against the cached arrival arrays, with every window’s span count checked against the packet count already in the cache — and re-ran the entire sweep with service that depends on it.

Three service models are compared. All three hold the *mean* service per window at T , so the utilisation ρ is identical across them and only the distribution of work across packets differs; any difference in the tail is attributable to the span dependence.

$$\begin{array}{ll} \text{const :} & S_i = T \quad \text{(the Part II assumption)} \\ \text{span :} & S_i = A(1 + r(\sigma_i - 1)), \quad r = 0.0432 \quad \text{(measured slope/floor)} \\ \text{prop :} & S_i = T \sigma_i / \bar{\sigma} \quad \text{(upper bound)} \end{array}$$

with A set so that the window mean of S_i is T . The *span* model uses the ratio of slope to floor measured on NQ book in Section 6.3, $0.312/7.23$. The *prop* model removes the per-packet fixed cost, making service proportional to the message count; this overstates the dependence, since the measured floor exceeds the slope by more than twenty to one, and it is included as an upper bound. As a correctness check the *const* arm reproduces the published constant-service grid of Table 2 to zero difference in all 28 cells.

Message-weighted quantiles. On this corpus, pooled over its 3.30×10^9 packets, 95.9% of packets carry one message and the mean span is 1.078 (per-window medians 96.0% and 1.077), so a packet-weighted quantile is dominated by singletons and changes little under any service model. Weighting by message, which counts a 20-message datagram twenty times, the in-packet position distribution is concentrated at zero with a long upper tail:

message quantile	p_{50}	p_{90}	p_{95}	p_{99}	$p_{99.9}$	max
in-packet position	0	0	1	4	19	84

89.0% of messages are alone in their packet and pay no serialisation; the remaining 11.0% constitute the tail. The live measurement reports the same shape. The comparison below is message-weighted.

T	const			span			span/const		prop/const	
	p_{50}	p_{99}	$p_{99.9}$	p_{50}	p_{99}	$p_{99.9}$	p_{99}	$p_{99.9}$	p_{99}	$p_{99.9}$
2	2.00	2.00	2.00	1.99	2.34	3.54	1.17	1.77	4.64	23.35
4	4.00	4.00	4.00	3.99	4.67	7.08	1.17	1.77	4.69	26.73
8	8.00	8.56	10.10	7.97	9.63	14.87	1.12	1.47	5.21	24.64
16	16.00	34.00	53.55	15.95	35.08	56.62	1.03	1.06	3.36	11.53
32	32.00	143.74	256.51	31.89	144.41	260.14	1.00	1.01	2.37	5.88
64	98.80	815.22	1557.72	100.02	820.05	1573.97	1.01	1.01	2.37	3.94
128	366.78	5440.22	12844.51	371.91	5550.42	13159.71	1.02	1.02	2.51	3.10

Table 16. Effect of the number of messages per packet (the packet’s *span*, not its byte length) on the single-stage tail. Each row is one service floor T ; every entry is the corpus median over 3512 windows of a message-weighted latency quantile, in μs , under real (Hawkes) arrivals with a single stage ($N = 1$). The *const* block charges every packet the same service T , the Part II assumption. The *span* block charges service that grows with the number of messages in the packet, at the slope measured on the live system. Both do the same mean work per window, so any difference between the two blocks is due to message count alone. The *span/const* columns give that difference directly as a ratio; a value near 1 means message count adds nothing. The *prop/const* columns repeat the ratio for a deliberately exaggerated model in which service is fully proportional to message count, as an upper bound. Read across: message count inflates the tail only at $T \leq 8 \mu\text{s}$, at or just above the tight-gap floor, where clustering builds little or no queue, and is negligible above it.

The two tail mechanisms occupy disjoint regimes. The *span/const* ratio is largest where the queueing tail is absent and smallest where the queueing tail dominates. At $T = 2$ and $4 \mu\text{s}$ the constant-service model has no tail ($p_{50} = p_{99} = p_{99.9} = T$, Section 4.1) while the span-dependent model has $p_{99} = 1.17T$ and $p_{99.9} = 1.77T$. Below the tight-gap floor there is a tail, but it is not a queueing tail. By $T \geq 32 \mu\text{s}$, where cluster queueing produces a p_{99} of $4.5T$ and rising, span contributes 0–2% on top.

Decomposition at the production operating point. The live system’s measured floor is $6.81\text{--}7.23 \mu\text{s}$ (Section 6.3), so the comparable row is $T = 8 \mu\text{s}$. Decomposing the message-weighted p_{99} excess over the floor into the part the constant-service model already produced (queueing) and the remainder introduced by span-aware service (serialisation):

T	floor	queueing excess	serialisation excess	serialisation share
2	1.99	0.00	0.34	100%
4	3.99	0.00	0.69	100%
8	7.97	0.56	1.09	66%
16	15.95	18.00	1.14	6%
32	31.89	111.74	0.78	1%
64	100.02	716.42	3.61	1%
128	371.91	5073.43	105.07	2%

At $T = 8 \mu\text{s}$ serialisation supplies 66% of the p_{99} excess and queueing 34%. The live measurement reports the same ordering: evaluating the position ladder at each stream’s own p_{99} position accounts for 71% to 80% of the observed p_{99} on four of its six streams, with the remainder left to queueing and to stalls. Applying that definition to the simulator at $T = 8$ gives 97%, higher than the live figure because the only residual the model contains is queueing; it has no stall population. Both

measurements attribute the larger share of the tail at the production service time to in-packet serialisation.

This is consistent with Part II. The queueing claim stands: below about $7.5\mu\text{s}$ of service, arrival clustering does not build a queue and splitting the servicing chain removes nothing. The span-dependent run adds that a second mechanism, in-packet serialisation, is active in the same regime; it is deterministic within the packet and a stage cut does not shorten it, while the queue that a large packet opens behind itself is taken up in Section 8.3. N^* , the p_{99} -minimising stage count, is unchanged at all seven service floors when service is made span-dependent: 1, 1, 1, 2, 4, 8, 8 under both models. The design rule of Section 10 is unaffected.

The in-packet ladder. Median latency by in-packet position at $N = 1$ reproduces the linear form of the live measurement. At $T = 8\mu\text{s}$ the corpus-median latency by position fits

$$\text{median} = 7.98 + 0.344 \times \text{position} \quad (\mu\text{s}),$$

recovering the $0.345\mu\text{s}$ per-message cost that was input to the model, which confirms that queueing contributes nothing at the median at this service time. The live NQ-book ladder is $7.23 + 0.312 \times \text{position}$; the forms agree and the levels differ by the difference between the $7\mu\text{s}$ measured decode path and the $8\mu\text{s}$ nominal service floor.

The two channels are additive, not coupled. In plain terms, there are two separate ways a message can end up slow. One is queueing: it waits behind a burst of earlier packets. The other is serialisation: it sits deep inside a large packet and waits for the messages ahead of it in that same packet to be decoded first. The question here is whether these two tend to happen to the same message at the same time. If large packets arrived mostly during bursts, for instance because the gateway coalesced more messages per datagram under load, a message could suffer both delays at once, the two would compound rather than add, and the Part II tables, which model queueing alone, would understate the tail. If instead large packets arrive no more often during bursts than at any other time, the two delays are independent, they simply add, and the Part II tables are correct as they stand. The two tests below find the second case.

Two tests were made. First, a control arm in which each window's span sequence is randomly permuted across packets, preserving the span marginal and the arrival times exactly and destroying only the association between them. If large packets arrived preferentially inside clusters, breaking that association would lower the tail. It does not: the ratio of the real to the span-shuffled p_{99} is 1.000 at every $T \leq 8\mu\text{s}$ and lies between 0.99 and 1.02 at every larger T , with no systematic sign. Second, measuring the association directly, the correlation between a packet's span and the logarithm of the gap that preceded it has corpus median -0.014 ; packets arriving after the tightest decile of gaps have mean span 1.010 against 1.043 for those arriving after the loosest decile, a ratio of 0.97. The association is weakly negative.

The live measurement reports the same instrument dependence: mean span rises with queue depth on ZN, where a quiet book is filled only by genuine bursts, but *falls* with queue depth on ES and NQ, whose rings back up with many small packets. Part II's corpus is NQ, and it behaves like the live NQ streams. The queueing and serialisation channels are therefore close to independent on this corpus, and the constant-service tables of Part II measure one of the two. Whether the channels couple on an instrument like ZN is open.

Part III

Best practices: implications for HFT systems design

7 The reference instance: KASPAR-HFT and its `fast_send` pipeline

This section introduces the system that the rest of Part III works on, and applies the design rule of Section 4.6 to it: cut a chain only where the cut shortens its slowest stage, and never once that stage is already below the feed’s tight-gap floor, because below the floor there is no tail left to remove. It first says what KASPAR-HFT is and names the two ways its actors hand messages to one another, one synchronous and one queued, because the choice between those two at each boundary is exactly what the design rule decides. It then lays out how the production pipeline is actually wired: which boundaries are queued and why, and which stages run as one unbroken chain on a single thread. Finally it runs the design equation on that chain. On the chain as it stands, about seven microseconds long, the rule says do not split it. On a longer chain, sixteen microseconds, it says make one balanced cut and no more. The section after this one turns those verdicts into a procedure for changing the real system one boundary at a time and measuring the result.

The recommendations in this Part are stated for HFT systems in general and worked out on one system. KASPAR-HFT [35, 36] is an open-source C++20 trading system for CME futures built on a shared-nothing actor framework, available at <https://github.com/vincent212/kaspar-hft>: every component (socket reader, feed arbitrator, SBE decoder, order book, strategy, order manager, iLink session) is an actor with private state and a mailbox, and components interact only by passing messages. It is at once a production system (MDP3 multicast in, full order books reconstructed order by order, authenticated iLink 3 sessions out) and, with the same actor code driven from recorded packet captures, a deterministic replay simulator in its own right. Part II does not run on that simulator: its measurements are made on a standalone tandem-Lindley model of the pipeline (Section 3.3), driven by packet-arrival timestamps extracted from the same class of capture. The framework supplies the architecture under study and the calibration of h ; it is not in the measurement loop. The `fast_send` paper documents the four extensions that make the actor model usable at HFT latencies: `fast_send` itself, a synchronous delivery in which the sending thread runs the receiver’s handler inline and takes the reply as a value; actor `Groups`, which co-schedule a set of actors on one thread behind a single mailbox; mailbox queue implementations selectable per actor; and a memory pool that removes per-message allocation and its tail. The defining property of `fast_send` is *receiver transparency*: a handler is written once and has no way to tell whether it was invoked synchronously or asynchronously, on which thread, or whether its sender is blocked. Section 8 builds on that property.

The `fast_send` paper [35] built and measured a production trading pipeline; that pipeline is the design this paper re-examines. It has three sections joined by exactly two cross-thread hops. Two socket-reader actors, one per redundant multicast channel, do nothing but drain their UDP sockets and hand datagrams to a buffer actor by an asynchronous `send`; that hop is asynchronous because a reader that blocked on its callee would drop packets under a burst. From the buffer, arbitration between the A and B feeds, SBE decode, order-book construction, and signal generation run as a single chain of `fast_sends` on one thread, with no queue and no context switch between stages. When the signal decides to act it hands the order to an order-management actor by a second asynchronous `send`, because the iLink write takes microseconds and blocking the decode thread on it would stall the market-data queue. The measured cost of each delivery path is given in Section 8. Against a

servicing chain of $\sim 7\mu\text{s}$ measured tick-to-book on live ES, NQ, and ZN feeds, the framework’s own per-message contribution is under 1% of the floor, and the same measurement attributed the latency tail to the arrival stream rather than to the framework; the stream was strongly clustered, with a Fano-implied branching-ratio ceiling of 0.85–0.97. That observation motivated the present paper.

Read against Part II, this pipeline is the single-thread servicing chain of Section 1.2 realised with actors: the whole T sits on one thread between two hops that were placed where blocking was unacceptable. The present paper adds that the same logic applies inside the chain once the chain is long enough for the feed to queue behind it, and the recommendations that follow are stated as changes to this pipeline: where to cut it, in what order to place the stages, which boundaries to leave on `fast_send`, and how to decide each by measurement.

The design rule applied to this pipeline. The rule of Section 4.6, cut only where the cut shortens the slowest stage and never below the feed’s tight-gap floor, can be run on the reference chain directly. It needs one input: the tail excess that a single stage of a given length s would carry, $\Delta_{\text{single}}(s)$. Table 9 supplies it. Section 4.6 showed that the excess depends only on the per-stage service T/N , so every entry of that table is the excess of one stage of length T/N , and read that way the table is a lookup: zero at $s \leq 4\mu\text{s}$, 0.61 at 8, 18.59 at 16, 114.14 at 32 and 721.82 at $64\mu\text{s}$.

Whatever the split of the $\sim 7\mu\text{s}$ chain across its natural boundaries (arbitration, SBE decode, book application, signal), every partition has $s_{\text{max}} \leq 7\mu\text{s}$, where Part II finds $\Delta_{\text{single}} = 0$: 0.00 at $s \leq 4\mu\text{s}$, and at $s = 7\mu\text{s}$ a tail on only 4.5% of windows with a median excess of 0.00 (Section 4.8). The chain has no queueing excess to remove, every cut returns exactly its hop tax, and $N^* = 1$. The tail it does carry comes from the few packets that carry many messages: their in-packet serialisation, which a stage cut does not shorten, and the queue their long service opens behind them (Section 8.3).

The design rule applied to a longer chain. The rule applies once the chain exceeds the floor, for instance when a strategy stage is added, or on a feed with a lower gap floor. Take a $16\mu\text{s}$ chain with the same four boundaries, decode and book now at $6\mu\text{s}$ each and the two small stages at $2\mu\text{s}$:

partition of the $16\mu\text{s}$ chain	s_{max} (μs)	hops	$\Delta_{\text{single}}(s_{\text{max}})$ (μs)	p_{99} excess + hop tax
single thread (16)	16	0	18.59	18.59
8 + 8 (arb+decode book+signal)	8	1	0.61	2.31
8 + 6 + 2 (three stages)	8	2	0.61	4.01
2 + 6 + 6 + 2 (all four split)	6	3	0.00	5.10

Every entry is read off Table 9 and the refined sweep of Section 4.8: $\Delta_{\text{single}}(16) = 18.59$, $\Delta_{\text{single}}(8) = 0.61$, $\Delta_{\text{single}}(6) = 0$. Three points follow. The balanced two-way cut is the best option: it removes $18\mu\text{s}$ of p_{99} excess for $1.7\mu\text{s}$ of hop cost, an eightfold improvement in the tail objective on these illustrative stage times. Three stages do not lower s_{max} , since the $8\mu\text{s}$ stage remains the bottleneck, so the second hop costs $1.7\mu\text{s}$ and removes nothing (Remark 2.3, third consequence). Splitting all four lowers s_{max} to 6 and removes the remaining $0.61\mu\text{s}$ of excess at a cost of 3.4, and is worse. The recommendation for a chain of this length is one cut, placed so that the two halves are as equal as the natural boundaries allow, with the larger half at ingress (Remark 2.4). The numbers are a prior for the end-of-build measurement of Section 8.4, not a substitute for it. The per-stage service times are illustrative and are replaced by the operator’s profile of the chain.

8 Implementation on KASPAR-HFT: two dispatch paths, one handler

The practical claim of this paper is that on Hawkes-clustered feeds the choice of `fast_send` versus asynchronous `send` at each actor boundary is a late-stage optimisation decision, made per boundary from measured end-to-end median and tail latency of the assembled pipeline on the target feed, rather than an early architectural commitment. KASPAR-HFT supports this workflow. This section states the property that makes it possible and the procedure it enables.

The design equation maps onto framework choices in a specific way. The paper’s model of an N -stage tandem assumes each stage’s actor runs on its own thread and sends its output to the next stage through an asynchronous mailbox; the hop cost h is the wall-clock latency of that handoff.

Two hop paths in the shipping KASPAR-HFT framework. Of the delivery mechanisms described in Section 7, two matter for the design equation. The default asynchronous `send` uses `BQueue`, a blocking mailbox built on `std::mutex + std::condition_variable + boost::circular_buffer`; a producer thread enqueues under the mutex and a consumer thread waits on the condition variable. This is the hop path the paper’s Section 3.3 model corresponds to, and halving its measured $\sim 3.4\mu\text{s}$ round trip gives the one-way hop cost $h \approx 1.7\mu\text{s}$ used throughout Part II.

The `fast_send` path is different: it is a *synchronous* dispatch that acquires a per-actor mutex, calls the receiver’s handler inline on the sender’s thread, and returns the reply. There is no queue and no cross-thread wake-up. Its per-call cost is much smaller than the `BQueue` path’s, but it is not a pipelined hop: the sender blocks until the receiver finishes, so a chain of `fast_sends` serialises through all stages on one thread and offers none of the throughput speedup of Proposition 1. The paper’s design equation does not apply to `fast_send` chains.

8.1 Receiver transparency

Every actor in KASPAR-HFT registers its handlers through the `MESSAGE_HANDLER(MsgType, handler_method)` macro. A handler is a function of one message argument; nothing in its body can access, observe, or condition on the delivery mode by which it was invoked. This property is called *receiver transparency* in Maciejewski [35] and is what makes the send-mode choice a purely sender-side decision.

Concretely:

- **Synchronous delivery:** `target->fast_send(m, this)`. The sender’s thread acquires `target`’s per-actor mutex, calls the receiver’s handler inline on the sender’s thread, and takes the reply as a return value. No queue, no cross-thread wake-up. Round-trip cost $\sim 30\text{ ns}$ per Table 4 of Maciejewski [35].
- **Asynchronous delivery:** `target->send(m, this)`. The message is enqueued on `target`’s `BQueue` mailbox (`mutex + condition_variable + boost::circular_buffer`); `target`’s own thread wakes up on the condition variable and runs the handler. Round-trip cost $\sim 3.4\mu\text{s}$ across threads, or $\sim 90\text{ ns}$ same-thread within a `Group`, per the same reference.

Both paths deliver the same handler with the same message. Neither the handler’s body nor any actor downstream of it can tell which path fired. Switching a boundary between the two is a one-line change at the sender’s call site — `target->fast_send(m, this)  $\leftrightarrow$  target->send(m, this)` — with no code change on the receiver side, no header change, no interface change, no test change.

8.2 Measurement, not local reasoning, decides

The end-to-end latency profile of an HFT pipeline is a joint property of (i) which stages exist and where they are cut, (ii) which pairs of adjacent stages are joined by `fast_send` versus `send`, and (iii) the arrival law of the input feed. Proposition 1 and Theorem 1 both fold across multiple boundaries: the throughput speedup is N -fold and the tail contraction is $N^{-\gamma}$ with $\gamma \geq 1$, so the joint optimum is not deducible from local reasoning at a single boundary. A designer who reasons one boundary at a time underestimates the aggregate tail benefit of adding hops; this is the reasoning behind the single-thread convention of Section 1.2.

Getting the joint optimum therefore requires end-to-end measurement of the assembled pipeline against a representative feed, and iteration over per-boundary flips. The observable objective is a two-tuple (p_{50}, p_{99}) of end-to-end latency under the target feed, with an operator-chosen trade-off between the two (κ of Section 4.6). The design equation of Section 4.6, $N^* = \arg \min_N [(N-1)h + \kappa \Delta(N)]$, or in its partition form the cut with the smallest reachable $s_{\max}$ and then the fewest hops, supplies a prior on where the optimum lies under a Hawkes feed; the measurement decides.

8.3 The production tail: a few large packets

This subsection measures the shipping serial configuration of the production system, one thread decoding every packet inline; it flips no boundary. It asks one question in plain terms. On the live feed, why is the slowest message in a thousand so much slower than the typical one, and what is the mechanism? The answer, on the ZN channel, argued from the fitted floor and slope rather than measured message by message, is that the tail is made by a very small number of packets that carry many messages. Such a packet is decoded one message after another on the one thread, so its last message waits for every message ahead of it, and every packet that lands behind it waits for the whole of it. The typical message, alone in its packet with nothing queued in front of it, sits on the floor.

Setup. Channel 344 (ZN), book stream, one 900 s window, recovery windows excluded, the shipping serial configuration (`cme_decode_workers 0`). Latency is the leg-1 socket-read-to-book-publish time of Section 6.3, per message. ZN is the instrument on which Section 6.6 left the coupling of span and bursts open. Two populations are reported: every message, and the *hot path*, the messages that arrived to an empty queue and were first in their packet, which is the decode with neither queueing nor in-packet serialisation in it.

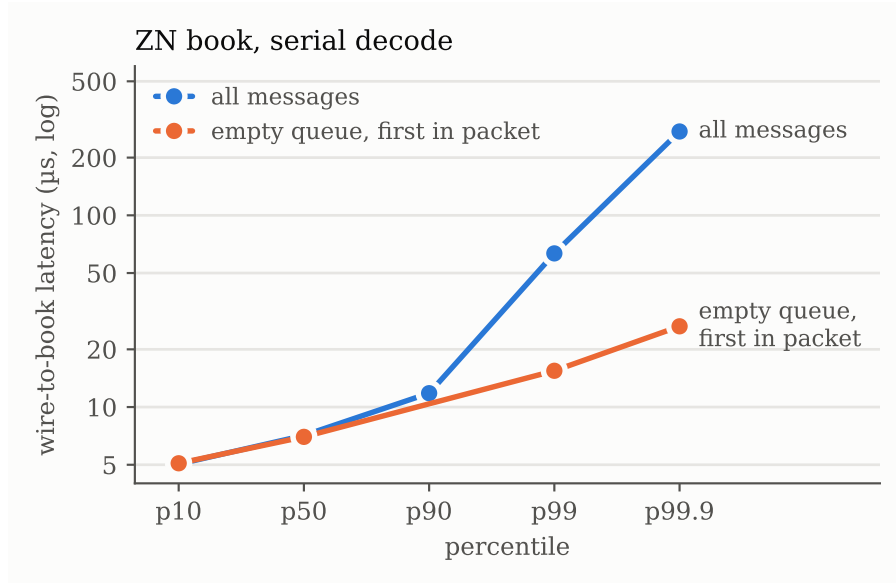


Figure 6. Wire-to-book latency by percentile on the ZN book stream, serial decode, for every message and for the hot path alone (empty queue, first in packet; its p_{90} was not reported). Log latency axis. The two populations share a floor near $5\text{--}7\mu\text{s}$ and separate only in the tail: the hot path reaches $26\mu\text{s}$ at $p_{99.9}$, all messages $274\mu\text{s}$.

population	messages	p_{10}	p_{50}	p_{90}	p_{99}	$p_{99.9}$	mean
all messages	157,699	5.04	7.11	11.81	63.30	273.69	9.95
empty queue, first in packet	122,500	5.09	6.99	—	15.46	26.37	—

Table 17. The data behind Figure 6, in μs . The hot path is 78% of all messages. The window maximum for all messages was $3046\mu\text{s}$; the report that records this measurement (tech_reports/serial_vs_parallel_decode.md in the KASPAR-HFT repository) flags a few such isolated spikes as possibly residual recovery, so it is not used here.

The spread. The slowest message in a thousand takes $274\mu\text{s}$ against $5\mu\text{s}$ for the fastest tenth and $7\mu\text{s}$ for the median: fifty-four times the one and thirty-eight times the other. On the hot path the same ratio is five. Conditioning on an empty queue and first position in the packet removes nine tenths of the far tail, $274 \rightarrow 26\mu\text{s}$, without changing the floor. The difference between the two tails is the contribution of queueing and of position inside the packet. The hot path itself is not at the floor in its tail (15.5 and $26.4\mu\text{s}$ at p_{99} and $p_{99.9}$ against a $7\mu\text{s}$ floor); that residual, present with neither queueing nor serialisation, is not resolved by these records.

The mechanism: span raises the service time linearly, and the queue behind it grows faster than that. Serialisation on its own cannot make the tail this long. Section 6.3 gives the ZN cost: a floor of $6.8\mu\text{s}$ plus about $1\mu\text{s}$ for each further message in the packet, so a packet’s service time grows in proportion to its span, and even the largest packet seen, forty-five messages (Section 6.5), costs its last message some $50\mu\text{s}$: a fifth of the $p_{99.9}$. The rest is queueing, and it is the large packets that start the queue. A deterministic server begins to build a queue when two arrivals are separated by less than its service time (Section 4.9). At one message per packet the service is $7\mu\text{s}$, at the tight-gap floor measured on NQ (ZN’s own gap floor was not measured), and a

queue almost never forms; the hot path in Table 17 is that regime. A forty-five-message packet holds the thread for $50\mu\text{s}$, seven times the floor, so every packet arriving in that window waits behind it, and each of those adds its own service to the wait of the ones behind it. The wait of the k -th packet in the queue is the sum of the service times ahead of it, so a run of two or three large packets, or one large packet landing inside an ordinary burst, produces a wait that grows with the number of packets queued, not with any one packet’s span. That is the non-linear step: span raises the service time linearly, and the raised service time lets a queue form whose wait compounds. The 0.47% of ZN packets carrying ten or more messages are few, but each one opens such a window, and the hundreds of microseconds at $p_{99.9}$ are the queues they open. This is argued from the fitted floor and slope and the reported percentiles, not from a per-message decomposition of the window.

Across the six streams. The same measurement covers three instruments and, on each, a book stream and a trade stream. Figure 7 puts the six serial tails side by side and Table 18 adds, for each stream, its packet rate, the size of its packets and the cost of each extra message. Three things are visible at once. First, the tail does not follow the packet rate: NQ book is by far the busiest stream and has the smallest tail, ZN book carries a third of its rate and a tail an order of magnitude larger, and the trade streams, thirty to a hundred times quieter than the books, have larger tails than the books on every instrument. Second, on every instrument the trade stream carries more messages per packet than the book stream and has the larger tail. Third, ZN, with the largest trade packets and the highest cost per extra message, has the largest tails on both streams; its book packets are not large at the 99th percentile, but at three times NQ’s cost per message and a maximum span of 45 the excursions they produce are the longest of any book stream. The arrival clustering itself is similar across the three instruments: the Fano-implied branching ceilings of Section 6.2 span 0.85–0.97 across all six streams, and the gateway floor is the same. What separates a twenty-fold spread of tails across streams that share the same clustering is therefore not the arrival rate and not the clustering; it is the service excursion a packet imposes, the number of messages it carries times the cost of each, and the queue that excursion opens behind it. On the live system, the far tail is mostly span.

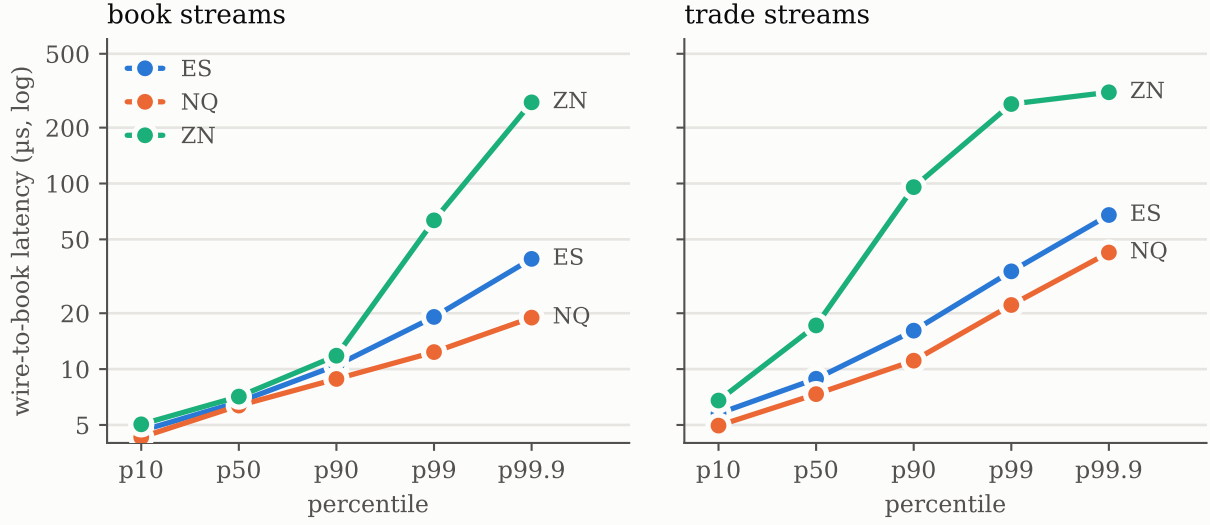


Figure 7. Serial wire-to-book latency by percentile on the six live streams, one 900 s window each, recovery excluded (book streams left, trade streams right, log latency axis). The tails order by messages per packet and cost per message, not by packet rate: the busiest stream (NQ book) has the smallest tail.

stream	packets	span p_{99}	$\mu\text{s}/\text{msg}$	p_{50}	p_{99}	$p_{99.9}$
NQ book	2,506,205	11	0.31	6.36	12.35	18.94
ES book	2,123,042	7	0.57	6.65	19.10	39.25
ZN book	824,536	6	0.97	7.11	63.30	273.69
ES trade	83,140	17	0.71	8.87	33.62	67.65
NQ trade	50,279	16	0.53	7.33	22.15	42.47
ZN trade	22,864	71	0.96	17.19	268.04	310.13

Table 18. The six streams of Figure 7, ordered by packet rate. Packets and span p_{99} are from the 53-minute capture of Section 6.5; the cost per extra message is the fitted slope of Section 6.3; the latency quantiles, in μs , are from the 900 s serial windows. Rate falls down the table while the tail rises; span p_{99} and cost per message, taken together, track the tail.

Consequences. The tail is therefore the mechanism the live data had already exposed. Section 6.5 measured that a packet’s decode time is not a constant but the floor plus one slope for each message beyond the first, with packets of up to 45 messages on the book streams and 85 on the trade streams, so that the largest packets take a disproportionate share of decode time. Section 6.6 re-ran the sweep with that span-dependent service and found four things: the message count inflates the tail only at service times at or just above the tight-gap floor, where it supplies about two thirds of the p_{99} excess at $T = 8 \mu\text{s}$; it adds a few percent at most once clustering takes over above $16 \mu\text{s}$; it leaves N^* unchanged at every service floor; and it does not compound with the bursts, because large packets are no more likely to arrive inside a burst than outside one. That is the serialisation channel. The ZN measurement adds the queue it induces, and places the tail exactly where Section 6.6 put it: in the far tail, at the production service time, on the largest packets. The typical packet, at the floor, is not the problem, and the verdict of Section 7 for it (do not split at the floor) stands. The large packet is a different regime: for the few packets that carry many messages the effective service is several times the floor, the regime in which the design rule of Section 4.6 admits a cut; the span-aware sweep of

Section 6.6, calibrated on NQ’s slope, a third of ZN’s, found N^* unchanged, so whether a cut pays on ZN is for the measurement to decide, while the per-message slope is the lever on the serialisation part (Section 10). Which lever to pull is decided by the end-of-build measurement of Section 8.4.

8.4 Operational discipline

The procedure is: draft the pipeline as a chain of actors at the natural cut points; wire each boundary with whichever of `fast_send` or `send` is convenient during development; once the pipeline runs end-to-end, measure the corpus-median p_{50} and p_{99} on a representative tape as a baseline; flip one boundary at a time, re-measure, and accept the flip if (p_{50}, p_{99}) improves under the operator’s trade-off; iterate until no single flip improves the objective. The full design flow, including the steps that precede this, is given in Section 10.

Under the arrival law of a CME MDP3 feed the design equation of Section 4.6 predicts what the iteration converges to: `async send` at every boundary whose removal would raise the largest stage, and `fast_send` everywhere else, in particular between short sub-tasks below the floor of Section 4.6 and between sub-tasks downstream of the bottleneck whose combined service stays at or below $s_{\max}$ (Remark 2.3). The measurement decides; the prediction indicates where to expect it to converge.

8.5 Separation of code construction from latency optimisation

A consequence of Section 8.4: in KASPAR-HFT two distinct topology decisions are not knowable at design time. The first is the *actor-to-thread mapping* — which actors share a thread by being placed in a common `Group`, and which run on their own thread. The second is the *per-boundary send-mode choice* — whether each pair of adjacent actors is joined by synchronous `fast_send` or asynchronous `send`. Neither decision is deducible from the source code alone. Both are functions of the target feed’s arrival statistics and the operator’s median-versus-tail trade-off, both of which are properties of the deployment environment. The optimum can only be measured from the assembled system running against a representative feed.

Under receiver transparency this is not a limitation. Actor code, message-type headers, and the full suite of unit and integration tests can be written and run before either the actor-to-thread mapping or the send-mode choice has been decided. Code construction and testing therefore proceed independently of the latency-optimisation consideration — both decisions are deferred to the end of the build, informed by measurements that could not have been taken any earlier because the system did not yet exist to measure. Correctness is established first; latency is optimised against the specific feed the system will run on, second. Two developers building two actors on the two sides of a boundary do not need to agree on whether their actors will ultimately live on the same thread or on different threads, nor on whether their handoff will be synchronous or asynchronous; they need to agree only on the message type. The message-type agreement is checked at compile time; the two topology decisions are parameters set at deployment.

This decoupling of correctness from latency optimisation is the practical consequence of receiver transparency, and distinguishes the design cycle from that of frameworks in which either topology decision enters the receiver’s interface (Section 8.6). Design proceeds without a commitment to a topology; the commitment is made once, near deployment, on the basis of measurements the code did not need to know about while it was being written.

8.6 Contrast with non-fungible frameworks

Most low-latency actor frameworks bake the sender’s choice of delivery mode into the receiver’s type. An Akka actor’s `tell` versus `ask`, a CAF actor’s `send` versus `request`, and an Erlang process’s `cast`

versus call each expose different receiver-side surfaces, so flipping a boundary from sync to async is a small refactor rather than a one-line change. Under those frameworks the design flow of Section 8.4 is not tractable, and the operator’s incentive is therefore to lock in the send-mode choice early — reasoning locally — rather than to defer it to an end-of-build measurement pass. This incentive is a plausible contributing cause of the single-thread convention: frameworks that penalise deferral push designers toward the topology that is least sensitive to a wrong early guess, which is the collapsed single-thread servicing chain.

KASPAR-HFT’s joint mailbox API of `fast_send` and `send`, with receiver transparency, is what makes measurement-driven send-mode optimisation practical. The design rule depends on this property operationally.

9 Other architectural considerations to re-evaluate under Hawkes

The paper’s central claim is that the single-thread hot-path recommendation does not hold above a service-time threshold on Hawkes-clustered feeds (Section 1.2, Section 4). Two adjacent conventions, pre-spinning receivers and pre-warming pipelines, are also affected by the arrival law, and this section examines them. Neither intervention is measured here; both are future work. The argument is that the arrival law provides much of the benefit these interventions are intended to deliver, on the tail quantile.

Pre-spinning receivers. The busy-poll recommendation of Rigtorp [40], Belay et al. [62], and the Aeron/Chronicle lineage (Section 1.2) pins a spinning thread per receiver so the wake path from a sleeping condition variable is never taken. On a Poisson feed, any arrival can find the receiver asleep, and the wake latency is paid on average; pre-spinning eliminates it. On a Hawkes feed with branching ratio $n \approx 0.8$, sleep/wake transitions happen only at cluster boundaries. Every p_{99} message is intra-cluster (Section 4.1: the p_{99} message waits behind a cluster, since at these service times the Poisson null produces no wait at all), so by the time a tail-setting message arrives, the receiver has been continuously processing for many multiples of the per-message service time and is already effectively spinning as a natural consequence of the load. Under the model used here, pre-spinning therefore saves little or no wake latency at the tail; this is a prediction of the queueing model, not a measurement of a production receiver’s state, and should be evaluated directly. It saves wake latency for the median and for the minimum-latency floor (inter-cluster arrivals which do find the receiver asleep after seconds-scale gaps), but the saving is bounded by $\sim 1.4\mu\text{s}$ per hop (the $1.7\mu\text{s}$ wake-path cost less the few-hundred-nanosecond mutex and cache-line floor of Section 4.10) and is much smaller than the tail win from splitting. There is also an operational cost: in the present author’s production experience, pinning spinning threads across the servicing chain has caused overall system degradation — steady core burn at 100%, cache thrash between the spin loop and the actor’s own working set, and pressure on shared resources like the LLC and memory bandwidth — that offset the median-latency gains; this is anecdotal and not measured here.

Pre-warming. Pre-warming is a subset of the wake-up cost discussed above. The $h = 1.7\mu\text{s}$ one-way hop cost that this paper calibrates already captures the entire process-wake-up path on the receiver side: the kernel context switch, the reload of the receiver’s instruction cache, the L1/L2 misses on its actor state and message pool, the TLB refill, and the scheduler putting the thread back on a core. To the extent the thread hop is dominated by cluster-wait on Hawkes feeds — because every p_{99} arrival is intra-cluster and the receiver is, under the model, already awake — the cache-warmth components inside that h should follow the same pattern, since they are the same

h decomposed. This is a consequence of the model rather than a measured result. Pre-warming interventions that fire synthetic traffic to keep instruction caches, data caches, TLBs, and branch predictors hot are therefore attacking the same h that Section 4 finds is dominated by cluster-wait. What such interventions can meaningfully deliver is confined to the median and the minimum — the inter-cluster first-arrival case where the receiver truly did cool down — and even there the delivery is bounded by the wake-up cost itself at the cost of continuous L1/L2 pollution from the warmup traffic. Under Hawkes the tail is unaffected.

This needs measurement. Pre-spinning and pre-warming are first-order tail-latency techniques when arrivals are uncorrelated. On self-exciting exchange feeds the arrival law itself provides the receiver-awake and cache-warm state on the tail quantile, so both interventions are second-order there and can be counterproductive once their steady costs are included. Both claims are corollaries of the paper’s main result rather than separate contributions, but each deserves a dedicated empirical evaluation on Hawkes-clustered feeds, analogous to the single-thread evaluation this paper carries out. The two open questions are:

- What is the actual ROI of pinned busy-poll receivers on the Hawkes-load $p_{50}, p_{99}, \text{max}$ triple, net of the steady core-burn cost and the cache pressure the spin loop imposes on the actor’s own working set?
- What is the actual ROI of periodic pre-warming traffic on the same triple, net of the L1/L2 pollution it creates and the residual advantage it delivers only on inter-cluster first-arrivals?

Both are direct extensions of the present paper’s methodology (measure under real CME MDP3 tapes; use a matched Poisson null to isolate the arrival-law dependence) and both are candidate follow-ons.

10 Recommendations for HFT system designers

The paper’s results qualify one architectural default (the single-thread hot path as an unconditional choice) and question two adjacent ones (busy-poll receivers, pre-warming). They do not contradict per-message code optimisation. This section separates what the paper leaves unchanged from what it argues against, and states the design flow the design equation implies.

Design principle (measure the crossover on the target feed). Whether a stage should be split is decided by how many packets arrive closer together than its service time, not by the packet rate or the branching ratio as such. That fraction is a property of the feed and its gateway. On CME MDP3 it is about 1% at a $7.5 \mu\text{s}$ floor and does not vary with rate or with the fitted branching ratio at HFT service times, so a $7\text{--}8 \mu\text{s}$ stage is not split. On a feed whose packing is set by the arrival process it can be tens of percent at the same fitted branching ratio, and the same stage is split (Appendix D). The crossover cannot be read off fitted parameters. Measure the end-to-end latency of the assembled pipeline under the load the system will see, and split where the measured tail excess exceeds the hop cost.

10.1 Recommendations left unchanged

Per-message hot-path optimisation — branch-free decoders, cache-line-aligned actor state, atomic-free single-writer queues, SBE decode in place, memory pools, `fast_send` at $\sim 30 \text{ ns}$, kernel-bypass NICs (Solarflare Onload, DPDK, VMA), `isolcpus`, `SCHED_FIFO`, huge pages, RTC counters instead of

`clock_gettime`, non-blocking IO on the socket, avoidance of allocation on the hot path, careful use of the branch predictor — remains necessary. All of it is contained in the service floor T , which the paper takes as given. The lower T is, the better the splitting rule performs: Proposition 1 gives an N -factor throughput speedup at any T , and the $1/N$ bound of Theorem 1 applies to a single-stage wait $\Delta(1)$ that itself scales with T . The splitting rule is therefore complementary to per-message optimisation: a $5\mu\text{s}$ -servicing-chain implementation of a two-stage tandem beats a $10\mu\text{s}$ -servicing-chain implementation of the same two-stage tandem at every quantile, and the difference is entirely in the per-message layer.

10.2 Recommendations the paper argues against

Two conventions in the layer above per-message optimisation. First, the single-thread servicing chain as the unconditional topology default (Section 1.2): on Hawkes-clustered feeds with T above the floor of Section 4.6, the tail is dominated by cluster wait and adding threads along the chain reduces it at a median cost of $(N - 1)h$. Second, pre-spinning receivers and pre-warming pipelines as always-on interventions (Section 9): on Hawkes-clustered feeds the receiver is already busy and warm at the arrival of every p_{99} message, so these interventions affect the median and minimum, not the tail, at a steady CPU and cache cost. The claim that a faster inner loop implies a single thread is preferable conflates two claims; the paper contradicts only the second.

10.3 Recommended design flow

The design flow that follows from the paper’s results, and that we recommend as the canonical HFT infrastructure workflow on self-exciting exchange feeds:

1. **Retain per-message optimisation.** This is orthogonal to the paper’s recommendation and determines absolute latency.
2. **Measure the target feed’s tight-gap floor, and use it as the split/don’t-split test.** Take the corpus of packet arrivals, form the per-window interarrival-gap distribution, and read off a low quantile of it — we use p_1 . This single number is the service time at which a deterministic stage begins to accumulate a tail, and on CME MDP3 channel 318 it is $\approx 7.5\mu\text{s}$ (Section 4.9). A stage whose service time sits comfortably below it will not develop a queue and must not be split: the split buys nothing and costs h at every quantile. A stage above it should be split, subject to the tail excess exceeding h , which on this corpus happens about $1.5\mu\text{s}$ further along (Section 4.8). Two cautions. First, use the tight end of the gap distribution, never the mean: on this corpus the mean gap is $2215\mu\text{s}$, some $300\times$ larger, and a rule stated on it would say never to split anything. Second, measure it under representative and peak-load conditions. On this corpus the floor is a property of the exchange’s packet-emission granularity and did not move between quiet and volatile sessions (Section 4.9), so it was measured once per feed; that invariance is a finding about this feed’s gateway, to be verified on another feed rather than assumed. At longer service times the tail does depend on intensity and branching ratio, and a chain in that regime needs its threshold measured under the load it will actually see.
3. **Draft the pipeline as a chain of actors at natural cut points, then balance it.** On a CME MDP3 handler: `SocketReader`, SBE decoder, order book, strategy actor, order manager — four or five stages, no artificial decomposition (Section 8). The tail factor is $s_{\max}/T$ (Remark 2.3), so the goal of the decomposition is the smallest largest stage, and the largest stage belongs at ingress so that all queueing sits in one mailbox (Remark 2.4).

4. **Wire boundaries with whichever of `fast_send` or `send` is convenient during development.** Receiver transparency in KASPAR-HFT makes this a one-line-per-boundary decision that can be deferred (Section 8).
5. **Do not commit to an actor-to-thread mapping at design time.** Grouping decisions and per-boundary send-mode decisions are both scalar dials at deployment, not code-structure decisions (Section 8.5).
6. **Measure the assembled pipeline against a representative feed.** Corpus-median p_{50} and p_{99} of end-to-end latency, per operator-chosen trade-off (Section 8.4).
7. **Flip one boundary at a time and re-measure.** Accept the flip if (p_{50}, p_{99}) improves. Iterate until no single-boundary flip helps.
8. **Prior on where step 7 converges.** On Hawkes-clustered feeds, async `send` at every boundary whose removal would raise the largest stage; `fast_send` at every other boundary, in particular between sub-tasks downstream of the bottleneck whose combined service stays at or below $s_{\max}$, and between short sub-tasks below the floor. A hop that does not lower $s_{\max}$ costs h at every quantile and compresses nothing.
9. **Do not pre-spin or pre-warm by default.** If pre-spinning or pre-warming is used, measure the ROI on a Hawkes-representative tape, net of the steady core-burn and cache-pollution costs. If the measurement improves the tail materially, keep the intervention. The paper’s prior is that on Hawkes feeds it will not (Section 9).

The workflow prioritises measurement over local reasoning because the arrival law of the target feed is a first-order determinant of the optimum and is not a property of the code.

Part IV

Conclusions

11 Conclusion

The paper put two questions to the prevailing view in HFT system design, which holds that the hot path should run on a single-thread event loop because every thread hop adds per-message latency. The answers qualify that view: it is correct below a service-time threshold and incorrect above it, and the threshold is set by the arrival process of the feed, which on CME is the self-exciting stream of matching-engine transactions.

Question 1: what produces the queueing tail. The self-excitation of the matching engine's transaction stream, and nothing else that was tested. The candidates were the packet rate, the clustering of arrivals and the number of messages per packet. The rate does not: utilisation is below 0.02 at every service time an HFT hot path occupies, and a window with four to five times the corpus-median packet rate has the same tail relative to T (Section 4.9); rate matters only from $64\mu\text{s}$, where ordinary utilisation queueing begins. The message count is a second, separate mechanism: decode time grows about $1\mu\text{s}$ per message, so a large packet delays its own later messages and holds the thread behind it; that is the live ZN far tail (Section 8.3), it dominates at or below the tight-gap floor and adds a few percent above $16\mu\text{s}$ (Section 6.6). The clustering is the cause, and Section 4.3 locates it. The arrival stream is the matching engine's transaction stream relayed one packet per transaction (98.3% of transactions); the packets of a split transaction arrive $15\mu\text{s}$ apart and 99.9% of gaps inside the floor separate different transactions; transaction starts are self-exciting with $n = 0.795$; and redrawing their times removes 75–98% of the tail while altering their sizes or packetisation leaves it within 15%. The clustering acts on the receiver through two channels whose weight depends on the service time: the surplus of near-simultaneous transactions over a rate-matched Poisson stream, which a permutation of the gap sequence preserves and which carries 79–88% of the tail at $8\text{--}16\mu\text{s}$, and runs of transactions, which the permutation destroys and which carry 63–78% at $64\text{--}128\mu\text{s}$ (Section 4.2). Second-scale load variation carries none of it.

Question 2: are two queues better than one. Above a service-time threshold, yes; below it, no; and the threshold is a property of the feed. On 3512 windows at $h = 1.7\mu\text{s}$ the corpus-median single-thread p_{99} is 2.2, 4.6, 12.8 and 41.5 times the service floor at $T = 16, 32, 64, 128\mu\text{s}$, a tail excess of 11 to 2900 times the hop cost, so above the floor the hop is small against what it lets the pipeline compress; below the floor the hop is cost only. Splitting a task into two half-length actors on separate threads joined by one hop lowers p_{99} on 100% of the windows (nested in 276 sessions) at every service floor from 16 to $128\mu\text{s}$, cutting the corpus-median tail excess to between 0.03 and 0.16 of its single-thread value for a median cost of exactly h ; the trade favours the split for any tail weight $\kappa \gtrsim 0.095$ at $T = 16\mu\text{s}$ and $\kappa \gtrsim 0.0004$ at $T = 128\mu\text{s}$, and at $T \leq 8\mu\text{s}$ the split loses on p_{99} . An N -stage tandem contracts the excess by a factor of N or better on every sample path (Theorem 1), in practice by far more where a split drives the per-stage service under the floor, while paying $(N - 1)h$ at every quantile; only its largest stage matters. The p_{99} -minimising stage count rises from 1 at $T \leq 8\mu\text{s}$ to 2 at 16, 4 at 32 and 8 at 64 and $128\mu\text{s}$, up to the number of natural cut points in the decode chain. Spending the same N cores as a dispatch pool that ignores ordering matches or beats the tandem on p_{99} and wins the median by $(N - 1)h$ (Section 4.4); the tandem's advantage is that it keeps an ordered stream in order without a resequencing stage, whose

cost in a dispatch design is not charged here. At the production operating point of the reference system, a socket-to-book chain of $6.8\text{--}7.2\,\mu\text{s}$, $N^* = 1$ and the chain should not be split (Section 7); the recommendation applies to stages above about $10\,\mu\text{s}$ on this feed.

The single-thread rule is a threshold rule. The single-thread rule is correct whenever the service time of the chain is below the feed’s threshold, and on this corpus that includes the reference system’s socket-to-book path. The rule holds in both directions: below the floor, splitting adds hop cost and removes no tail; above it, the cluster-wait tail exceeds the hop cost and splitting reduces it. The threshold is set by the feed’s arrival statistics. On this corpus it is set by the tight-gap structure of the transaction stream and is invariant to intensity and branching ratio at HFT service times; at longer service times both intensity and branching ratio raise the tail (Section 4.9). Near-critical branching ratios on CME futures ($n \approx 0.9$ on ES trades in the literature, $n \approx 0.8$ on NQ transactions and packets in this corpus) have been documented for over a decade, and those clusters are what produce the tail. These results are for one feed and one exchange gateway. Appendix D constructs a feed on which both dependences appear at $T = 8\,\mu\text{s}$ and on which splitting pays where on CME it does not. On another feed the threshold, and its dependence on load and on clustering, must be measured.

Practical consequences. The recommendation is that HFT designers measure end-to-end latency of the assembled pipeline under actual market conditions, including peak load, and split the servicing chain wherever the measurement shows a tail excess larger than the hop cost. Splitting the hot path is a design option of equal standing with optimising it. For a chain above the threshold, one cut at a natural boundary reduces the p_{99} tail by more than any per-message optimisation is likely to: on this corpus a balanced cut of a $16\,\mu\text{s}$ chain reduces the p_{99} excess plus hop cost from 18.6 to $2.3\,\mu\text{s}$ (Section 7), whereas per-message work moves the floor, and the per-message slope, by tenths of a microsecond. Below the threshold, for the typical packet, per-message optimisation is the lever; the far tail that a few many-message packets open on a live channel (Section 8.3) is a separate regime, whose remedy the end-of-build measurement decides. Framework designers should measure busy-poll dequeue on pinned isolated cores against the wait/notify mailbox on a Hawkes-representative tape rather than assume it is faster (Section 9), drive the mailbox mutex uncontended, keep memory-pool discipline, and cut the decode pipeline at whichever of its natural boundaries (4–5 on CME) minimises the largest stage, with that stage at ingress and everything downstream merged up to its service time; the tail factor is $s_{\max}/T$, and a hop that does not lower $s_{\max}$ is pure cost. Strategy designers of both POV-style and race-sensitive algorithms benefit for the same reason.

Revision of a prior design context. An earlier paper by the present author [35] did not itself argue for the single-thread servicing chain; it took the collapsed single-thread servicing chain as the widely accepted design context of the time and offered `fast_send` as the mechanism that makes actor composition inside it free of context-switch overhead. What the present paper retracts is the design context, not the mechanism: on Hawkes-clustered feeds the single-thread servicing chain should no longer be the unconditional default. The corrected position is that `fast_send` (synchronous, inline, sub-microsecond) and asynchronous `BQueue` handoffs (a real pipelined hop at a few microseconds) are *complementary* tools, and the choice between them is a per-boundary design decision that depends on the service time being chained and the arrival law of the feed the pipeline sits on.

In shipping KASPAR-HFT the `fast_send` versus asynchronous-send choice at each actor boundary is normally left open until the end of the build, and is resolved by measuring the end-to-end median

and tail latency of the assembled pipeline under the target feed. A convenient property of the KASPAR-HFT actor API is that the receiving actor’s `MESSAGE_HANDLER` is agnostic to how it was invoked: swapping a synchronous `fast_send` at a call site for an asynchronous `send` (or vice versa) is a one-line change on the sender side and requires no code change on the receiver side. The choice therefore lives entirely at each individual call site and can be flipped per boundary at end-of-build with minimal engineering cost. The paper’s design equation now supplies the arrival-law-aware default value of that end-of-build measurement: on Hawkes-clustered feeds, tasks above the threshold should be split into shorter actors joined by asynchronous sends, on the measured result that splitting reduces the tail by more than the hop cost raises the median. The rule presupposes that the task being split has a service time T comfortably above the floor of Section 4.6; a task of order h or below cannot be usefully split (a $1\ \mu\text{s}$ task cannot be turned into two 500 ns actors when a single hop alone costs $1.7\ \mu\text{s}$). Short synchronous chains inside one actor’s servicing of one message therefore continue to use `fast_send` unchanged. The end-of-build measurement decides any specific boundary.

Main caveats. The corpus is a single instrument (NQ front-month, channel 318); ES and BTC replications are follow-on work. A transaction is identified by equality of `transactTime`; whether the matching engine can publish one economic event under more than one `transactTime` is not known to the paper, and if it can, the transaction-level clustering reported in Section 4.3 includes that effect and the packetisation result is unaffected. The live measurements of Sections 6 and 8.3 come from one host on one afternoon, a 53-minute capture and one 900s window, without replication, and the 3512 windows are nested in 276 sessions, so uncertainty on the aggregates is session-clustered and no interval is reported. Constant per-packet service is a simplification; Section 6.6 removes it for the one source of variability the live data lets us measure, the message count per datagram, and finds the design rule unchanged on NQ, but per-message work that varies with book state is still not modelled. The cache-pollution cost of spinning receivers on many cores is bounded but not measured. And Theorem 1 is a bound whose slack depends on the arrival law; the paper measures that slack ($\gamma = 3.3$ and 2.7 where it is defined, against the guaranteed $\gamma \geq 1$) rather than deriving it, and a closed form for the slack on Hawkes input is open.

None of these caveats reverses the two answers on this corpus. The two tests deferred in an earlier draft, the three-null comparison of Section 4.2 and the equal-core comparison of Section 4.4, have been run, and Section 4.3 identifies the clustered process.

12 Future work

- Dispatch with its costs charged: an ingress stage, a hop to and from N whole-packet servers, and an in-order resequencer, under constant and span-dependent service, reported beside the tandem and the uncharged $M/D/N$ bound of Section 4.4.
- Multi-instrument replication (ES, BTC).
- Closed form for the slack in Theorem 1 under Hawkes input, i.e. the exponent $\gamma - 1$ as a function of $(\rho, n, \beta T)$. By the reduction this is a single-server question about $M^{[K]}/D/1$ with Borel batches at two service levels, which is tractable; the paper measures γ instead.
- Span/clustering coupling on a coalescing-heavy feed. The span-aware rerun of Section 6.6 finds the serialisation and queueing channels close to independent on NQ, matching the live NQ streams; the live ZN streams show span rising steeply with queue depth instead. Whether the two channels compound on an instrument of that kind, and what that does to the design rule, is not answered here.

- Cache-pollution empirical measurement per spinning-receiver stage count.
- Power-law-kernel Hawkes fit versus exponential.
- Overnight-session regime.
- Own-capture pcap validation of the network sub-stage.
- Load-conditioned hop cost $h(\rho)$. This paper calibrates $h = 1.7 \mu\text{s}$ from the low-load wake-latency figure in Maciejewski [35]; under sustained-load conditions inside a Hawkes cluster the receiver never sleeps and the effective h collapses toward the mutex + cache-line-transfer floor of a few hundred nanoseconds (Section 4.10). Measuring $h(\rho)$ directly and re-running the sweep against the load-conditioned h would sharpen the reported tail-win, which the current single- h simulation understates.

13 Reproducibility

All results in Part II are reproducible from the KASPAR-HFT repository [36], <https://github.com/vincent212/kaspar-hft>, directory `arrival_paper/`, which holds the packet extraction, the per-window Hawkes fit, the (T, h, N) tandem Lindley sweep with its Poisson null and per-window bound check, the span-aware rerun of Section 6.6, the synthetic feed of Appendix D, and the code that regenerates every table and figure from the cache. The Poisson null is seeded per window from an MD5 of `(session, window_id)`, so the cache is byte-reproducible.

The live measurement of Section 6 uses the `LatencyProbe` actor in the same repository and the readers under `kaspr/perf/`; its latency results are published in Maciejewski [35].

The 281 session tapes are derived from licensed CME MDP3 channel 318 pcaps (NQ front month, 2025-01 through 2026-02) and are not redistributed; the extraction script and the cached per-window metadata are released.

References

- [1] Hawkes, A. G. (1971). Spectra of some self-exciting and mutually exciting point processes. *Biometrika* 58, 83–90.
- [2] Hawkes, A. G., Oakes, D. (1974). A cluster process representation of a self-exciting process. *Journal of Applied Probability* 11(3), 493–503.
- [3] Tanner, J. C. (1961). A derivation of the Borel distribution. *Biometrika* 48(1–2), 222–224.
- [4] Asmussen, S. (2003). *Applied Probability and Queues*, 2nd ed. Springer. Chapter X (heavy-tailed and subexponential asymptotics for single-server queues).
- [5] Gao, X., Zhu, L. (2018). Functional central limit theorems for stationary Hawkes processes and application to infinite-server queues. *Queueing Systems* 90, 161–206.
- [6] Jackson, J. R. (1957). Networks of waiting lines. *Operations Research* 5, 518–521.
- [7] Burke, P. J. (1956). The output of a queuing system. *Operations Research* 4, 699–704.
- [8] Kelly, F. P. (1979). *Reversibility and Stochastic Networks*. Wiley.

- [9] Kleinrock, L. (1976). *Queueing Systems, Vol. II: Computer Applications*. Wiley.
- [10] Whitt, W. (1983). The queueing network analyzer. *Bell System Technical Journal* 62, 2779–2815.
- [11] Friedman, H. D. (1965). Reduction methods for tandem queueing systems. *Operations Research* 13(1), 121–131.
- [12] Lindley, D. V. (1952). The theory of queues with a single server. *Mathematical Proceedings of the Cambridge Philosophical Society* 48(2), 277–289.
- [13] Foss, S., Korshunov, D. (2000). Steady-state asymptotics for tandem, split-match and other feedforward queues with heavy-tailed service. *Queueing Systems* 35, 65–80.
- [14] Baccelli, F., Foss, S., Lelarge, M. (2005). Tail asymptotics for monotone-separable networks. *J. Appl. Probability* 42(3), 793–812.
- [15] Heindl, A. (2001). Decomposition of general tandem queueing networks with MMPP input. *Performance Evaluation* 44(1–4), 5–23.
- [16] Kim, Y. H., Chydzinski, A. (2002). Connection-wise end-to-end performance analysis of queueing networks with MMPP inputs. *Performance Evaluation* 47(2–3), 137–156.
- [17] Dębicki, K., Mandjes, M. (2015). *Queues and Lévy Fluctuation Theory*. Springer.
- [18] Daw, A., Pender, J. (2018). Queues driven by Hawkes processes. *Stochastic Systems* 8(3), 192–229.
- [19] Koops, D. T., Boxma, O. J., Mandjes, M. R. H. (2018). Infinite-server queues with Hawkes arrivals. *Queueing Systems* 92, 27–52.
- [20] Chen, X., Blanchet, J. (2021). Perfect sampling of Hawkes processes and queues with Hawkes arrivals. *Stochastic Systems* 11(3), 264–283.
- [21] Sheldon, D., et al. (2023). Steady-state analysis and online learning for queues with Hawkes arrivals. arXiv 2311.02577.
- [22] Gao, X., Zhu, L. (2024). Single-server queues with state-dependent Hawkes arrivals. *Mathematics of Operations Research*.
- [23] Ozaki, T. (1979). Maximum likelihood estimation of Hawkes’ self-exciting point processes. *Ann. Inst. Statist. Math.* 31(1), 145–155.
- [24] Filimonov, V., Sornette, D. (2012). Quantifying reflexivity in financial markets: toward a prediction of flash crashes. *Physical Review E* 85, 056108.
- [25] Filimonov, V., Sornette, D. (2015). Apparent criticality and calibration issues in the Hawkes self-excited point process model: application to high-frequency financial data. *Quantitative Finance* 15(8), 1293–1314.
- [26] Achab, M., Bacry, E., Gaïffas, S., Mastromatteo, I., Muzy, J.-F. (2018). Uncovering causality from multivariate Hawkes integrated cumulants; analysis of order book flows using a nonparametric estimation of the branching ratio matrix. *Journal of Machine Learning Research* 18(192), 1–28; arXiv 1706.03411.

- [27] Hardiman, S. J., Bercot, N., Bouchaud, J.-P. (2013). Critical reflexivity in financial markets: a Hawkes process analysis. arXiv 1302.1405.
- [28] Bacry, E., Mastromatteo, I., Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity* 1(1).
- [29] Yoon, J. (2026). The world’s fastest matching engine algorithm. arXiv 2606.01183.
- [30] Byrd, D., Hybinette, M., Balch, T. H. (2020). ABIDES: Towards high-fidelity market simulation for AI research.
- [31] Frey, S., Li, K., Nagy, P., Sapora, S., Lu, C., Zohren, S., Foerster, J., Calinescu, A. (2023). JAX-LOB. arXiv 2308.13289.
- [32] Aquilina, M., Budish, E., O’Neill, P. (2022). Quantifying the high-frequency trading “arms race”. *Quarterly Journal of Economics* 137(1), 493–564.
- [33] Dean, J., Barroso, L. A. (2013). The tail at scale. *Communications of the ACM* 56(2), 74–80.
- [34] Zhao, K., Alizadeh, M., Pan, R., Ports, D. R. K., Perry, J., Yang, X., et al. (2023). Parsimon: scalable tail latency estimation for data center networks. NSDI 2023.
- [35] Maciejewski, V. (2026). Adapting the actor model of concurrency for high-frequency trading: synchronous message delivery (`fast_send`) and a tick-to-book latency study. arXiv:2609.21173. <https://arxiv.org/abs/2609.21173>
- [36] Maciejewski, V. (2026). KASPAR-HFT: an open-source C++20 actor framework and trading system for CME futures. Source repository. <https://github.com/vincent212/kaspar-hft>
- [37] Thompson, M. (2011). The Single Writer Principle. *Mechanical Sympathy*, 29 September 2011. <https://mechanical-sympathy.blogspot.com/2011/09/single-writer-principle.html>
- [38] Fowler, M. (2011). The LMAX Architecture. 12 July 2011. <https://martinfowler.com/articles/lmax.html>
- [39] LMAX Exchange (2011). Disruptor: high-performance alternative to bounded queues for exchanging data between concurrent threads. <https://lmax-exchange.github.io/disruptor/>
- [40] Rigtorp, E. (2020). Low latency tuning guide. <https://rigtorp.se/low-latency-guide/>
- [41] Chronicle Software (2023). Building fast trading engines: Chronicle’s approach to low-latency trading. <https://chronicle.software/insights/blogs/building-fast-trading-engines-chronicles-approach-to-low-latency-trading>
- [42] Real Logic (2016). Aeron: efficient reliable UDP unicast, UDP multicast, and IPC message transport. <https://aeron.io/>
- [43] Kwan, A., Wang, H., Luk, W. (2023). C++ design patterns for low-latency applications including high-frequency trading. arXiv 2309.04259.
- [44] Hu, W. (2024). The deterministic event-driven sequencer architecture: a competitive edge for high-frequency trading. Medium, October 2024. <https://medium.com/@hu.wenzhe124124/the-deterministic-event-driven-sequencer-architecture-a-competitive-edge-for-high-frequency-371cbfbe9c2f>

- [45] Schmidt, D. C. (1995). Reactor: an object behavioral pattern for concurrent event demultiplexing and event handler dispatching. In *Pattern Languages of Program Design*, J. O. Coplien and D. C. Schmidt (eds.), Addison-Wesley.
- [46] Kegel, D. (1999). The C10K problem. <https://www.kegel.com/c10k.html>
- [47] Sysoev, I. (2004). nginx: a high-performance HTTP server and reverse proxy. <https://nginx.org/>
- [48] Dahl, R. (2009). Node.js: JavaScript runtime built on the V8 engine. Presented at JSConf EU, Berlin, November 2009. <https://nodejs.org/>
- [49] Welsh, M., Culler, D., Brewer, E. (2001). SEDA: an architecture for well-conditioned, scalable internet services. *Proc. 18th ACM Symposium on Operating Systems Principles (SOSP)*, 230–243.
- [50] Welsh, M. (2010). A retrospective on SEDA. <https://matt-welsh.blogspot.com/2010/07/retrospective-on-seda.html>
- [51] Kohler, E., Morris, R., Chen, B., Jannotti, J., Kaashoek, M. F. (2000). The Click modular router. *ACM Transactions on Computer Systems* 18(3), 263–297.
- [52] Murray, D. G., McSherry, F., Isaacs, R., Isard, M., Barham, P., Abadi, M. (2013). Naiad: a timely dataflow system. *Proc. 24th ACM Symposium on Operating Systems Principles (SOSP)*, 439–455.
- [53] Marty, M., de Kruijf, M., Adriaens, J., Alfeld, C., et al. (2019). Snap: a microkernel approach to host networking. *Proc. 27th ACM Symposium on Operating Systems Principles (SOSP)*, 399–413.
- [54] Kaffes, K., Chong, T., Humphries, J. T., Belay, A., Mazières, D., Kozyrakis, C. (2019). Shinjuku: preemptive scheduling for μ second-scale tail latency. *Proc. 16th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 345–359.
- [55] Ousterhout, A., Fried, J., Behrens, J., Belay, A., Balakrishnan, H. (2019). Shenango: achieving high CPU efficiency for latency-sensitive datacenter workloads. *Proc. 16th USENIX NSDI*, 361–378.
- [56] Fried, J., Ruan, Z., Ousterhout, A., Belay, A. (2020). Caladan: mitigating interference at microsecond timescales. *Proc. 14th USENIX OSDI*, 281–297.
- [57] Chen, H., Delimitrou, C., Kozyrakis, C. (2021). Perséphone: a microsecond-scale time-sharing scheduler for wide service-time distributions. *Proc. 28th ACM SOSP*, 653–668.
- [58] DPDK Project (2024). DPDK Programmer’s Guide: run-to-completion and pipeline modes. https://doc.dpdk.org/guides/prog_guide/
- [59] Zhang, X. (2016). Pipelined FAST protocol market-data decoder for high-frequency trading. Chinese patent CN105654383A.
- [60] Vinnikov, A. (2014). Batch latency analysis and phase transitions for a tandem of queues. arXiv 1403.2400.

- [61] David, T., Guerraoui, R., Trigonakis, V. (2013). Everything you always wanted to know about synchronization but were afraid to ask. *Proc. 24th ACM Symposium on Operating Systems Principles (SOSP)*, 33–48.
- [62] Belay, A., Prekas, G., Klimovic, A., Grossman, S., Kozyrakis, C., Bugnion, E. (2014). IX: a protected dataplane operating system for high throughput and low latency. *Proc. 11th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 49–65.

Appendices

A The Borel batch-size tail at fixed branching ratio

This appendix records the calculation behind Equation (8) and the numerical check that the critical $k^{-1/2}$ survival tail of the Borel distribution is confined to a finite window at every $n < 1$.

Stirling's formula $k! = \sqrt{2\pi k} k^k e^{-k} (1 + O(1/k))$ in Equation (7) gives

$$\Pr(K = k) = \frac{(nk)^{k-1} e^{-nk}}{k!} = \frac{n^{k-1} k^{k-1} e^{-nk}}{\sqrt{2\pi k} k^k e^{-k}} (1 + O(1/k)) = \frac{1}{n\sqrt{2\pi}} k^{-3/2} (ne^{1-n})^k (1 + O(1/k)).$$

Writing $n = 1 - \epsilon$, $\log(ne^{1-n}) = \log(1 - \epsilon) + \epsilon = -\epsilon^2/2 - \epsilon^3/3 - \dots$, so the geometric factor is $\exp(-k\epsilon^2/2 (1 + O(\epsilon)))$. The pmf is therefore $\propto k^{-3/2}$ for $k \ll k^* = \epsilon^{-2}$ and geometric beyond; summing, $\Pr(K > k) \asymp k^{-1/2}$ on the same window and $\asymp k^{-3/2} e^{-k\epsilon^2/2}$ beyond it. At $n = 1$ exactly, $k^* = \infty$, $\Pr(K > k) \sim \sqrt{2/\pi} k^{-1/2}$ (Borel–Tanner), and $E[K] = \infty$; that is the unstable case $\rho = \infty$, which no stationary queue attains.

Because the tail is geometric, K has finite exponential moments $E[e^{\theta K}] < \infty$ for $\theta < \epsilon^2/2 (1 + O(\epsilon))$ and is therefore outside the subexponential class $\mathcal{S}$. Heavy-tail queueing results that assume subexponential batch work, including the single-big-batch principle for $M^{[K]}/G/1$ and the monotone-separable tail theorem of Baccelli et al. [14], do not apply, and the stationary waiting time of the batch-arrival queue has a Cramér–Lundberg (exponential) tail at every fixed $n < 1$.

Table 19 reports the local log-log slope of the exact pmf, $\Delta \log \Pr(K = k) / \Delta \log k$ between successive k , which a pure $k^{-3/2}$ law would hold at -1.5 .

n	k^*	$10 \rightarrow 30$	$30 \rightarrow 100$	$100 \rightarrow 300$	$300 \rightarrow 1000$	$1000 \rightarrow 3000$
0.80	25	−1.92	−2.84	−5.71	−14.96	−43.63
0.90	100	−1.59	−1.81	−2.48	−4.62	−11.26
0.95	400	−1.52	−1.57	−1.73	−2.25	−3.85
0.99	10000	−1.50	−1.50	−1.51	−1.53	−1.59

Table 19. Local log-log slope of the Borel pmf, Equation (7), between successive cluster sizes k . The critical slope -1.5 holds while $k \ll k^* = (1 - n)^{-2}$ and steepens geometrically beyond.

The practical reading is that at the branching ratios fitted on the corpus ($n_{\text{pkt}} \approx 0.8$, Section 3.2), $k^* \approx 25$: clusters of order a hundred packets are the largest that occur with non-negligible frequency, and the single-stage wait behind the largest clusters has scale $Tk^* \approx 25T$, against the corpus-median p_{99} tail excess of Section 4, which is zero below the tight-gap floor and rises from $1.2T$ at $T = 16 \mu\text{s}$ to $39T$ at $T = 128 \mu\text{s}$. This is the object that Theorem 1 divides by N , and the reason the $n = 0.80$ row of Table 19 is the relevant one.

B Numerical check of Theorem 1

A direct simulation of the model of Theorem 1 checks both the identity Equation (4) and the bound Equation (6). Immigrants arrive as a Poisson process at rate $\mu = 0.004 \mu\text{s}^{-1}$, each spawning a Galton–Watson cluster with Poisson(0.9) offspring, so $K \sim \text{Borel}(0.9)$ and $E[K] \approx 10$; clusters are placed as instantaneous batches, giving $\rho = \bar{\lambda}T \approx 0.40$ at $T = 10 \mu\text{s}$. The tandem has $N = 4$ equal stages of $2.5 \mu\text{s}$ and $h = 1.7 \mu\text{s}$; 2×10^5 messages are simulated with an explicit per-stage FIFO recursion

(independent of the paper’s sweep code). Over all messages, $\max_j |W_j^{(4)} - (w_j(T/4) + 3h)| < 10^{-9} \mu s$ and Equation (6) is violated by no message. The quantiles are

q	$W_q^{(1)}$	$W_q^{(4)}$	$W_q^{(1)}/4 + 3h$	$(W_q^{(4)} - 3h)/W_q^{(1)}$
0.5	350.0	55.1	92.6	0.14
0.9	2066.6	355.1	521.8	0.17
0.99	5342.1	1127.6	1340.6	0.21
0.999	7586.2	1585.1	1901.7	0.21

in microseconds. The tandem lands below the bound at every quantile, with the ratio in the last column between 0.14 and 0.21 against the bound’s 0.25; the slack is the ρ/N utilisation effect described in the remark after Theorem 1. Under the Poisson null at the same message rate ($\rho = 0.40$, which is far above the corpus utilisation and is used here only to make the Poisson wait visible) the single-stage wait quantiles at $q = 0.9, 0.99, 0.999$ are $10.3, 24.8, 38.5 \mu s$ and the tandem’s are $5.1, 7.5, 9.2 \mu s$, again inside the bound. At the corpus utilisation ($\rho = 0.0045$ at this appendix’s $T = 10 \mu s$) the same two collapse to 0 and $3h$ at $q = 0.9$ and $q = 0.99$, which is Corollary 2; at $q = 0.999$ they do not, since ρ there exceeds $1 - q = 0.001$ and the corollary does not apply.

C Numerical check of the unequal-stage claims

The same generator as Appendix B (Borel(0.9) clusters, $\rho = 0.40$, $T = 10 \mu s$, $h = 1.7 \mu s$), with each cluster’s members spread uniformly over $3 \mu s$ instead of placed as an instantaneous batch and 4×10^5 messages, run through the partitions of Remark 2.3 with an explicit per-stage FIFO recursion. The last two columns are the tail-excess ratio $(\Delta_{99}(P) - \text{hops} \cdot h)/\Delta_{99}(\text{single})$ and the bound $s_{\max}/T$ from Theorem 1.

partition	$s_{\max}$	hops	p_{50}	p_{99}	$p_{99.9}$	ratio	bound
10	10.0	0	388.2	5719.0	8249.3	1.00	1.00
7 + 3	7.0	1	213.6	3429.5	5297.1	0.60	0.70
3 + 7	7.0	1	213.6	3429.5	5297.1	0.60	0.70
5 + 5	5.0	1	134.9	2265.5	3574.2	0.39	0.50
7 + 2 + 1	7.0	2	215.3	3431.2	5298.8	0.60	0.70
4 + 3 + 3	4.0	2	104.6	1746.5	2714.6	0.30	0.40
4 + 3 + 2 + 1	4.0	3	106.3	1748.2	2716.3	0.30	0.40
2.5×4	2.5	3	66.4	1049.8	1562.2	0.18	0.25

Latencies are end-to-end in microseconds. The three claims of Remark 2.3 read off directly: the tail excess depends on the partition only through $s_{\max}$ and sits below the bound at every row; $7 + 3$ and $3 + 7$ are identical to the reported precision; and $7 + 2 + 1$ differs from $7 + 3$, and $4 + 3 + 2 + 1$ from $4 + 3 + 3$, by exactly one hop cost $h = 1.7 \mu s$ at every quantile. The $3 \mu s$ intra-cluster spread, which is above $s_{\max}$ for the last row, is why the equal-stage ratio here (0.18) sits slightly below the batch-input value of Appendix B (0.21): a cluster that arrives spread out lets the faster tandem drain part of it before the next member lands.

D A feed on which the tail depends on intensity and criticality

Section 4.9 finds the normalised tail on CME MDP3 invariant to packet rate and branching ratio at HFT service times. This appendix constructs, with the same simulator, a feed on which both dependences appear at $T = 8 \mu\text{s}$, and identifies what distinguishes it from CME.

Arrivals are generated as a Hawkes process through its cluster representation: immigrants arrive as a Poisson process of rate $\mu = \bar{\lambda}(1 - n)$, and each spawns a Galton–Watson tree with $\text{Poisson}(n)$ offspring at exponential delays of mean $1/\beta$. Three conditions are run. In the first the kernel is fast, $1/\beta = 1 \mu\text{s}$, and there is no emission floor. In the second the same fast kernel is followed by a $7.5 \mu\text{s}$ emission floor: each arrival is delayed until at least $7.5 \mu\text{s}$ after its predecessor, modelling a gateway that serialises packet emission at the spacing measured on CME. In the third the kernel is slow, $1/\beta = 80 \mu\text{s}$, with the same floor; this is the condition closest to the CME gap structure. The grid is $n \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$ and $\bar{\lambda} \in \{500, 2000, 8000, 32000, 64000\}$ packets per second, so that $\rho = \bar{\lambda}T$ at $T = 8 \mu\text{s}$ runs from 0.004 to 0.51; 2×10^5 events per cell, three seeds, median reported. The last column is the fraction of interarrival gaps at or below the floor, measured at $\bar{\lambda} = 8000$; on the CME corpus that fraction is about 1% (Section 4.9).

	p_{99}/T at $T = 8 \mu\text{s}$, $N = 1$					$\text{gaps} \leq 7.5 \mu\text{s}$
n	$\bar{\lambda}=500$	2000	8000	32000	64000	at $\bar{\lambda}=8000$
<i>fast kernel, no floor</i>						
0.50	12.2	12.5	13.2	15.5	22.6	52%
0.70	38.9	40.7	41.2	47.3	66.6	71%
0.80	91.4	100.0	97.1	105.0	140.7	80%
0.90	421.3	429.4	411.3	413.7	620.4	90%
0.95	1544.2	1482.6	1721.3	1802.4	2487.2	95%
<i>fast kernel, $7.5 \mu\text{s}$ floor</i>						
0.50	1.75	1.75	1.88	2.44	4.38	53%
0.70	3.50	3.62	3.88	5.44	11.87	72%
0.80	6.81	7.38	7.56	10.88	23.12	81%
0.90	28.06	28.50	30.44	41.25	100.19	91%
0.95	98.06	94.75	116.36	193.96	382.18	95%
<i>slow kernel, $7.5 \mu\text{s}$ floor</i>						
0.50	1.12	1.12	1.19	1.51	3.56	13%
0.70	1.25	1.31	1.50	3.18	10.44	20%
0.80	2.38	2.71	3.12	8.12	19.56	28%
0.90	23.50	22.57	26.25	37.25	95.75	47%
0.95	90.12	86.19	112.19	188.78	364.30	67%
<i>Poisson, matched rate</i>						
—	1.00	1.39	1.87	2.69	4.45	—
ρ at $T = 8$						
	0.016	0.064	0.256	0.512		

Table 20. Synthetic Hawkes feeds at $T = 8 \mu\text{s}$. The ρ row gives $\bar{\lambda}T$ for the last four columns; the first column is $\rho = 0.004$.

Three observations.

Intensity. In every Hawkes condition the tail is flat in $\bar{\lambda}$ at $\rho \leq 0.02$ (the Poisson row rises from 1.00 to 1.39 over the same range) and rises from $\rho \approx 0.06$, steeply by $\rho = 0.5$. The CME corpus

at $T = 8\mu\text{s}$ spans $\rho = 0.002$ to 0.008 between its quietest and busiest quintiles, which is inside the flat region of every synthetic condition. The intensity invariance of Section 4.9 is therefore what a Hawkes model predicts at that utilisation, and it ends where utilisation queueing begins, as the CME sweep also finds at $T \geq 64\mu\text{s}$.

Criticality. The synthetic feeds depend strongly on n in every condition, by a factor of 55 to 130 between $n = 0.5$ and $n = 0.95$, and by a factor of 1.9 to 2.6 between $n = 0.7$ and $n = 0.8$ alone. The CME corpus shows no dependence over its observed range $n = 0.77\text{--}0.83$. The last column explains the difference. A Hawkes process at $n = 0.8$ places 28% to 81% of its interarrival gaps at the floor, depending on the kernel; the CME corpus places 1% there. On the synthetic feeds the branching ratio controls how many arrivals are packed at the minimum spacing, and that packing is what queues on an $8\mu\text{s}$ server. On CME the packed fraction is set by the gateway, is about 1%, and does not vary with the fitted n over the observed range; the fitted n describes the ordering of arrivals at longer timescales, consistent with Section 3.2, where the corpus is found to be self-exciting with heavy-tailed gaps rather than a pure exponential-kernel Hawkes process.

The design rule is feed-dependent. At $T = 8\mu\text{s}$, $n = 0.8$, $\bar{\lambda} = 8000$, a two-stage tandem has the lower p_{99} on every synthetic condition: $97.1 \rightarrow 47.4$ without a floor, $7.56 \rightarrow 1.21$ with the fast kernel and floor, $3.12 \rightarrow 1.21$ with the slow kernel and floor. On CME at the same T , $N^* = 1$ (Section 4.8). The service time at which splitting begins to pay is not a constant; it is a property of the feed.

In plain terms. Whether splitting a stage reduces its tail depends on how many packets arrive closer together than the stage’s service time, not on the packet rate or the branching ratio as such. On CME the gateway does not emit packets closer than about $7.5\mu\text{s}$ apart, and about 1% of gaps sit at that floor. That fraction does not change when the market becomes busier, since a busier market produces more clusters with the same spacing inside them, nor when the fitted branching ratio moves within its observed range. At the $7\text{--}8\mu\text{s}$ service time of a production decode path there is therefore little to queue behind, and one thread is the right design. On a feed on which the arrival process itself sets the packing, the result inverts: a Hawkes process at the same $n = 0.8$ places 28–81% of its gaps at the floor instead of 1%, the tail depends strongly on the branching ratio, splitting reduces the tail at $T = 8\mu\text{s}$ where on CME it does not, and at high enough load the packet rate matters as well. The Poisson null shows only the load effect, and only at high utilisation, as a null should. The rule is therefore neither one thread nor two. The crossover is a property of the feed’s packet spacing, it cannot be read off the fitted parameters, and it must be measured under the load the system will see. The CME gateway places this corpus below the crossover at $7\mu\text{s}$; another exchange, or the same exchange after a gateway change, may not. Section 10 states this as a design principle.